TITLE	DOCUMENT #	REVISION
8. FIELD SERVICE BULLETINES		
New Installation Procedure for Infinity 40-100	282	n/a
Intensity Fluctuations in the IR	287	n/a
Infinity Installations	290	n/a
DPSS Laser Electronics	292	n/a
Infinity Noise Specifications	293	n/a
Harmonic Mount Upgrades	305	n/a
Part Number Changes for Head and Power Supply Boards	315	n/a
CE-MARK Infinity Power Supply Change	319	n/a
New Part Numbers for Pre-mounted Optics	321	n/a
Power Supply Control Board Part Numbers	322	n/a
Faraday Isolator Design Change	324	n/a
Infinity Packing Procedure	333	n/a
Intermittent Shutdown of the Infinity Laser System	340	n/a
Infinity Water Level Low Fault	382	n/a
9. SCHEMATICS		
Cable, D.I. Water Sensor	0166-630-00	B
Power Supply Control Board	0167-823-00	A
Power Supply Control Board	0168-863-00	B
Triggering Board	0167-849-00	B
Booster Board	0164-777-00	D
Photocell Board	0166-267-XX	C
Signal Distribution Board	0166-343-00	C
Head Board #1	0168-755-00	A
Head Board #2	0168-757-00	A
Energy Modulator Board	0166-920-00	D

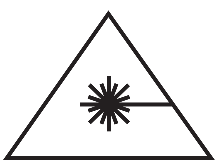
TITLE	DOCUMENT #	REVISION
QS-DPSS Controller Board	0164-777-00	D
DPSS Q-Switch Power Supply	0166-918-00	B
QS-DPSS Prelase Autolock Board	0167-781-00	A
Infinity Trigger BD	0167-849-00	D

Introduction

The Infinity laser is a high performance laser and has undergone extensive testing to ensure that, with proper usage, it is a safe and reliable system. Due to the nature of laser light and the performance level of the Infinity laser there are some important safety considerations that the you must be aware of. These relate to optical safety, electrical safety, and for the Infinity system, chemical safety. The chemical safety issues are due to CFC 113 used in the cell in which the phase conjugate return beam is generated. In order to use the Infinity system in a safe way, it is important that all users, and everyone near the laser system, are aware of the dangers involved.

Optical Safety

Laser light, because of its special properties, poses safety hazards not associated with light from conventional sources. The safe use of lasers requires that all laser users, and everyone near the laser system, are aware of the dangers involved. The safe use of the laser depends upon the user being familiar with the instrument and the properties of coherent, intense beams of light.



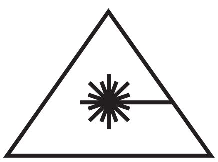
Direct eye contact with the output beam from the laser will cause serious damage and possible blindness.

The greatest concern when using a laser is eye safety. In addition to the main beam, there are often many smaller beams present at various angles near the laser system. These beams are formed by specular reflections of the main beam at polished surfaces such as lenses or beamsplitters. While weaker than the main beam, such beams may still be sufficiently intense to cause eye damage.

Laser beams are powerful enough to burn skin, clothing or paint. They can ignite volatile substances such as alcohol, gasoline, ether and other solvents, and can damage light-sensitive elements in video cameras, photomultipliers and photodiodes. The laser beam can ignite substances in its path, even at some distance. The beam may also cause damage if contacted indirectly from reflective surfaces. For these reasons, and others, the user is advised to follow the precautions below.

1. Observe all safety precautions in the preinstallation and operator's manual.

2. Extreme caution should be exercised when using solvents in the area of the laser.
3. Limit access to the laser to qualified users who are familiar with laser safety practices and who are aware of the dangers involved.
4. Never look directly into the laser light source or at scattered laser light from any reflective surface. Never sight down the beam into the source.
5. Maintain experimental setups at low heights to prevent inadvertent beam-eye encounter at eye level.

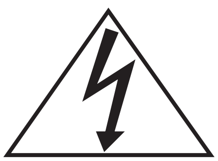


Laser safety glasses can present a hazard as well as a benefit; while they protect the eye from potentially damaging exposure, they block light at the laser wavelengths, which prevents the operator from seeing the beam. Therefore, use extreme caution even when using safety glasses.

6. As a precaution against accidental exposure to the output beam or its reflection, those using the system should wear laser safety glasses as required by the wavelength being generated.
7. Avoid direct exposure to the laser light. The intensity of the beam can easily cause flesh burns or ignite clothing.
8. Use the laser in an enclosed room. Laser light will remain collimated over long distances and therefore presents a potential hazard if not confined.
9. Post warning signs in the area of the laser beam to alert those present.
10. Advise all those using the laser of these precautions. It is good practice to operate the laser in a room with controlled and restricted access.

Electrical Safety

Lethal voltages and currents are present in both the Infinity power supply and laser head. For every laser pulse, up to 50 Joules of energy is deposited into the flash lamp, and the voltage across the capacitor bank can be as high as 570 V. High voltages are also present at various points on the circuit boards. These units are therefore designed to be operated with the protective covers in place. Certain procedures such as changing the flash lamp, water filter, or cleaning optical components, require removal of the protective covers. It is important that all safety precautions contained in this manual are observed by anyone using the laser. The most important rule when performing work on any part of the laser is first to switch it off completely!



There are no user serviceable parts in the electrical side of the power supply. Service procedures on system electronics must be carried out by a Coherent certified field service engineer. Always switch off the laser when working on any part of it.

Troubleshooting and adjustments on the circuit boards are strictly service procedures and should only be done by a Coherent field service technician. Unauthorized attempts by the user to modify or repair the circuit boards will void any warranty!

Chemical Safety

The Infinity laser incorporates a cell filled with CFC 113, which is used to generate the backward traveling return beam going through the power amplifiers. The choice of CFC 113 is motivated by its superior physical properties with respect to reflectivity, chemical stability and threshold for the phase conjugation process. Under normal operating conditions there are no situations in which the user will be exposed to the liquid. In the event of a cell leak or breakage, avoid direct skin and eye contact with the liquid. Ensure good ventilation until it has evaporated completely and the environment is odor free (CFC 113 has an ethereal and a faint sweetish odor). Call Coherent technical support for information about replacement cells and handling of damaged cells.

Safety Features and Compliance to Government Requirements

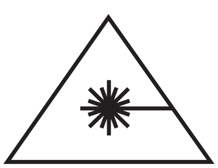
The following features are incorporated into the instrument to conform to several government requirements. The applicable United States Government requirements are contained in 21 CFR, subchapter J, part II administered by the Center for Devices and Radiological Health (CDRH). The European Community requirements for product safety are specified in the “Low Voltage Directive” (published in 73/23/EEC and ammended in 93/68/EEC). The “Low Voltage Directive” requires that lasers comply with the standard EN 61010-1 “Safety Requirements For Electrical Equipment For Measurement, Control and Laboratory Use” and EN60825-1 “Radiation Safety of Laser Products”. Compliance of this laser with the requirements is certified by the CE mark.

Classification

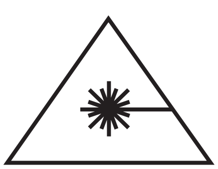
The governmental standards and requirements specify that the laser must be classified according to the output power or energy and the laser wavelength. The Infinity is classified as Class IV based on 21 CFR, subchapter J, part II, section 1040-10 (d). According to the European Community standards, the Infinity classified as Class 4 based on EN 60825-1, clause 9. In this manual and other documentation of the Infinity, the classification will be referred to as Class 4.

Protective Housing

The Infinity head is enclosed in a protective housing that prevents human access to radiation in excess of the limits of Class I radiation as specified in the Federal Register, July 31, 1975, Part II, Section 1040.10 (f) (1) and Table 1-A/EN 60825-1, clause 4.2 except for the output beam, which is Class IV.



Operation of laser with cover removed will allow access to hazardous visible and invisible radiation. The laser housing should only be opened for the purposes of maintenance and service by trained personnel cognizant of the hazards involved.



Use of controls or adjustments or performance of procedures other than those specified in the manual may result in hazardous radiation exposure.

Use of the system in a manner other than that described herein may impair the protection provided by the system.

Safety Interlocks

Electrical safety interlock breakers are positioned at four places in the Infinity system to prevent accidental exposure to dangerous levels of electricity or laser radiation. (21 CFR 1040.10 (f)(2)). The four interlocks are: head cover (1), flash lamp terminal housing (1) and amplifier housing latches (2).

External Interlock Connector

A RCA plug on the rear of the power supply is provided for use as an external interlock connection. This connector must be short circuited for the laser to operate. The laser is shipped with a plug installed. If this is removed then an external interlock circuit can be connected in its place (21 CFR 1040.10 (f)(3)).

Key Control

The instrument cannot be turned on or operated until the system key switch, located at the bottom of the power supply beneath the BNC connectors, is turned on (21 CFR 1040.10 (f)(4)).

Laser Radiation Emission Indicator

The appropriately labeled indicator lights on the laser head are turned on at least five seconds before laser emission can occur. It is visible without exposing the operator to laser emission. White light is used so that it may be seen regardless of the type of safety glasses employed (21 CFR 1040.10 (f)(5)).

Beam Attenuators

A mechanical shutter is provided with the Infinity. The shutter is actuated through the computer interface (21 CFR 1040.10 (f)(6)).

Manual Reset

After an interruption in laser activity because of remote interlock activation, or from an interruption due to loss of electrical power, the laser will not restart until the operator presses the On button in the main control window in the computer interface (21 CFR 1040.10 (f)(10)).

Location of Safety Labels

Refer to Figures 1.1-1 through 1.1-11 for descriptions and locations of all safety labels. These include warning labels indicating removable or displaceable protective housings, apertures through which laser radiation is emitted and labels of certification and identification [CFR 1040.10(g), CFR 1040.2, and CFR 1010.3/EN61010-1 EN60825-1, Clause 5].

References

More detailed information on laser safety can be found in the following publications:

"LIA Laser Safety Guide"
Laser Institute of America, 12424 Research Parkway, Suite 125, Orlando, FL 32826-3274, Telephone: (800)-345-2737 or (407)-380-1553

"*American National Standard for the Safe Use of Lasers*"
American National Standards Institute, 1430 Broadway, New York, NY 10018

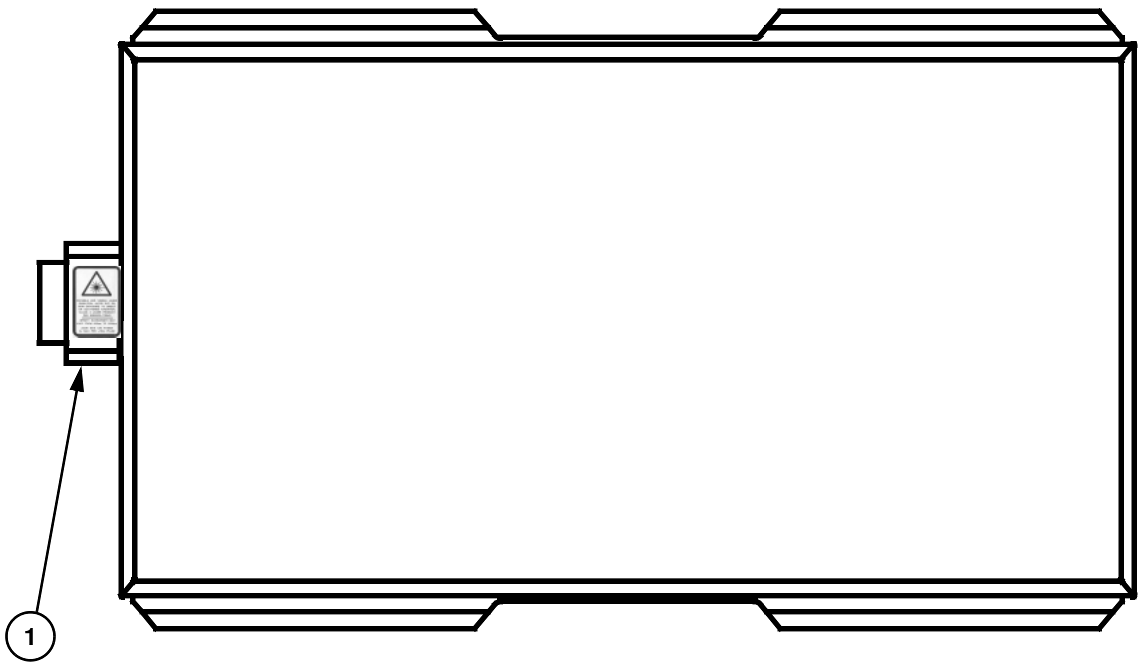
"Safety with Lasers and Other Optical Sources"
D. Sliney and M. Wolbarsht, Plenum Publishing Company, New York, N.Y. 1980

Additional safety information and material can also be found through the following sources:

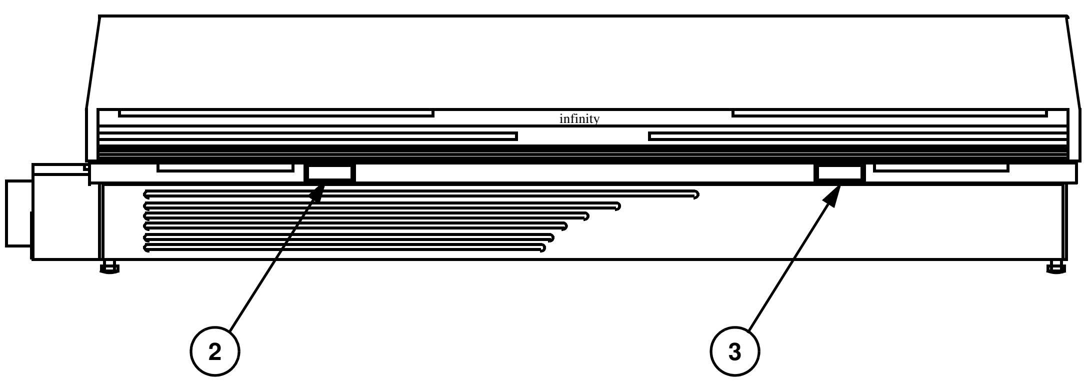
Laser Institute of America
Books, Safety standards, Training courses, Warning signs.
12424 Research parkway, Suite 125
Orlando, FL 32826-3274
Telephone: (800)-345-2737 or (407)-380-5588

Coherent, Inc.
"Laser Safety Comes to Light", Video
5100 Patrick Henry Drive
P.O. Box 54980
Santa Clara, CA 95056
Telephone: (408)-764-4800 or (800)-527-3786

Laser Focus World Buyer's Guide
List of suppliers of safety materials, e.g. signs, goggles etc.
One Technology Park Drive
P.O. Box 989
Westford, MA 01886
Telephone: (508)-692-0700



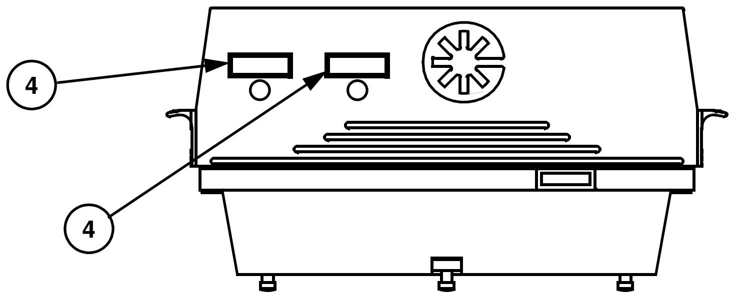
Laser Head Exterior, Top View



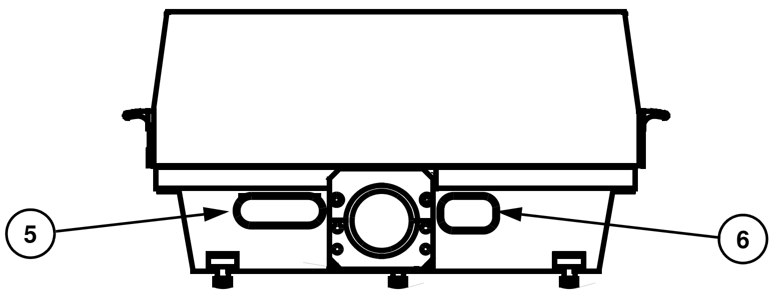
Laser Head Exterior, Side View

Note: Key is located on sheets 7 through 11.

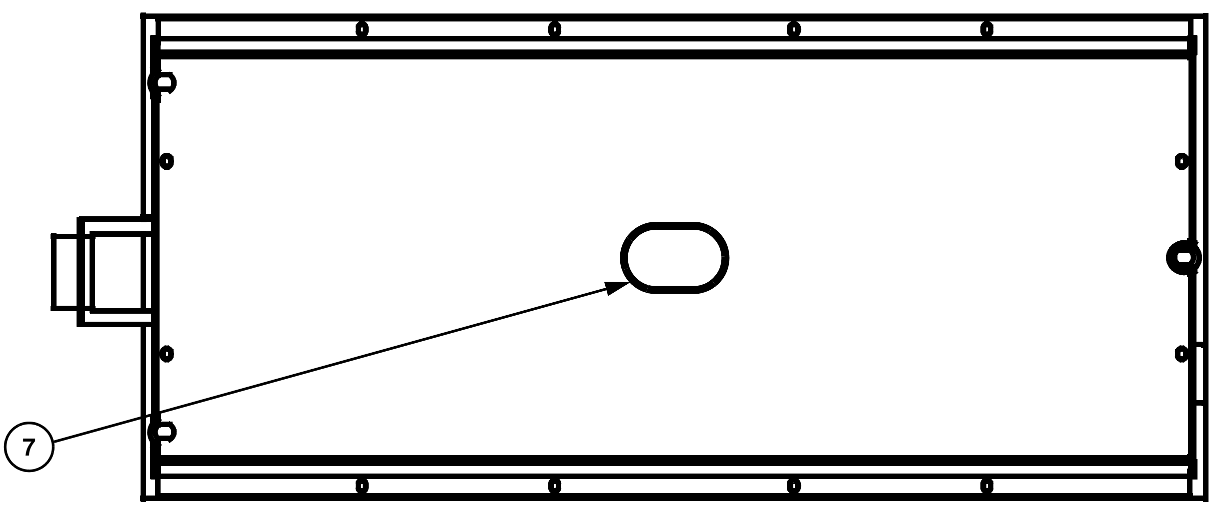
Figure 1.1-1. Location of Safety Labels (Sheet 1 of 11)



Laser Head Exterior, Front View



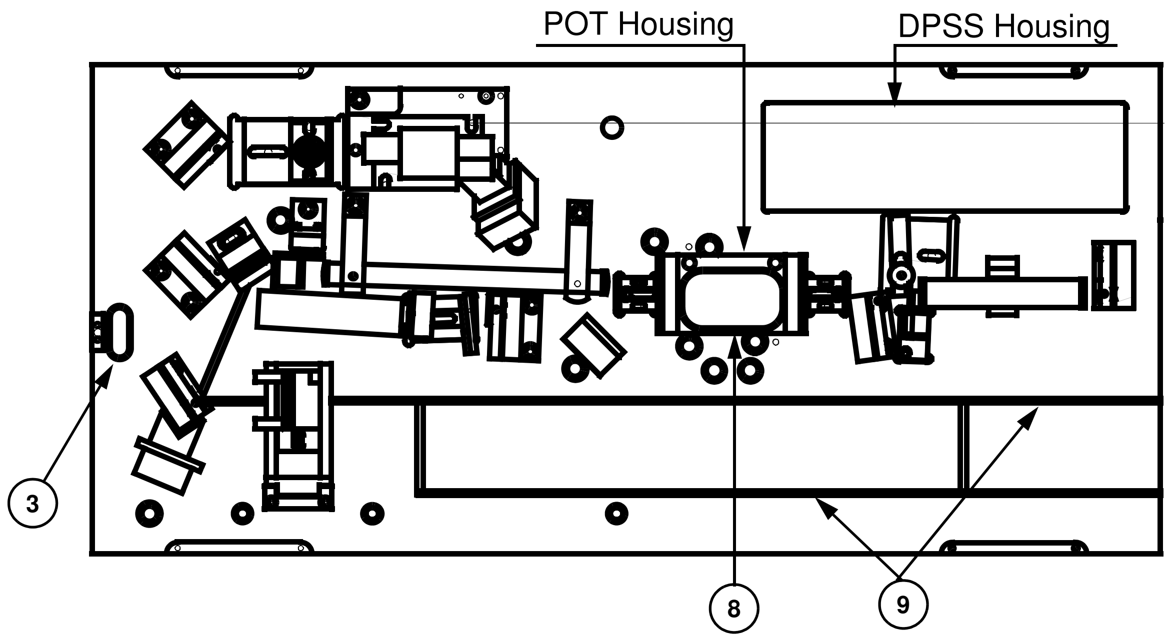
Laser Head Exterior, Rear View



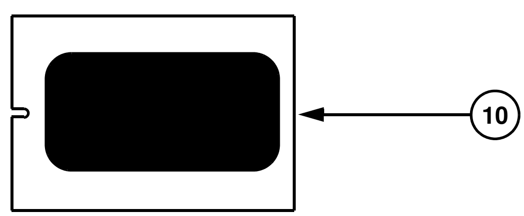
Laser Head Exterior, Bottom View

Note: Key is located on sheets 7 through 11.

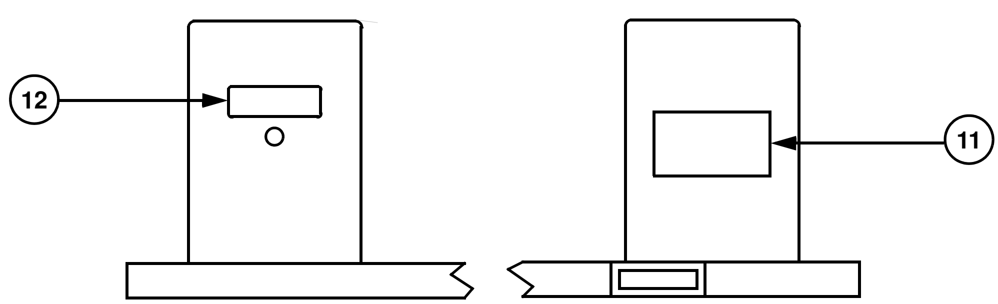
Figure 1.1-2. Location of Safety Labels (Sheet 2 of 11)



Laser Head Interior Without Cover, Top View



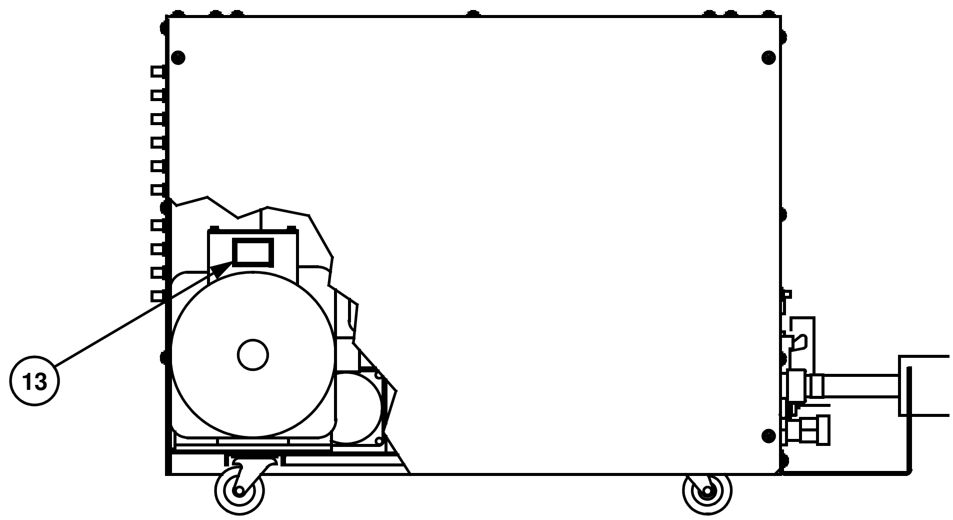
Laser Head Interior, POT Terminal Housing Cover



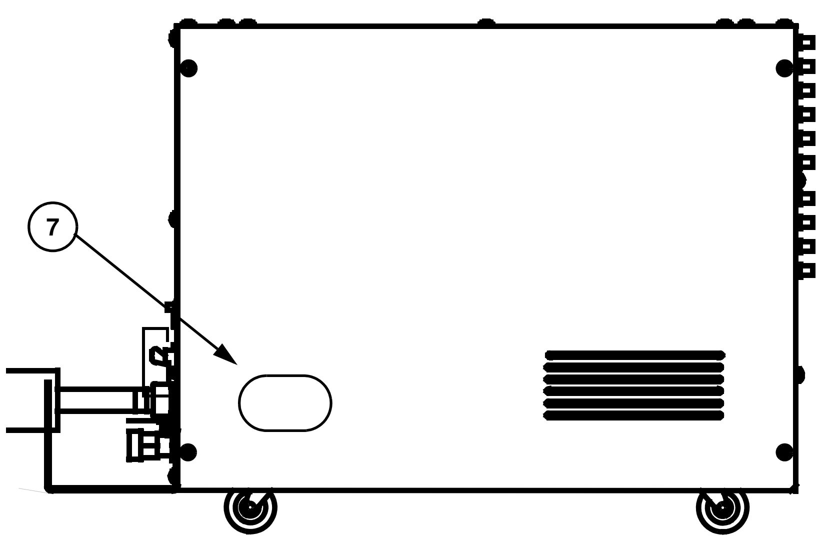
Laser Head Interior, DPSS Housing, Exit Aperture (12) and Rear (11)

Note: Key is located on sheets 7 through 11.

Figure 1.1-3. Location of Safety Labels (Sheet 3 of 11)



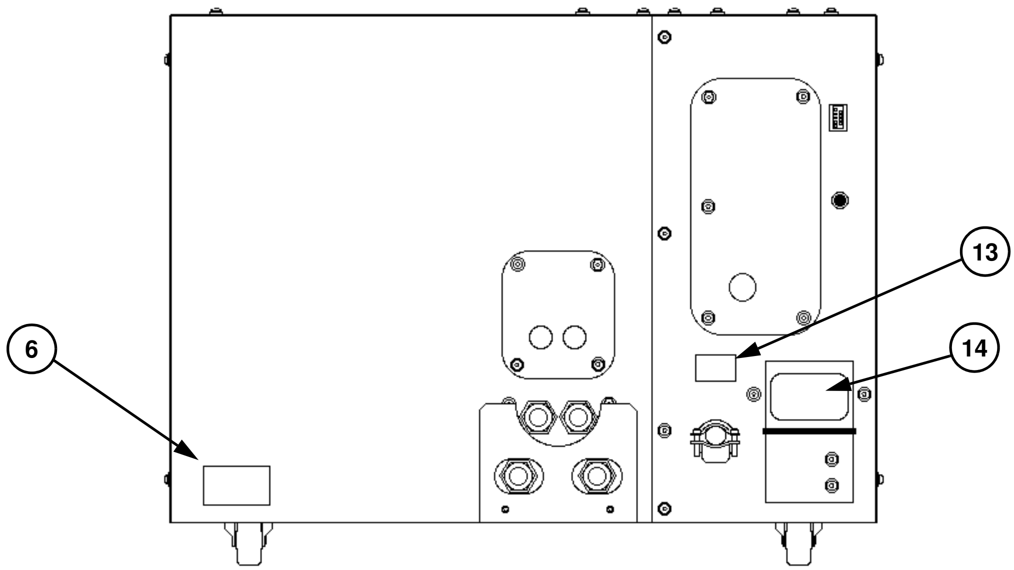
Power Supply, Cut-away, Right Side View, Water Pump



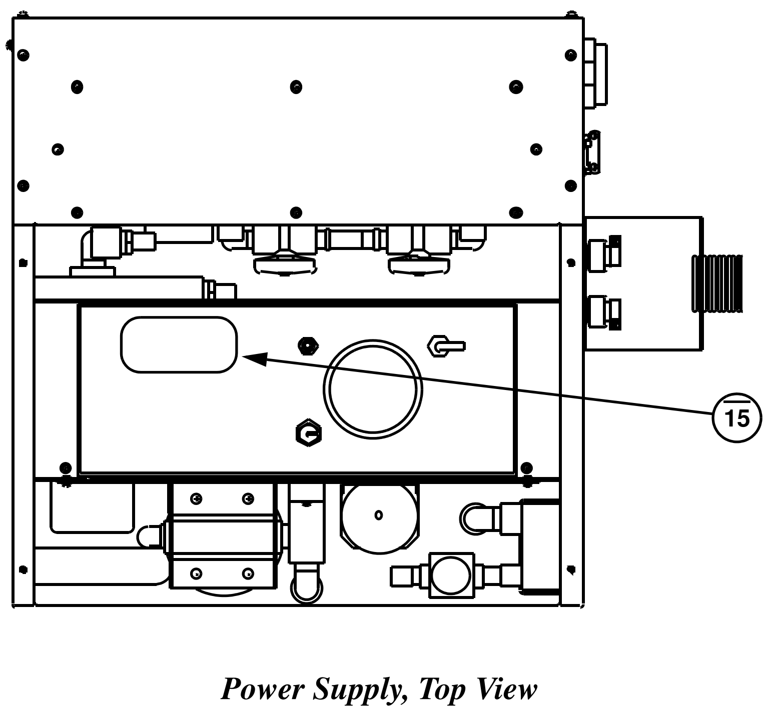
Power Supply, Left Side View

Note: Key is located on sheets 7 through 11.

Figure 1.1-4. Location of Safety Labels (Sheet 4 of 11)



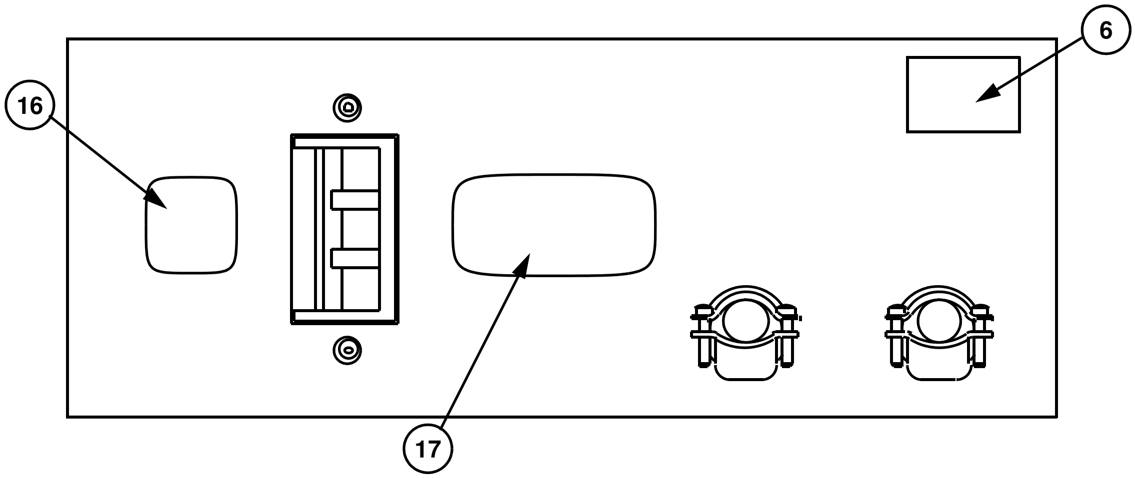
Power Supply, Rear View



Power Supply, Top View

Note: Key is located on sheets 7 through 11.

Figure 1.1-5. Location of Safety Labels (Sheet 5 of 11)



External EMI Filter (European Systems Only), Back View

Note: Key is located on sheets 7 through 11.

Figure 1.1-6. Location of Safety Labels (Sheet 6 of 11)

Label 1

HAZARDOUS
ELECTROMAGNETIC RADIATION
WHEN OPEN AND
INTERLOCK DEFEATED

Label 2

Label 3

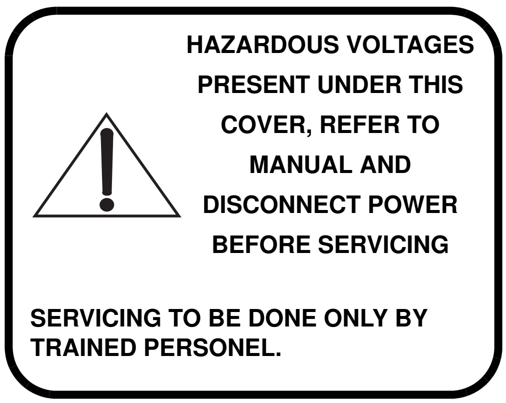
Label 4

WARNING: Contains CFC-113, a substance which harms public health and environment by destroying ozone in the upper atmosphere

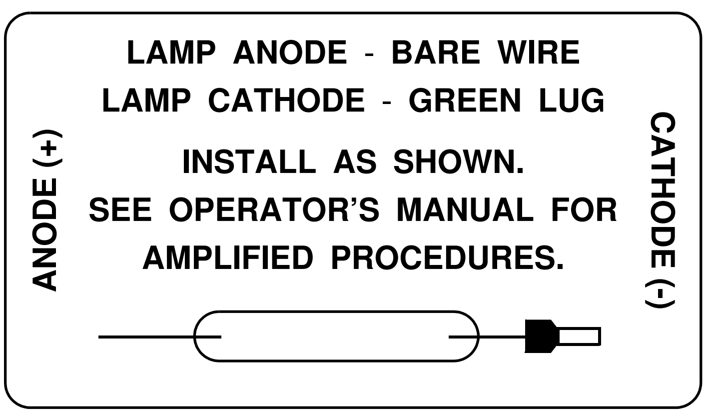
Label 5

Figure 1.1-7. Location of Safety Labels (Sheet 7 of 11)

Label 6



Label 7

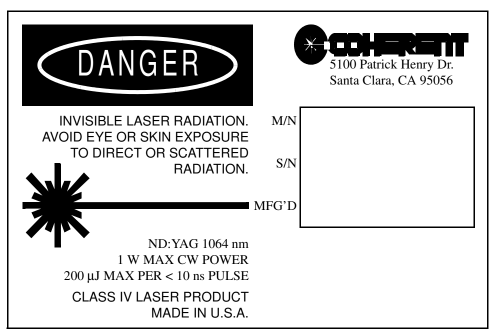


Label 8

Figure 1.1-8. Location of Safety Labels (Sheet 8 of 11)

Label 9

Label 10



Label 11

Label 12

Figure 1.1-9. Location of Safety Labels (Sheet 9 of 11)

CAUTION
THIS EQUIPMENT IS
CONFIGURED FOR
208V AC / 3-PHASE
OPERATION ONLY

CAUTION
THIS EQUIPMENT IS
CONFIGURED FOR
380V AC / 3-PHASE
OPERATION ONLY

Label 13

FUSE CARRIER ONLY

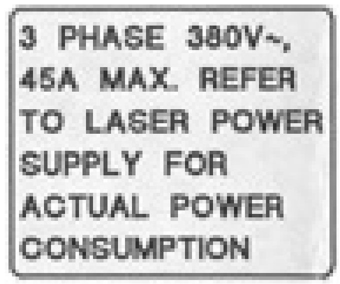
NOT TO BE USED AS
A DISCONNECT SWITCH

Label 14

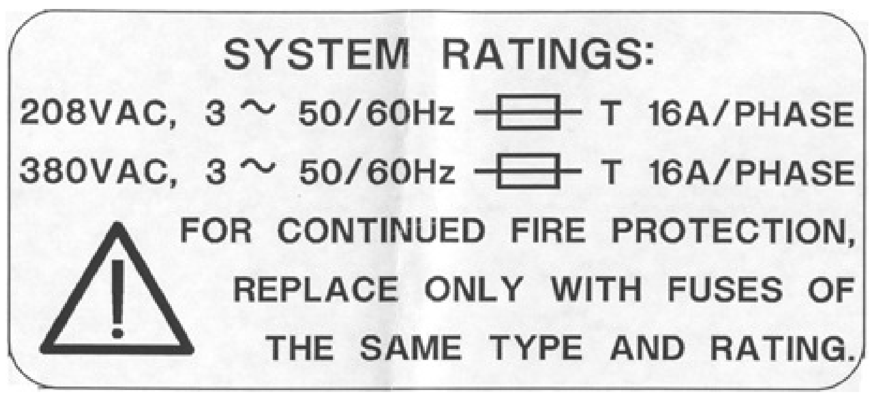
WARNING
Fill only with steam distilled water
with total dissolved solids of 3 mg/l or less,
and resistivity of 1 MΩ-cm or greater.
Deionized water is not sufficient.
Keep tank cover installed except when filling.
Do not allow deionization cartridge to dry out.

Label 15

Figure 1.1-10. Location of Safety Labels (Sheet 10 of 11)



Label 16



Label 17

Figure 1.1-11. Location of Safety Labels (Sheet 11 of 11)

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Installation Equipment

In addition to the standard maintenance hand tools and cleaning equipment, the following items should always be available when installing or servicing the Infinity laser system:

1. IR Viewing card.
2. LM-80V and LM-10 or 210 Power meters.
3. Beam expanding lens, f = -10 cm. For installations of IR and SHG systems the lens should be AR coated for 1064/532 nm. When installing a THG system an additional lens AR coated for 532/355 nm is needed.
4. Burn paper + plastic bags.
5. Weird angle mirror holder + HR IR mirror.
6. Photo diode.
7. CCD camera system.
8. Shear plate + holder.
9. Half wave plate for 1064 nm + holder (attenuator).
10. HR for 1064 nm + holder.
11. Two uncoated glass or quartz wedges + holders.
12. Set of neutral density filters + holder.
13. Faraday Isolator Apertures.
14. Isolation probe.

For a complete IR mode characterization (not part of a regular installation procedure) the following additional optic is required:

1. Lens, f = 50 cm + holder.

For Energy-in-the-Bucket measurements (not part of a regular installation procedure) the following additional equipment is required:

1. Energy meter, e.g. Gentec type.
2. Set of fixed size pinholes (The Coherent beam card).
3. XZ translator on which the pinholes can be mounted.
4. Focusing lens, f = 1.5 m, + holder.
5. Oscilloscope, to be provide by the customer.

Summary of Installation Procedure

The following is an outline of the Infinity installation procedure. Details and references for these steps are provided in the subsequent sections of this chapter.

1. Prepare the work space. Check facility water and electricity.
2. Uncrate the laser head and the power supply
3. Position the laser head on an optical table
4. Open the laser head cover and inspect for damage
5. Mount the umbilical holder to the power supply. Connect facility water to the power supply.
6. Remove the water side cover on the power supply.
7. Turn on the facility water and check for leaks inside the power supply and at the connections.
8. Connect the umbilical to the power supply. Connect the water hoses first to reduce stress on the electrical connectors.
9. Unpack the computer. Connect it to the power supply.
10. Turn on the computer and run the Infinity program
11. Install the new deionization cartridge. Flow arrow should point up.
12. Fill the reservoir with steam distilled water.
13. Open the Service window and start the pump by clicking on heat exchanger. Turn it off after a few seconds. Check for leaks in the laser head, at the umbilical connections and in the power supply. Run the pump for a few seconds again and check for leaks.

    Make sure that the pump is primed, i.e. circulates the water. This can be seen in the reservoir. If the water is circulating the water looks as if it “boiling”. Do not run the pump if the water is not circulated. The pump can cavitate which will contaminate the water! If the water does not start to circulate after the pump has been turned on and off a couple of times try:

    a. Open the latches holding the pot and tilt it slightly to the side so that air can enter. Close the latches and start the pump.
    b. Open the particle filter holder a few turns. Start the pump.

14. Turn on the DPSS laser and open the shutter. Follow the CW beam from the DPSS laser to the SBS cell.
15. Set the repetition rate to 10 Hz and the lamp voltage to 360 V (minimum).
16. Go to Standby. Check that the lamp is simmered.
17. Go to On. The lamp should now be flashing at 10 Hz.
18. Increase the lamp voltage manually, 5 V at a time, and watch for lasing after the thin film polarizer on the IR card.
19. When threshold is reached (very weak!) use the laser beam to align the power meter and expanding lens. Make sure that the beam is not reflected back into the laser by the lens or power meter!
20. Open the Monitor window.
21. Continue to increase the lamp voltage until 200 mJ/pulse is reached on the external meter. Verify the calibration on the internal meter. Check for arcing on the high energy side of the pinhole in the spatial filter. The depolarized energy should be below 2–3 mJ/pulse.
22. If the internal energy meter is calibrated, set the IR energy to 600 mJ in the main control window. Verify 6 W on the external energy meter. (Note that for older systems the maximum output energy is 5 W at 10 Hz.)
23. Check for arcing on the high energy side of the pinhole in the spatial filter. The depolarized energy should be below 1.5% of the pulse energy.
24. Take mode burns in the near field and outside the cover (base plate). Do not continue to operate the laser if there are signs of beam clipping or strong asymmetries in the mode!
25. Increase the repetition rate, in steps of 10 Hz, up to system maximum while monitoring the depolarized energy. Use the minimization procedure in Chapter 2 of this manual if the depolarization energy is higher than 1.5% of the pulse energy.

26. Check for arcing on the high energy side of the pinhole in the spatial filter. Verify system specifications to the customer; noise, IR power.

    If the noise is high, this could be an indication that one of the optical components moved during the shipment of the system. To determine which component misaligned, place the CCD camera behind the “relay mirror” mount and remove the mount face plate. Remove the vacuum cell and lens #1. If the intensity distribution is distorted, off center, or there is a strong diffraction pattern, one of the optics up stream of rod #1 has moved. Remove the optical components one at a time to determine which one is causing the problem. See the alignment procedure in Chapter 2 for viewing the DPSS output through lens and rod #2.

27. If previously arranged, demonstrate the Energy-in-the - Bucket measurements.
28. Install any harmonic and/or optical options and verify system specifications to the customer.
29. Instruct the customer on the functions of the Infinity software.
30. Demonstrate to the customer how to minimize the depolarization energy.
31. Demonstrate to the customer how the Infinity POT lamp is replaced.

Verification of Specifications

The IR specifications that are demonstrated to the customer at the time of installation are:

- 40 W average power at 100 Hz. Intensity noise ≤1.7%. (40-100 systems only)
- 25 W average power at 50 Hz. Intensity noise ≤1.0%. (40-100 and 50-25 systems)
- 15 W average power at 30 Hz. Intensity noise ≤1.0%. (15-30 systems only)
- 6 W average power at 20 Hz. Intensity noise ≤1.0%. (The specification for older systems is 5 W average power at 10 Hz.)

- Burn marks are taken in the near field (at the position of the SHG crystal) to verify that the mode is round and symmetrical with no side features. It is a good idea to show burn marks taken just outside the laser and explain that they are affected by diffraction (intermediate field).

Other specifications are only verified if this has been agreed on with the customer prior to the installation. Examples of such additional tests are mode characterization using a CCD camera and Energy-in-the-Bucket measurements.

Harmonics

For a systems equipped with harmonic options, the engineer should demonstrate that each of the options (i.e., SHG, THG, FHG,...) meets specification. Consult the Operator's manual for the latest power specifications.

When installing a FHG system a quartz beam expanding lens must be utilized. If such an expansion lens is not available the 266 nm output can be sent directly into the power meter for "short" periods of time.

Noise specifications are not verified for any of the harmonics, unless arrangements have been made prior to the installation date.

Computer Interface Demonstration

It is important that the customer is taught how to operate the system using the computer interface. A detailed description of the user interface is available in software chapter of the Operator's Manual. Utilizing this detailed discussion, go through each of the software windows with the customer.

Flash Lamp Replacement Procedure

Follow the procedure described in the Operator's Manual, and the sheet delivered with the flash lamp. Show how to reset the shot counter (under the lamp menu in the Service window). Start the laser and verify with the user that energy, noise and depolarized energy are the same.

Minimizing Depolarized Energy

For demonstration follow the procedure described in the Operator's Manual. Note that this procedure is not the same as the Service procedure. Explain some of the underlying theory of polarization control and relay imaging.

Energy-In-The-Bucket

This section describes how to perform an Energy in the Bucket measurement. The purpose of this measurement is to characterize the mode quality with respect to its divergence.

Set-Up Procedure

The experimental set-up is schematically displayed in Figure 1.3-1. A fraction of the beam exiting from the Infinity system is split off and focused through a pinhole. The transmitted power is measured using a energy meter connected to a oscilloscope. Since the beam is focused it is important that the pulse energy is kept low enough not to cause sputtering of the pinhole. Follow the set-up procedure below.

1. Adjust the lamp voltage so that the laser operates just above threshold for lasing.
2. Position a power meter or a beam dump in the beam path.
3. Insert wedge 1 as shown in Figure 1.3-1. Use a small angle (e.g., ~15°). A smaller angle reflects less energy.
4. Position wedge 2 in the path of the front surface reflection according to Figure 1.3-1. Make sure that it is the front surface reflection that is used! Locate the rear surface reflection and position a beam dump in its path.

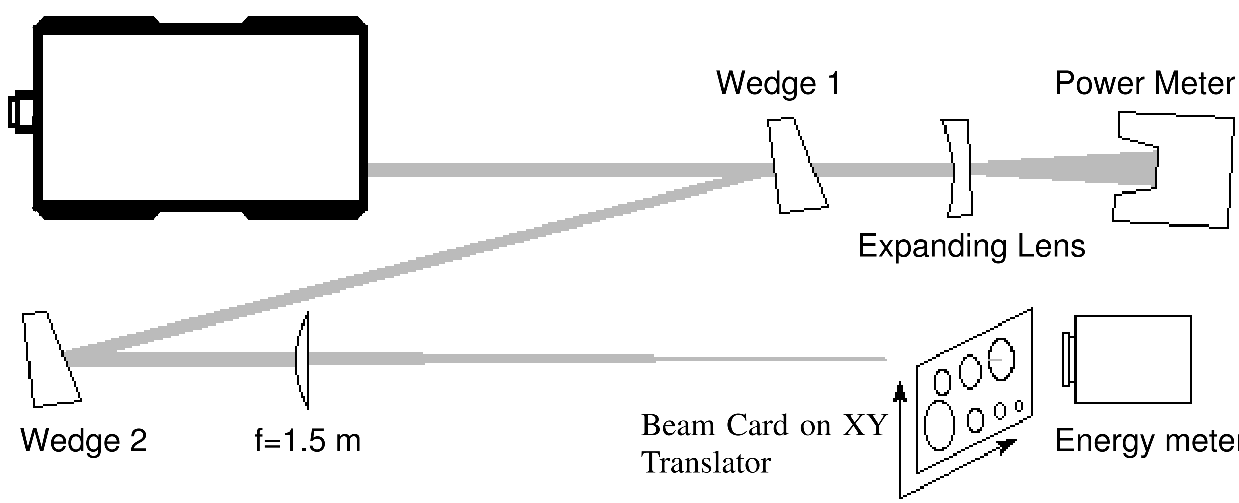

Figure 1.3-1. Experimental Set-Up for Energy-in-the-Bucket Measurement.

5. Direct the front surface reflection from wedge 2 to the energy meter, e.g. a Gentec. The energy meter needs to be a little further away than the focal length of the lens.
6. Locate the rear surface reflection from wedge 2 and position a beam dump in its path.
7. Position the lens so that the beam strikes the same spot on the energy meter. Remove the energy meter. Position a beam dump in the expanded beam after the focus.
8. Set the IR energy to the maximum allowed for the repetition rate chosen and run the laser until it has reached a stable operating temperature.
9. Position the XZ-translator so that the beam card is exactly one focal length away from the lens (for nominal f=1.5 m CVI lens this distance is 1.525 m). If necessary, position a neutral density filter at a slight angle in the beam after the focusing lens.
10. Position the energy meter right behind the beam card, ~0.5 cm. It is important for the accuracy of the measurement that all energy transmitted through the aperture is detected by the energy meter. When the focused beam passes through the smaller holes in the beam card the diffraction by the aperture edge will be significant. This will greatly increase the beams divergence. If the energy meter is positioned too far behind the beam card the energy striking the meter will not only be a function of the aperture size but also of how much it is diffracted by the aperture.
11. If a Gentec energy meter is used, connect the amplifier and the oscilloscope.

Data Acquisition Procedure

After the set-up procedure is completed, follow the steps below. Use the template provided in Table 1.3-1. Note that it only is correct for measurements of the IR beam and when using a focusing lens with a focal length of f=1.5 m.

Table 1.3-1 will only give meaningful data for measurements of the IR, 1064 nm, and when using a focusing lens with f = 1.5 m.

1. Remove the beam card from the focus and measure the signal. Enter this in the data acquisition sheet as the maximum signal, Tmax (normalizing point).
2. Insert the beam card at the beam waist. Position it so that the beam is centered on the first hole for which the transmission will be measured. Move the whole mount so the aperture is roughly centered on the beam and use the micro meter screws to optimize the transmission.
3. When the beam card is positioned so that the transmission is maximized, enter the energy meter signal in column B in Table 1.3-1. Repeat for all apertures in the table.
4. Repeat for all apertures in table.

Table 1.3-1. Template for Energy-in-the-Bucket Measurements for 1064 nm

A	B	C	D=B/TMAX
APERTURE [MM]	**MEASURED SIGNAL [MV]**	**NORMALIZED APERTURE [M]**	**NORMALIZED SIGNAL [%]**
No aperture, Tmax=		N/A	100
0.4		0.56	
0.5		0.71	
0.6		0.85	
0.7		0.99	
0.8		1.13	
0.9		1.27	
1.0		1.41	
1.1		1.55	
1.2		1.69	
1.3		1.84	
1.4		1.98	
1.5		2.12	

Data Evaluation

After measuring the transmission for all apertures evaluate the data according to the steps below.

1. Divide all entries in column B with Tmax, the signal measured with the beam card removed and enter the new, normalized, value in column D.
2. Plot the data points in column D as a function of the data in column C, i.e. Transmission as a function of Normalized Aperture.

Figure 1.3-2 shows a plot in which data points acquired at 10 Hz and 100 Hz repetition rates are entered. The solid line shows what diffraction theory predicts for a beam with a diameter of 5.5 mm and a top hat intensity distribution. The Infinity laser is specified to have a transmission of greater than 84 % for an aperture with M>1.5.

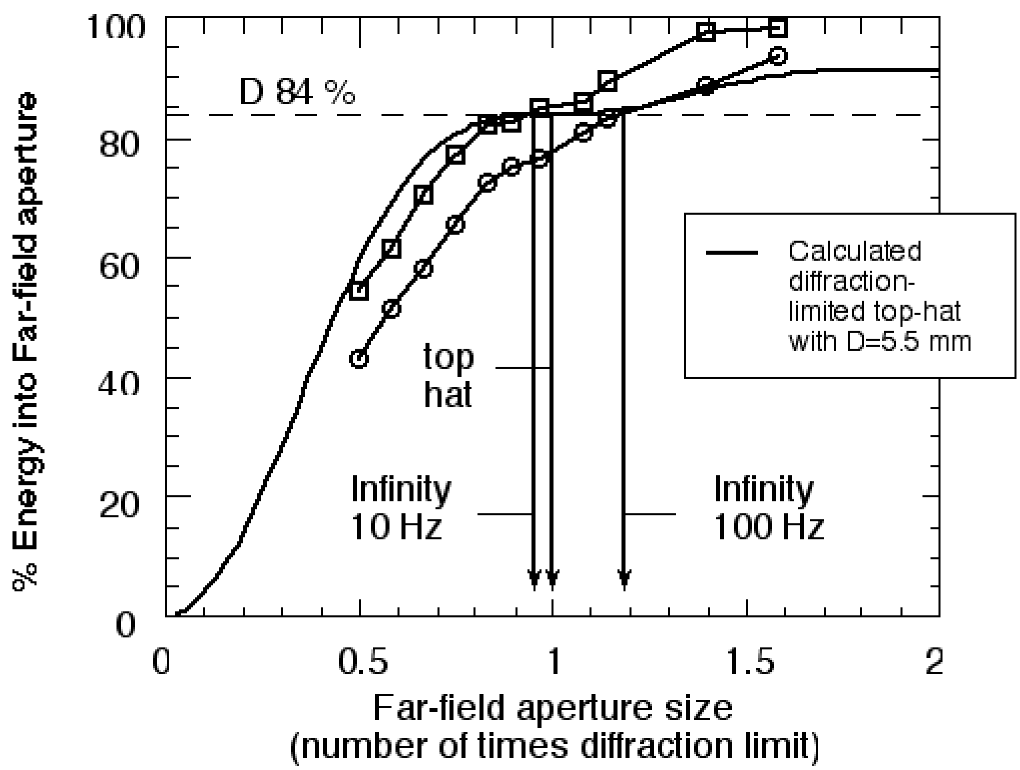

Figure 1.3-2. Results of Energy-in-the-Bucket Measurements at 10 Hz and 100 Hz.

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Introduction

This section presents checklists and procedures for the independent replacement of the Infinity laser head, power supply and computer.

To insure the most up to calibration and system information, the Infinity software data files (i.e., files with an "in" extension) should always be backed up onto a floppy disk after a system is serviced. This data disk should be stored in the system maintenance kit.

To prevent injury due to electrical shock, it is very important that the Infinity laser system be disconnected from the facility power before attempting to replace the laser head and/or the power supply.

Replacing the Infinity laser system head or power supply will require that the cooling water be siphoned from the heat exchanger. To prevent contamination of the cooling system it is recommended that the siphoned not be reused. To refill the heat exchanger, 6 gallons of steam distilled will be required.

If it has been more than 8 months since the cooling system was last maintenance it is highly recommended that the particle filter (P/N: 2603-0144) and the deionization cartridge (P/N: 2603-0141) be replaced during the component replacement.

Head Exchange Checklist

The following checklist presents the steps for exchanging an Infinity laser system head. For complete procedural details see the subsection following the power supply replacement checklist.

- ☐ Replace faulty laser head.
- ☐ Update computer calibration files for new laser head.
- ☐ Set the cooling water resistivity level.
- ☐ Run automated capacitor bank calibration routine.
- ☐ Verify system specifications.

Power Supply Exchange Checklist

The following checklist presents the steps for exchanging an Infinity laser system power supply. For complete procedural details see the subsection following this replacement checklist.

- ☐ Replace faulty power supply.
- ☐ Set the water resistivity level.
- ☐ Run automated capacitor bank calibration routine.
- ☐ Verify system specifications.
- ☐ Set the operational water temperature.

Head & P/S Exchange Procedure

1. Disconnect the laser system from the facility power. Replace the laser head or power supply as follows:
    - Remove the P/S heat exchanger cover. Drain the cooling water from heat exchanger and disconnect the head cooling lines.
    - Remove the cover on the “electronics” side of the power supply (P/S) and the two umbilical cable access panels on the back of the P/S.
    - Disconnect the head connectors, replace the head or power supply, and reconnect the umbilical connectors. Reinstall the two rear access panels.
    - Reconnect the head cooling lines, replace the water filter and deionization cartridge if desired, and refill the heat exchanger reservoir.
    - Fully open the deionization cartridge “by-pass” value.

- If the laser **"head"** is being replaced, obtain the system data file disk (the disk is marked with the laser head number) and copy the "state.in" and the "calconst.in" files into the Infinity folder on the laser system computer.
- If the laser **"power supply"** is being replaced, set the temperature regulation value position indicator to approximately 1.5.

2. On one of the P/S Booster boards, hook a DVM between B++ and Load minus. Restore the facility power to the Infinity laser system.
3. Open the Infinity software "Service" window and quickly click the heat exchanger on and off. Repeat this step until all of the air has been purged from the heat exchanger.
4. Through the Service window turn on the heat exchanger and open the Infinity software "Monitor" window. Once the water resistivity reaches 1.6 MΩ-cm close the resistivity by-pass valve and then reopen the valve approximately 1.5 turns.

From this point on it is important to periodically monitor the water resistivity and adjust the by-pass valve until the resistivity value stabilizes between 2.5 and 5 MΩ-cm.

5. Close the Service window. From the main Infinity software window set the system repetition rate to 10 Hz and place the Infinity in "Standby".

If the Infinity lamp does not ignite and simmer, perform the "Control Board Calibration" procedure located in Chapter 6 and then resume this procedure at step 8.

6. Open the Service window and set the system voltage to 480 V using the "Lamp voltage" slide bar. From the Service window select the "Calibration" menu item and then select "Cap bank voltage...". Press the "Start" button in the resulting dialog box.
7. After the calibration procedure is complete, using the Lamp voltage slide bar, increase and decrease the system voltage. The voltage values on the slide bar and the DVM should agree to within 5%. If they do not, the calibration procedure given in Section 6.1 will need to be performed.

8. Decrease the lamp voltage to 360 V and close the Service window. Open the Infinity Monitor window if it was closed in a previous step. In the main software window increase the repetition rate to the system maximum and set the IR energy to 1 mJ. Click the laser system "On" and slowly increase the IR energy while monitoring the system depolarized energy and noise. Verify that the system meets its long term noise specification and that the depolarized energy is less than or equal to 1.5% of the pulse energy.

If it is found that the system is not meeting noise specification, the technician is directed to Troubleshooting Table 3.3-1.

If it is found that the system depolarized energy is greater than 1.5% of the pulse energy, the technician is directed to minimization procedure located in Section 2.9.

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9. If the system **"power supply"** was replaced, while the laser system is running at its maximum repetition rate and pulse energy, adjust the water temperature regulation value to the temperature given on the original system "Performance Certification" sheet. If this sheet is not available, the system water temperature should be set in the 30-35 °C range.

The component replacement procedure is now complete. From the main Infinity software window turn the laser system off. Turn the P/S key switch to the off position. Disconnect the DVM from the Booster board and re-install the P/S covers (heat exchanger and electronics).

Computer Exchange

The following procedure outlines the steps for exchanging an Infinity laser system computer.

1. Replace faulty computer system and if required install the Infinity software on the new computer.
2. Obtain the laser system data file disk (the disk is marked with the laser head number) and copy the most recent data files (i.e., files with the extension "in") into the Infinity directory on new computer.

   See Chapter 5 for additional information on the Infinity software and the required data files.

3. Launch the Infinity software and open the “Monitor” window. Set the laser repetition rate to the system maximum and set the IR energy to 1 mJ. Click the laser system “On” and slowly increase the IR energy while monitoring the system depolarized energy and noise. Verify that the system meets its long term noise specification and that the depolarized energy is less than or equal to 1.5% of the pulse energy.

If it is found that the system is not meeting noise specification, the technician is directed to Troubleshooting Table 3.3-1.

If it is found that the system depolarized energy is greater than 1.5% of the pulse energy, the technician is directed to the minimization procedure located in Section 2.9.

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Introduction

A well-aligned system will deliver maximum pulse energy in a beam with minimum divergence, exhibit good pulse to pulse stability and is also less likely to suffer damage to its optical components.

The system is capable of producing specified energy at specified noise figures even if it is slightly misaligned. If this is the case it will, however, affect the mode quality. Mode quality is the most sensitive indicator of how well the system is aligned.

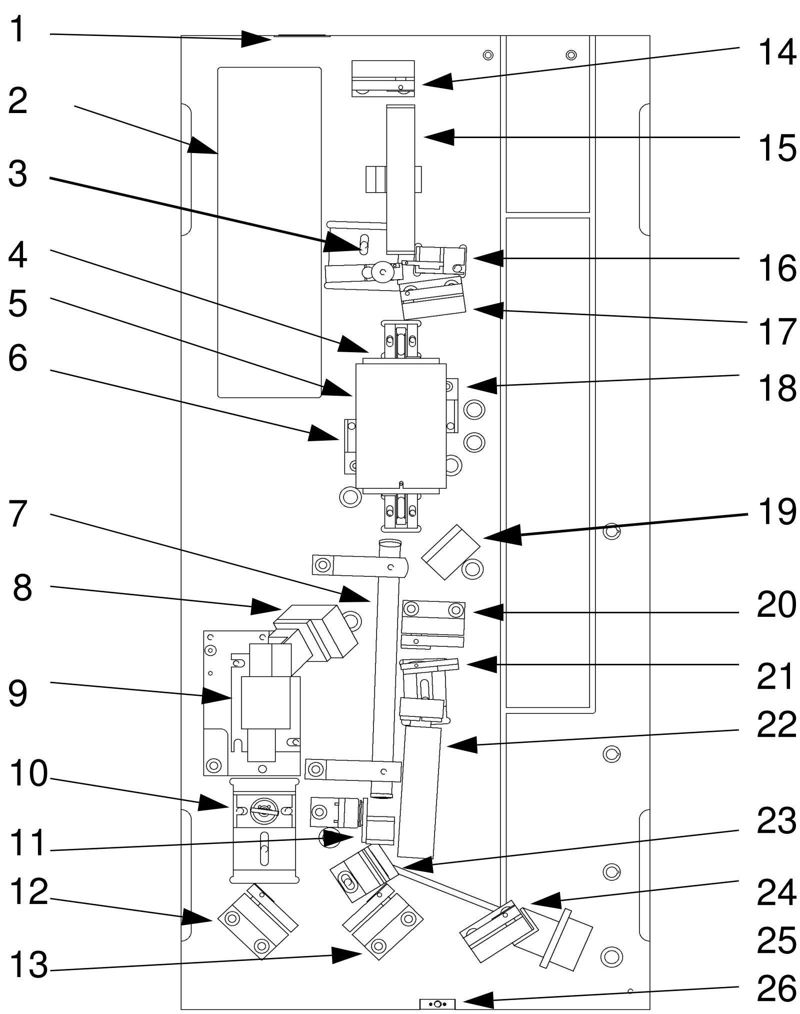

Figure 2.1-1. Positions of Laser Head Controls and Indicators

Table 2.1-1. Laser Head Controls and Indicators

CALLOUT	NAME
1	Laser emission indicator
2	DPMO, Diode Pumped Master Oscillator
3	Relay system Lens 1
4	POT, Amplifier assembly
5	Flash lamp terminal box
6	Safety interlock dowel pin
7	Vacuum spatial filter
8	Photo-diode monitoring depolarized energy
9	Faraday isolator
10	Collimating lens
11	Mechanical shutter
12	Folding mirror, M1
13	Folding mirror, M2
14	Concave mirror, relay system
15	Relay system vacuum cell
16	Relay system lens 2
17	90° Polarization rotation prism
18	Safety interlock dowel pin
19	Flash track photo-diode
20	Quarter wave plate
21	SBS lens
22	SBS cell
23	Thin film polarizer
24	Folding mirror, M3
25	Photo-diode monitoring IR energy
26	Head cover interlock

Initial Preparations

The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. It is strongly recommended that everyone present in the room wears appropriate safety glasses and that access to the room be restricted. Be aware that the DPSS laser itself is a class IV laser as well.

1. Open the laser head cover and defeat the interlock located on the baseplate where the umbilical is connected.
2. Run the Infinity program and click on the “Service...” button in the main window.
3. Turn on the oscillator by clicking in the “DPSS” box. Verify that it works by placing a IR viewing card in the beam path between the Faraday isolator and the DPSS unit. A bright orange fluorescent spot should appear on the card. Open the shutter. Using the IR card make sure that the beam is transmitted all the way to the SBS cell without being clipped.
4. Turn on the Heat exchanger. Check that there are no leaks and allow the water temperature to stabilize at approximately 26 to 30°C. This will allow the base-plate to reach the same temperature as when the laser is operated.
5. If the desired temperature is not obtained, open the heat exchanger side of the power supply and adjust the temperature valve. Make small adjustments until the 26 to 30°C temperature range is reached. Clockwise will decrease the temperature and counter clockwise will increase it.

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Alignment of Faraday Isolator

1. Remove the collimating lens assembly and the folding mirror assembly from the laser head, shaded grey in Figure 2.2-1.
2. Install a $\lambda/2$ waveplate between the DPMO and the Faraday. This will be used to attenuate the DPMO output so that the CCD camera is not saturated.
3. Position the CCD camera so that it monitors the beam exiting from the Faraday isolator.
4. There are two different Faraday isolators for the Infinity laser system. These assemblies can easily be distinguished by their color. The early model (which is now obsolete) is charcoal grey. The new model is red. If the system you are working on contains the red Faraday you should install the alignment apertures at this time. These apertures should be left in the Faraday until the system is completely aligned.

   Adjust the position of the Faraday isolator, as described below, until the diffraction patterns from the entrance and exit apertures are uniform and centered, see Figure A-1 in section 2.11. View the diffraction pattern in black and white mode and the energy distribution in color mode on the CCD camera. Figure A-2, section 2.11, shows the mode when the Faraday isolator is slightly misaligned. Instead of one center there are three, each corresponding to a separate diffraction pattern.
5. Loosen the translation screws, 2 and 4 in Figure 2.2-2, on the Faraday isolator and coarsely position it so that the beam is centered side to side in the aperture. Tighten either one of the screws and leave the other one loose enough to allow the assembly to be moved. Using the fixed screw as a pivot point move the assembly back and forth until the diffraction patterns overlap. Tighten the screw and release the other one. Repeat the back and forth movement for the other side. Iterate between the two sides until the optimum position is found.
6. Use the setscrews, 1, 3 and 5 in Figure 2.2-2, to change the tilt angle and the height of the assembly. Adjust the tilt angles first by adjusting 1 and 3. If the diffraction patterns can't be centered either rise or lower the platform by adjusting first 5 and then 1 and 3. Repeat the optimization using 1 and 3.

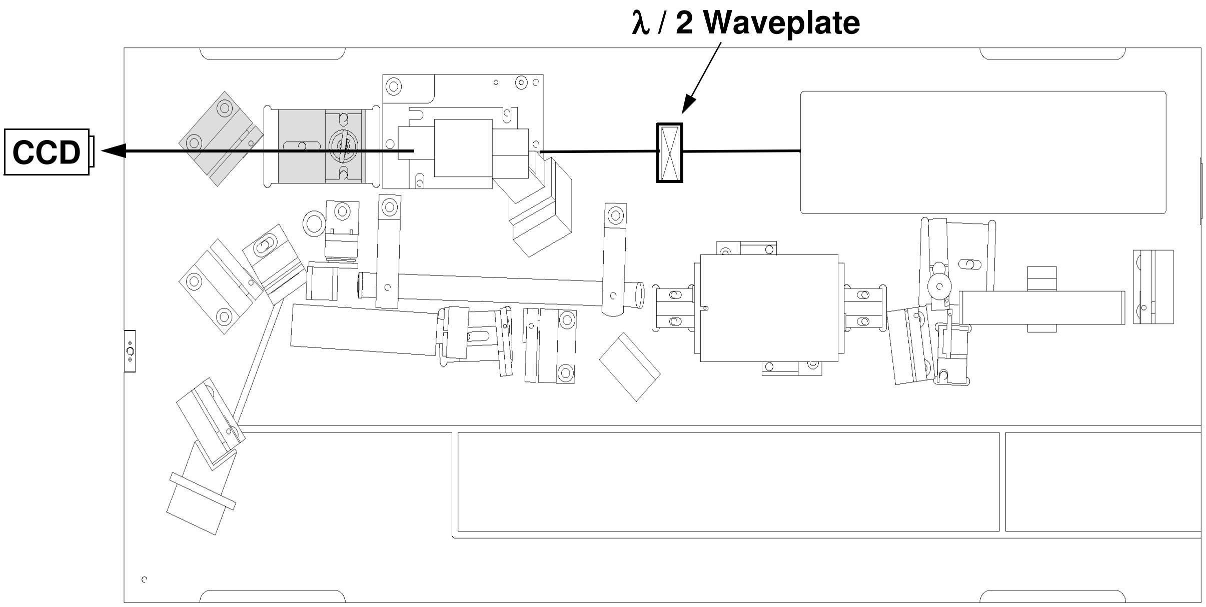
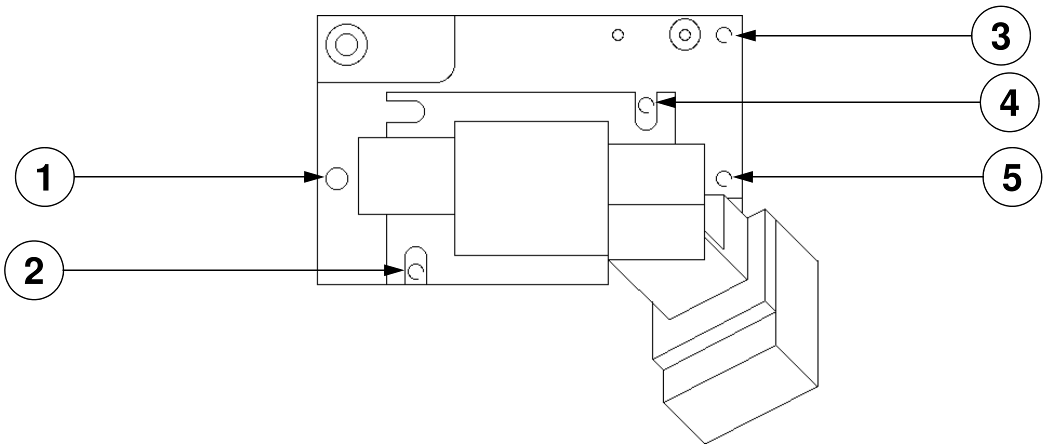


Figure 2.2-1. Layout for Faraday Isolator Alignment.



1, 3, 5 - Vertical tilt adjustment screws

2, 4 - Translation screw

Figure 2.2-2. Faraday Isolator and Photo-Diode Assembly

Alignment of Collimating Lens

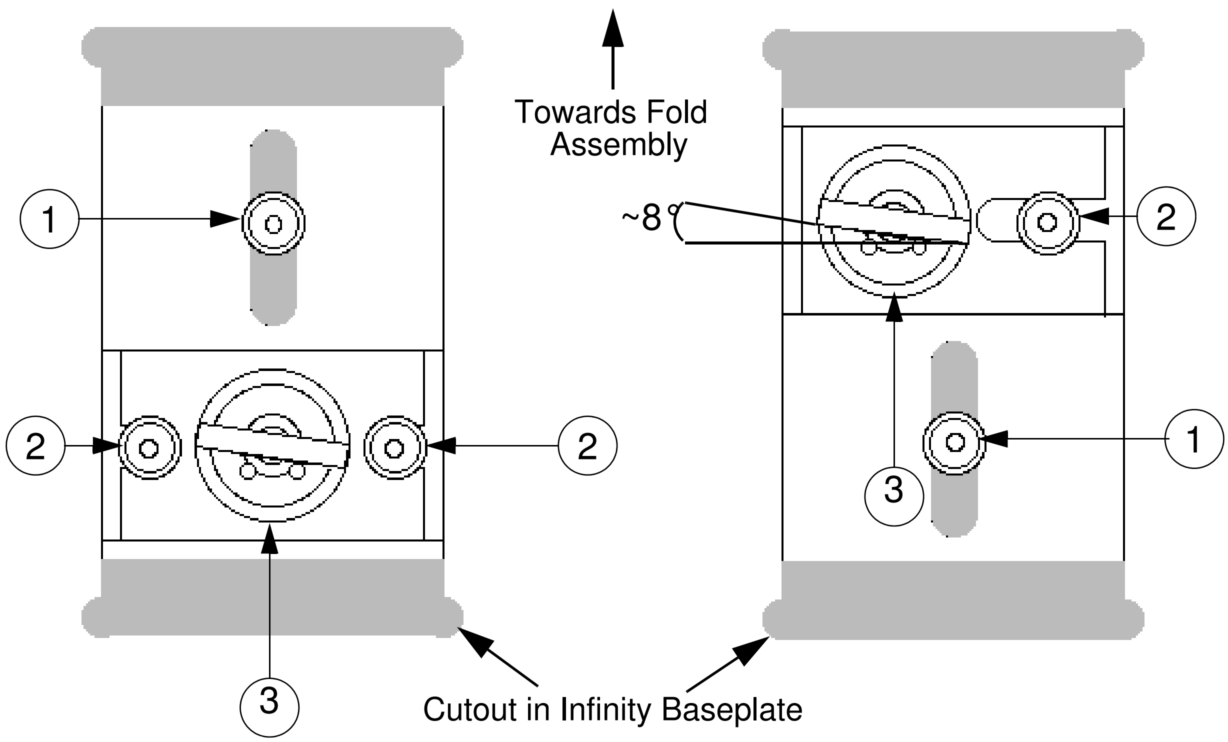
The next step is to adjust the position of the collimating lens so that the beam is centered through it and collimated. This is done as an iterative procedure. The reason for this is that the lens in its correct position will be slightly tilted about its vertical axis to compensate for the astigmatism induced by the optical components of the DPMO. There are hence two parameters that need to be optimized, the distance of the lens from the oscillator and its tilt angle. Consequently two measurements are needed.

The system is designed so that the lens compensates optimally when tilted close to 8° (true for most systems). It is therefore advisable to tilt the lens to close to this angle before going through the collimation procedure. It does not matter in which direction the lens is tilted. The alignment procedure is carried out for the vertical direction first since the cross talk between the lens-oscillator distance and the lens tilt angle is less in this direction than in the horizontal.

Center the DPSS mode pattern in the cross hairs of the CCD display. Reinstall the Lens/Aperture assembly, Figure 2.3-1. Note that there are two possible orientations for the collimating lens assembly. If the system contains the current version of the Faraday Isolator (i.e., the isolator assembly with red anodized metal) the collimating lens assembly should be installed as illustrated on the right hand side of Figure 2.3-1. If the laser system has an old style isolator (i.e., it is not red in color) than the assembly should be orientated as illustrated on the left hand side of Figure 2.3-1.

Translate the lens along the horizontal and vertical axes until the mode is centered on the CCD display. Vertical alignment is obtained by loosening the 3/32” set screw that secures the shaft of the lens holder to the mount. Slide the lens up and down until the circular interference patterns are overlapped in the horizontal plane. Horizontal alignment is obtained by loosening the translation adjust screw(s) that secure the lens assembly, item 2 in Figure 3.2-1. Slide the lens assembly side to side until the circular interference pattern are overlapped in the vertical plane.

Iterate between the vertical and horizontal alignment procedures until the beam is again centered in the CCD display cross hairs.



1. Lens/Aperture assembly focusing lock screw
2. Horizontal translation lock screw
3. Vertical adjust lock screw

Figure 2.3-1. Collimating Lens Assembly. (Left) Orientation for use with old style Faraday (Right) Orientation for use with new style (red) Faraday

For both the vertical and horizontal collimation of the DPMO output, the CCD camera should be positioned to view the "front" surface reflection from the shear plate.

Vertical Beam Collimation

Insert the shear plate in the beam path according to Figure 2.3-2 and adjust the CCD camera so that the interference pattern is clearly visible. Make sure that the wedge of the shear plate is oriented with the wedge angle perpendicular to the direction in which collimation is adjusted. The shear plate will have a reference line across it that indicates the fringe orientation for a collimated beam. The wire is perpendicular to the plane of the wedge.

Slide the lens focus adjustment back and forth until a position is found where the fringes of the interference pattern on the CCD camera are horizontal. An example of a pattern created by a collimated beam (section 2.11) is shown in Figure A-3 and that of a slightly non-collimated beam in Figure A-4. In Figure A-3 the image is symmetrical around a vertical axis through the center while Figure A-4 shows a little bit more intensity on the right hand side.

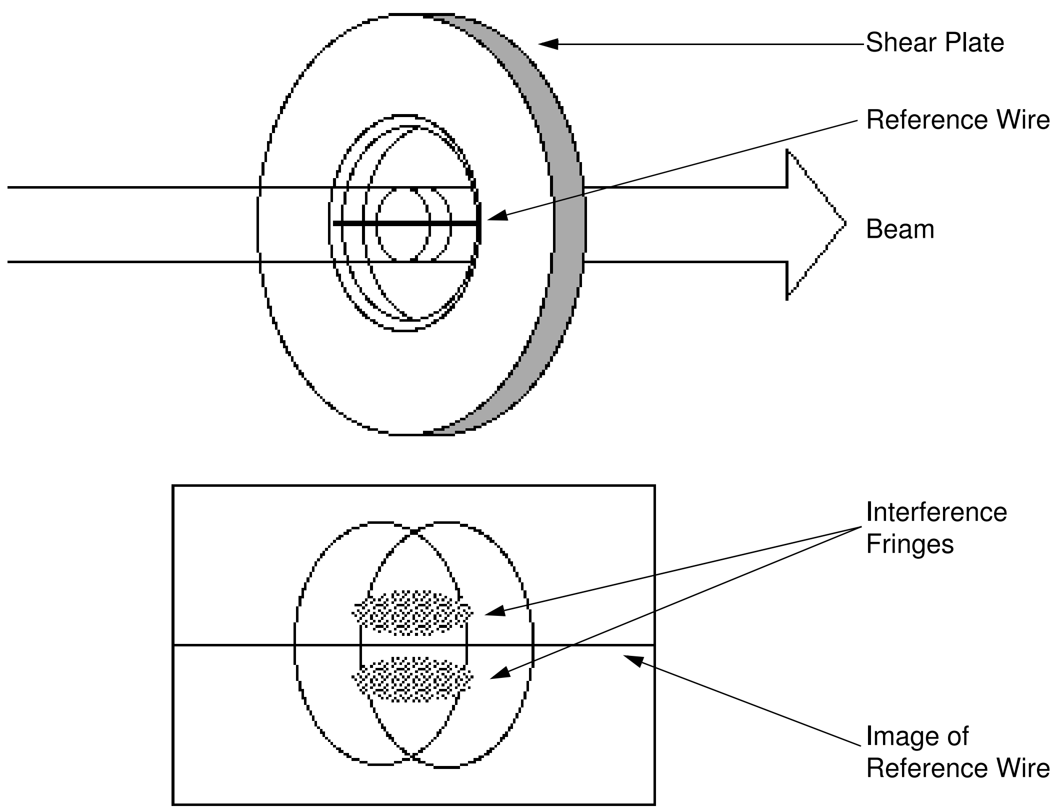


Figure 2.3-2. Shear Plate Aligned for Collimation of Beam in Vertical Direction, Viewed from Above.

Horizontal Beam Collimation

Reorient the shear plate so that it now monitors the collimation in the horizontal direction. This situation is illustrated in Figure 2.3-3. The next step is to carefully, without changing the height of the lens, tilt it so that the fringes are aligned with the reference line. In this position the lens will collimate the beam while correcting for the astigmatism imposed onto the beam by the DPMO optics.

Iterate between vertical and horizontal alignment until the beam is collimated in both directions. It is necessary to perform these steps since there is a slight cross talk between the tilt angle of the lens and the focus adjustment.

Finally check that the beam is centered through the lens. Mark the position of the lens mount on the baseplate. Remove the whole lens assembly, mark the position of the beam on the CCD camera screen, and then reinsert the assembly. Readjust the vertical and horizontal position of the lens to re-center the aperture/lens in the DPSS output, if required. If adjustment to the lens position is required, re-verify the vertical and horizontal collimation of the CW beam.

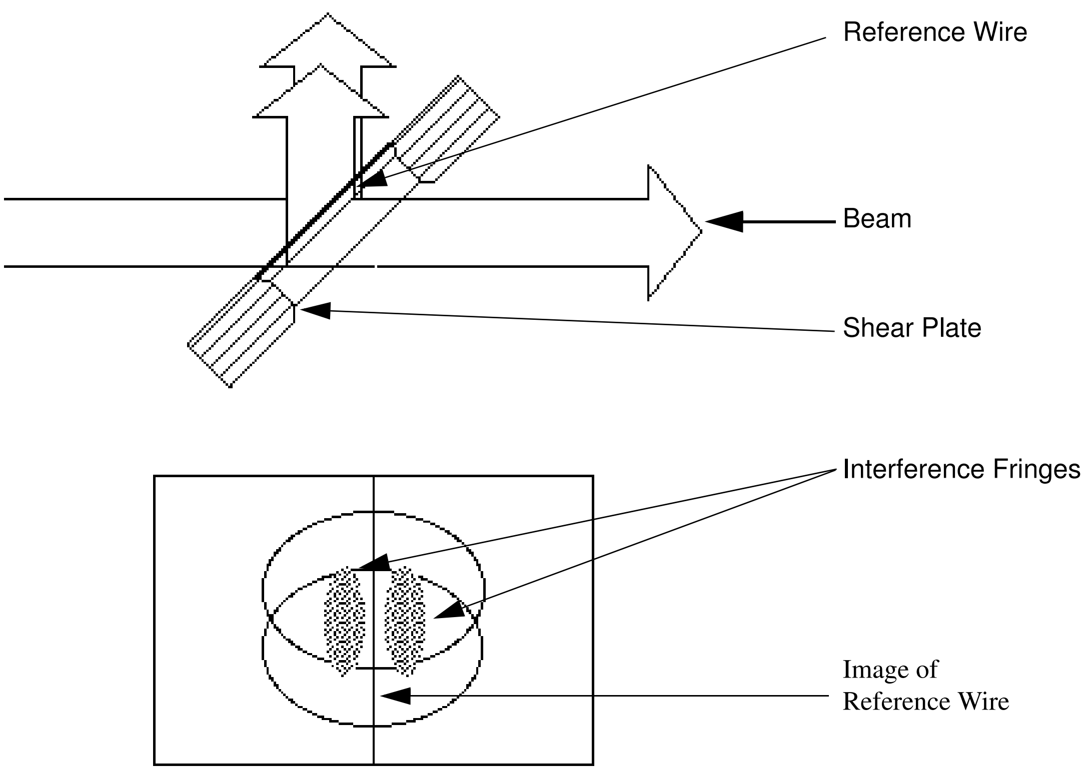


Figure 2.3-3. Position of Shear Plate for Collimation of Beam in Horizontal Direction, Viewed from Above.

Alignment of Thin Film Polarizer

If the thin-film polarizer needs to be aligned (i.e., has been moved) follow the procedure below.

1. Position the polarizer so that it is in the center of its rotational travel range.
2. Follow the procedure in the next section, Alignment through Rod #1.
3. Remove the SBS lens mount, the SBS cell and its mount, and the spatial filter, marked grey in Figure 2.4-1.
4. Position a mirror as shown in Figure 2.4-1. Retroreflect the beam back through the polarizer.
5. While monitoring the intensity of the beam reflected off the polarizer, rotate the polarizer counter-clockwise. Stop when the reflection becomes markedly brighter.
6. Rotate the mount clockwise until the reflection becomes dim again. Do not continue to rotate the mount clockwise once the reflected intensity has gone down!
7. Lock the polarizer in this position.

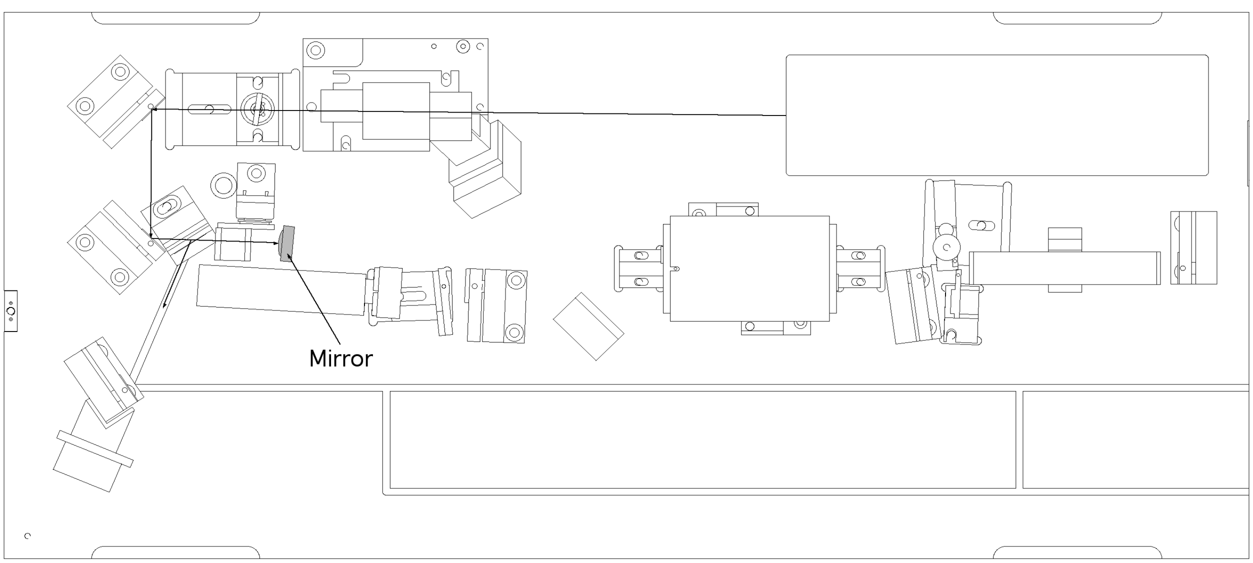


Figure 2.4-1. Alignment of Thin-Film Polarizer

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Alignment through Rod #1

After the beam has been collimated the next step is to center the DPSS output colinearly through the rod 1. The numbers in brackets refer to Figure 2.1-1.

1. Loosen the screw that holds the mechanical shutter and rotate it away from the vacuum spatial filter.
2. Remove the following components, grey in Figure 2.5-1. **Do not remove the Thin Film Polarizer!**
   - Vacuum spatial filter [7]. Loosen the set screws in the brass ring holders and gently move the cell, first away from the amplifier far enough to slide the end through the brass ring and then out of the second holder by moving it toward the amplifier.
   - Concave mirror assembly [14].
   - Relay lens #1 assembly [3]. If the system has been aligned once mark the position in the cut out so that the mount can be reinserted close to its original position.
   - Relay cell [15].
3. Position the CCD camera behind the mount that held the concave mirror, Figure 2.5-1.
4. Steer the beam through rod #1, using the horizontal and vertical controls of mirrors [12] and [13] to eliminate all diffraction patterns but one. The one remaining pattern should be symmetrically positioned with respect to the center (geometrical) of the rod aperture. An example of a well aligned system is shown in Section 2.11, Figure A-5. With the beam intensity appearing symmetrical on the CCD camera screen move it in the vertical direction, using the controls on fold mirror mount [13], so that it is pointing downwards. Move the beam until only the top part of it is visible and a flare shoots up from it. Adjust the horizontal controls of mirror mounts [12] and [13] so that the flare is symmetrical with respect to the beam as shown in Section 2.11, Figure A-7 (i.e., walk the beam). Move the beam back into the center using the vertical control of [13]. Repeat the procedure in the horizontal direction, Figure A-8, Section 2.11. It will be necessary to iterate between the two directions since there is a slight cross-talk in the mounts.

Due to slight cross talk between the fold assembly tilt axes, it is very important to verify the first steps in the alignment (vertical or horizontal) procedure before locking down the controls.

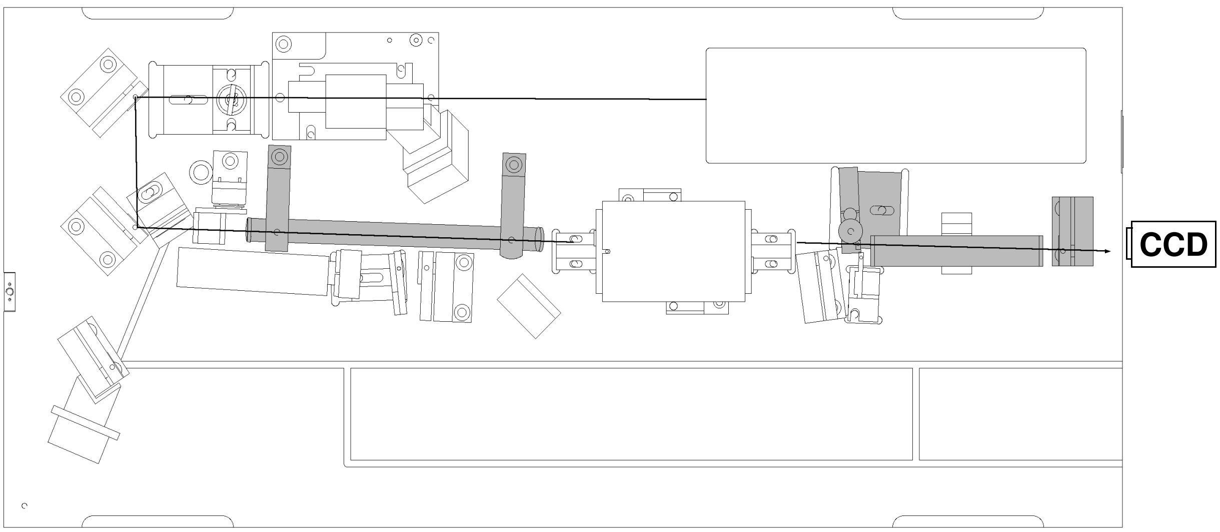

Figure 2.5-1. Alignment through Rod #1

Alignment of Spatial Filter

Reinstall the spatial filter mounts. Make sure that they are aligned along a straight line. An easy way of doing this is to press a steel ruler against the mounts while positioning them, as shown in Figure 2.6-1. Reinsert the vacuum spatial filter so that the stem points straight down and is closer to the shutter than the pot. The distance between the filter and the pot should be between 50 and 55 mm. Make sure that both set screws are firmly tightened and the brass ring holders are centered in their cradles.

The goal is, using the vertical and horizontal controls of the spatial filter mounts, to move the filter aperture so that the transmitted beam appears round and symmetrical. Note that because of the way the mounts are designed there will be some cross-talk between the two controls. Rather than moving the filter in a pure horizontal or vertical direction it will move it along an arc.

1. Move the end of the filter closest to the shutter so that the beam is centered on the lens. Use the IR viewing card to verify the beams position. After this has been accomplished scan the other end of the filter through a grid (horizontal and vertical motion) until transmission of the beam is achieved.

2. Once part of the beam goes through the pinhole use all four controls to optimize its transmission. Move only the filter. **Do not move the beam!**

3. The spatial filter aperture will remove the higher frequencies from the diffraction pattern. As a result the mode will appear “softer”. As the beam traverses the rod a diffraction pattern, transposed on the transmitted mode, is created. The goal in this alignment step is to minimize the contrast of this diffraction pattern and make the mode as smooth and symmetrical as possible. See Figure A-9 (B&W) and Figure A-10 (Color) in Section 2.11.

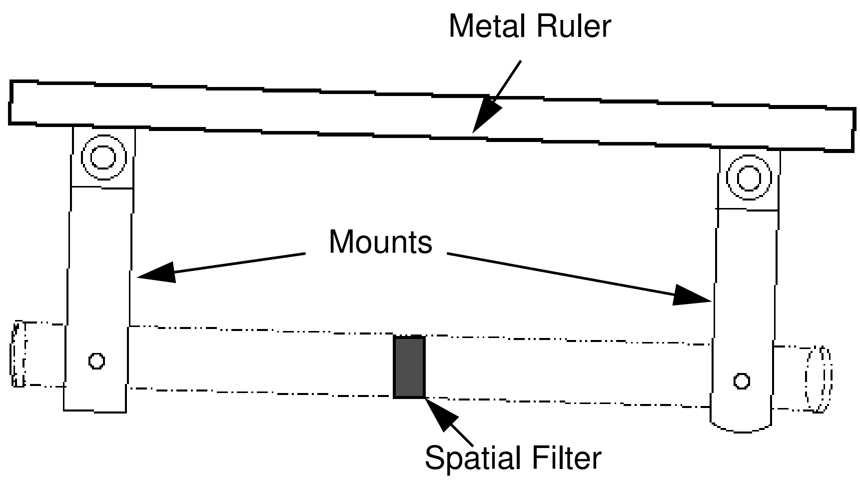

Figure 2.6-1. Alignment of Vacuum Spatial Filter Holders

Alignment of Relay System

After the beam is aligned through rod #1 the next step is to align the relay imaging system.

1. Mark the center of the mode on the CCD camera screen.
2. Install relay lens #1. The exact location of the mount will be determined later when the depolarized energy is minimized. A good starting point is achieved by first sliding the mount as far away from the pot as possible and then back about 5 mm towards the pot.

   Adjust the vertical and horizontal controls until the beam is centered around the same point as before on the CCD camera. An example image is shown in Figure A-11 (B&W) and in Figure A-12 (Color) in section 2.11.
3. Install the relay vacuum cell with the stem pointing down. Use the IR card to verify that the beam enters and exits the cell without clipping. If necessary rotate and/or translate the mount holding the vacuum cell. The distance between the cell and relay lens #1 should be close to 5 mm.
4. Install the concave mirror for the relay system.

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Alignment through Rod #2

1. Position the CCD camera so that it monitors the beam that is reflected off the steering optic, as shown in Figure 2.8-1. If necessary remove the Flash Track photo diode housing first. Put the camera as close as possible to the mirror. Since the beam exiting the rod is highly divergent the diameter will be larger than the CCD chip.

2. Remove the second relay lens. Center the beam through the second Nd:YAG rod using the controls of the concave mirror. The mode should, when centered in the rod, look similar to the one shown in Figures A-13 (B&W) and A-14 (Color) in Section 2.11.

3. Center the beam mode in the cross hairs on the CCD camera display. Reinstall the second relay lens. Position the second relay lens (i.e., vertically and horizontally adjust the lens mount) so that beam remains centered in the cross hairs on the CCD monitor with the second lens in place. The mode should, when centered in the rod, look similar to the one shown in Figures A-15 (B&W) and A-16 (Color) in Section 2.11.

   The part of the alignment performed in the CW mode is now completed. The next step, minimization of depolarized energy, requires the lamp to be flashing and the amplifier stage to be enabled. Reposition the FlashTrack and the SBS cell mounts. Install the quarter wave plate, the SBS lens and the SBS cell (see Figure 2.1-1 for the locations of these components). Verify, using the IR card, that the beam enters the SBS cell without being clipped. If the system is equipped with a new style (red) Faraday remove the alignment apertures now. All the mounts, optics and photo-diodes necessary to run the laser as an IR system should now be in position.

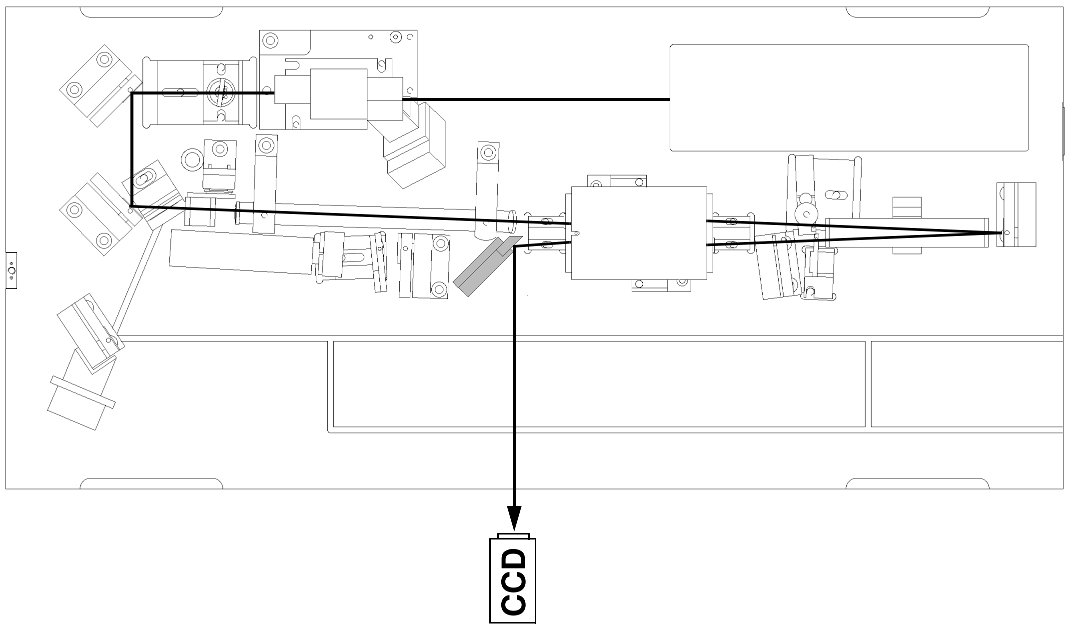
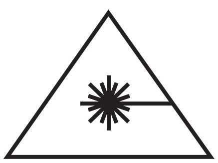


Figure 2.8-1. CCD Camera Set-up for Alignment through Rod #2

Minimization of Depolarized Energy

The laser energy which is not vertically polarized after amplification can not be coupled out of the amplifier stage by the thin film polarizer. Instead, it is transmitted back into the Faraday isolator and is called depolarized energy (Note that the radiation after the Faraday, propagating in the direction of the POT assembly, is horizontally polarized). It is desirable to keep this energy component to no more than 1.5% of the output energy, measured when operating the laser at its maximum pulse energy. If the energy going through the Faraday isolator, as measured by the photo diode attached to it, exceeds 2% of the output energy the on-board microprocessor will shut off the system. This is done in order to protect the master oscillator from being damaged. A bar graph display showing the depolarized energy is available in the Infinity System Monitor panel which can be accessed by clicking on the “Monitors...” button in the Infinity Control panel.



The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. It is strongly recommended that everyone present in the room wears appropriate safety glasses and that access to the room be restricted.

1. In order to access the depolarized energy, the photo-diode attached to the Faraday isolator needs to be removed. Loosen the screw holding the diode assembly in place, and pull the assembly up and to the side. Be careful not to bump the spatial filter.
2. In the main control window set the repetition rate to 10 Hz. Set the lamp voltage to 360 V in the Service window. Start the laser by clicking on On. Gradually increase the lamp voltage until the depolarized light can be detected using an IR fluorescent card.
3. Position the weird angle folding mirror so that the beam reflected from the exposed polarizer is directed between the two folding mirrors into an external power meter. Make sure, using the IR viewing card, that the beam goes into the power meter unobstructed. Use an energy meter that has sufficient dynamic range at the powers measured, such as the Coherent Power Meter Model 210.

Make sure that there is no ablation of the aperture in the vacuum spatial filter. Always continuously check the pinhole and record burn marks as the IR energy is increased for the first time. Go back and realign the system in cw mode if you see signs of ablation of the pinhole or if the burn marks show signs of diffraction or beam clipping. Never adjust the alignment of the spatial filter, or any other optical component, while running the amplifier stage!

Low Repetition Rate Minimization

4. Increase the lamp voltage until the depolarized energy is ~50 mW (i.e., 5 mJ/pulse). If the depolarized energy as a result of the adjustments falls below 50 mW, increase the pulse energy until 50 mW is again reached, or 6 W IR output power (5 W output power in older systems). **Never exceed this energy during the alignment process!**
5. While holding the Quarter wave plate in place with a finger, loosen the setscrew securing it to the mount (see Figure 2.9-1). Carefully, while monitoring the intensity of the depolarized light, make small rotational adjustments until the amount of depolarized energy has been minimized.
6. Adjust the tilt angle of the 90° polarization rotator, mount 2 in Figure 2.9-2. Use the long 5/32” ball driver supplied with the laser. Slowly tilt the crystal holder in the vertical and horizontal direction until a minimum in the depolarized energy is found.
7. Repeat steps 5 and 6 until the depolarized energy can no longer be minimized or 6 W output power is reached (5 W output power in older systems).

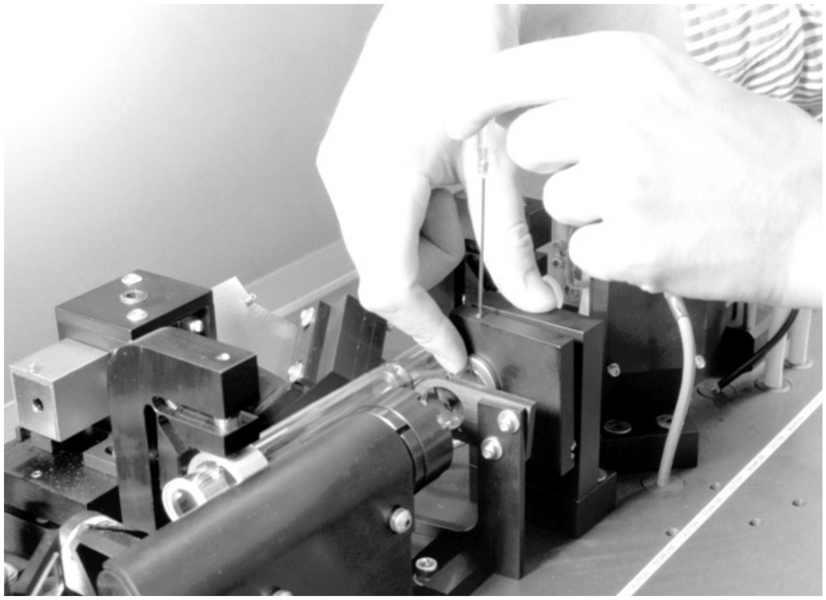


Figure 2.9-1. Alignment of the Quarter Wave Plate

High Repetition Rate Minimization

8. Set the lamp voltage to 360 V and the repetition rate to the system maximum (100, 50, or 30 Hz). Increase the lamp voltage until the depolarized energy is 5 mJ/pulse. If the depolarized energy as a result of the adjustments falls below 5 mJ/pulse, increase the lamp voltage until 5 mJ/pulse is reached or the maximum pulse energy for the system is obtained (40, 25 or 15 W).

9. Slowly adjust the horizontal and vertical translation controls of the first relay lens, mount 1 in Figure 2.9-2, while monitoring the amount of depolarized light. Iterate between the two controls until the minimum is reached.

10. Adjust the tilt angle of the 90° polarization rotator, mount 2 in Figure 2.9-2. Use the long 5/32” ball driver supplied with the laser. Slowly tilt the crystal holder in the vertical and horizontal direction until a minimum in the depolarized energy is found.

11. Carefully make vertical and horizontal tilt adjustments of the quarter wave plate holder, mount 3 in Figure 2.9-2, until the amount of depolarized energy has been minimized.

12. Repeat steps 9-11 until it is possible to run the system at its maximum pulse energy with less than 1.5% depolarized energy.

13. Reduce the lamp voltage to 360 V and the repetition rate to 10 Hz. Verify that the depolarized energy is still minimized at low repetition rates by repeating steps 4-7.

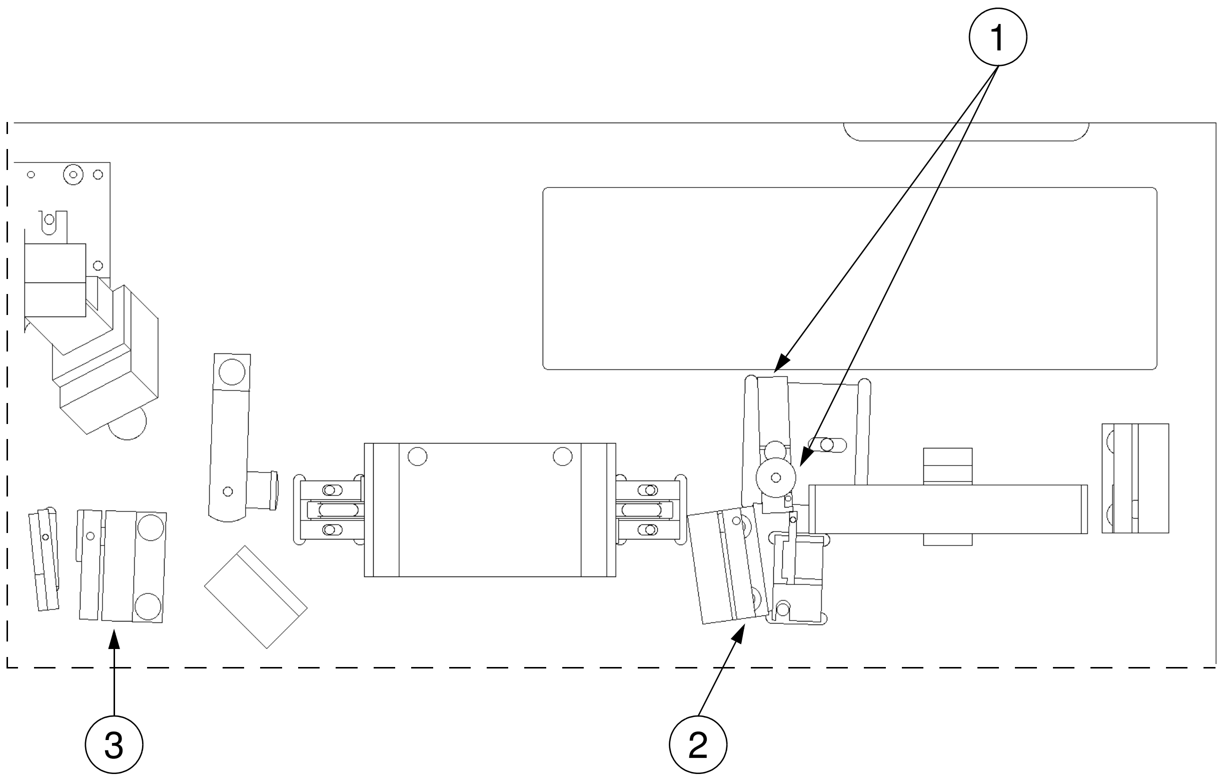


Figure 2.9-2. Order of Adjustments for Minimization of Depolarized Energy.

Troubleshooting

If the depolarized energy can not be minimized, try:

1. Move relay lens #1, mount 1 in Figure 2.9-2, a few millimeters to the left or right (further or closer to the POT assembly) and go through the minimization procedures again starting at step 8.

   If the depolarized energy increases after the first adjustment, try moving the mount in the other direction. If the depolarized energy decreases, continue in the same direction until a minimum is found.

Care must be taken when adjusting the relay mirror position. Only very small adjustments should be necessary, 1/16 of a turn at a time. Larger adjustments might result in less depolarized energy by moving the entire beam away from the Nd:YAG rods!

2. While running the system at its maximum repetition rate and the lamp voltage which produces no more than 2% depolarized energy, alternately adjust the vertical and horizontal controls of the concave mirror holder while looking at the depolarized energy. Look for a decrease in the depolarized energy which corresponds to no, or only a slight, decrease in the output power of the laser system.

   Go through all steps above, in the same order, until no further reduction of the depolarized energy is obtained.

3. If the depolarized energy is well above 2% at both low and high repetition rates, there might be stress-induced birefringence present. Take out the POT assembly and, on one side at a time, release the pressure from the endplate on the rods (be sure to re-seat the teflon rings). This requires the screws holding the two outermost endplates on each side to be loosened. Retighten the screws on one side before doing the other side. After reinstalling the POT assembly return to step 1 of this procedure.

4. If the depolarized energy is high at both low and high repetition rates it could be an indication that there is a problem with the alignment of the 90° rotator. To determine if this is the case try the following:

   A. Set the repetition rate to the system maximum and increase the lamp voltage until the maximum allow depolarized energy is read (i.e., <6 mJ/pulse).

   B. Increase the depolarized energy by 20-30 mW using the horizontal tilt (upper) control of the 90° rotator, mount 2 in Figure 2.9-2.

   C. Try to compensate with the horizontal translation control of relay lens #1, mount 1 in Figure 2.9-2.

   D. If the end result is less, or the same, amount of depol. energy go back to step "**A**" turning the tilt control in the same direction.

E. If the end result is an increase in the depol. go back to step "**A**", but turn the control in the opposite direction.

Continue steps **A-E** until a minimum is found. Note that after 2 or 3 adjustments to one of the translation controls of relay lens mount #1, the other control should be tweaked to make sure that it is still optimized. There is a small amount of cross talk between the horizontal and vertical translation controls of relay lens mount #1.

Once a minimum is found using the horizontal tilt control of the 90° rotator, repeat steps **A-E** using the vertical tilt control of the 90° rotator and the vertical translation control of lens mount #1. Once a minimum is found in this axis you will want to go back to 10 Hz and check the SBS lens. Increase the rep. rate back to the maximum and go through the standard "high repetition rate minimization" procedure.

Alignment of SBS Lens

In order to find the optimum position for the SBS focusing lens the mode needs to be imaged onto a CCD camera.

1. Use two wedges to split off a reflex from a reflex. Position the CCD camera so that it monitors the front surface reflection of the front surface reflection. Use neutral density filters if necessary.
2. Set up the CCD camera software for pulsed mode and auto trigger, threshold ~20%.
3. Run the laser at 10 Hz and 600 mJ/pulse (500 mJ/pulse on older systems). You should now be able to see the mode on the CCD camera. Adjust the angle of the neutral density filters to fine tune the intensity on the CCD camera.

Assume that there is a 10% loss in the average output power due to the IR laser beam going through the wedge optic.

---

4. Adjust the tilt of the SBS lens for maximum roundness and symmetry of the mode.
5. Block the DPSS output. Loosen the lock screw so that the SBS lens mount can be rotated around its vertical axis.
6. Unblock the beam. Rotate the mount back and forth until the mode appears round and symmetrical.
7. Block the beam and tighten the screw.
8. Utilizing hemostats, loosen the two screws that hold the SBS lens mount. Leave the screws tight enough to hold the lens in its position.
9. Unblock the beam. Move the lens up and down, left and right until a position is found in which the mode looks round and symmetrical on the CCD camera.
10. Block the beam and tighten the screws. The SBS lens alignment is now completed.

In Section 2.11, Figure A-17 shows the mode outside the laser (i.e., in the intermediate field) for a well aligned system. In some rare cases it might be possible to obtain a completely symmetrical mode in the intermediate field but mostly the mode will exhibit some asymmetries as illustrated in Figure A-17, this is normal and to be expected.

Also shown in Section 2.11, Figure A-18 shows the mode when the system is running at 50 Hz and Figure A-19 when running at 100 Hz.

Only adjust the lens position when running the system at 10 Hz.

Calibration of Energy Meters

At this point the alignment of the Infinity system is complete and the output mode has been optimized. The last step is to recalibrate the internal IR and depolarized energy meters. These procedures can be found in the Operator's Manual.

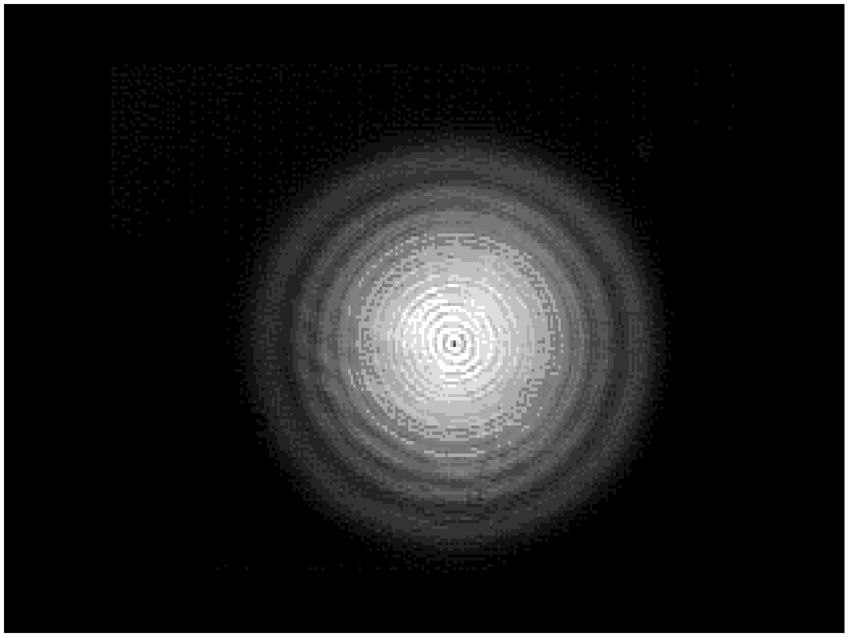


Figure A-1. Image when Faraday isolator is correctly aligned

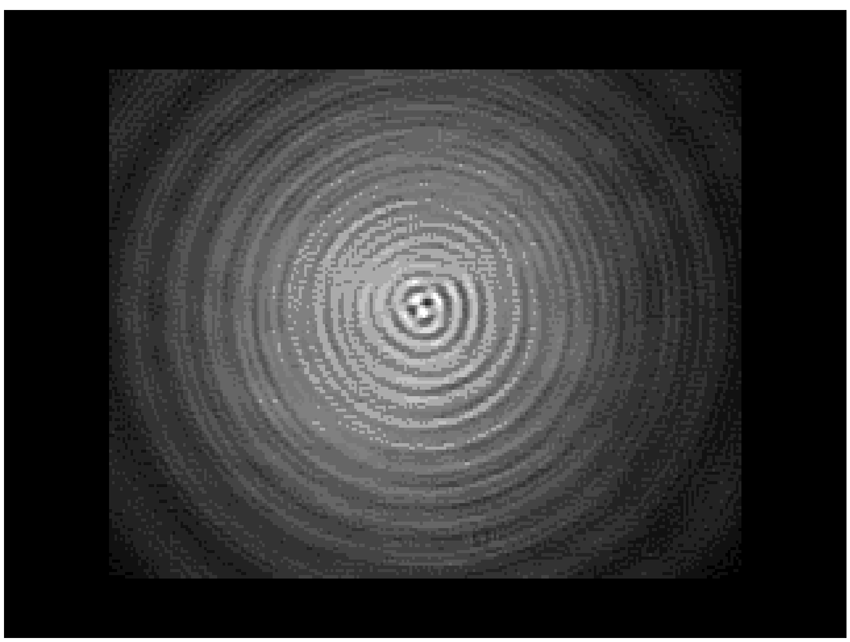


Figure A-2. Image when Faraday isolator is misaligned

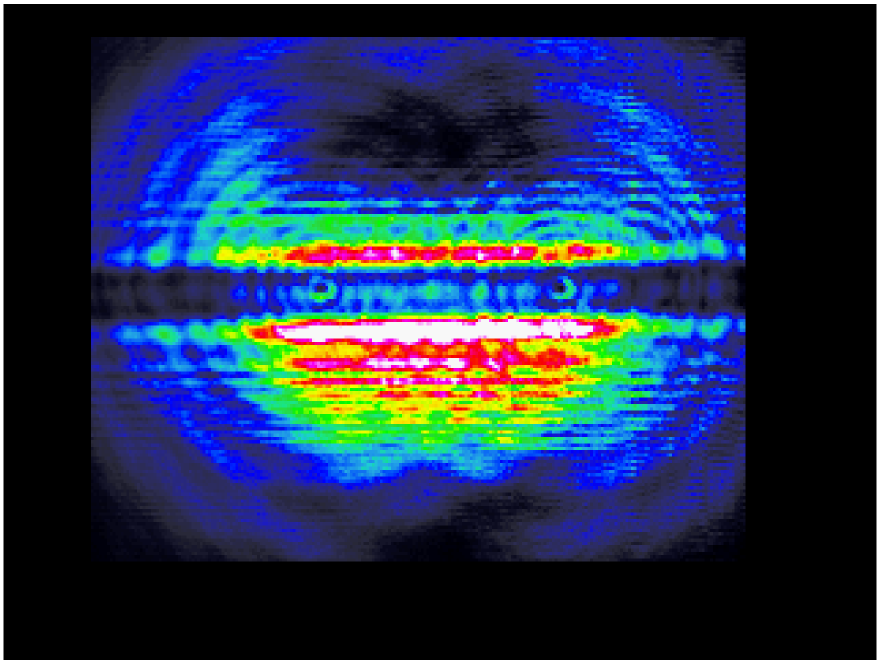


Figure A-3. Image from shear plate when collimating lens is correctly positioned

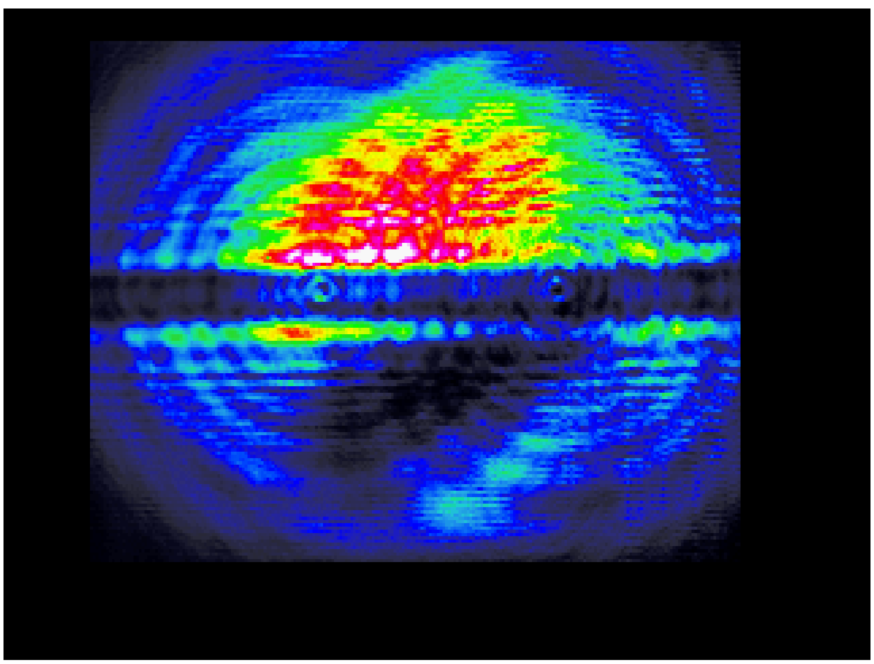


Figure A-4. Image from shear plate when collimating lens is incorrectly positioned

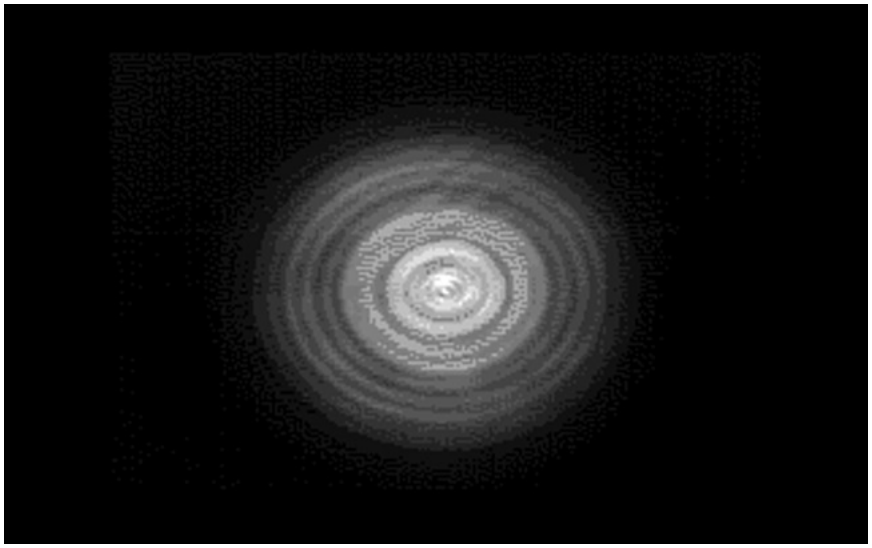


Figure A-5. Image with beam centered through rod 1

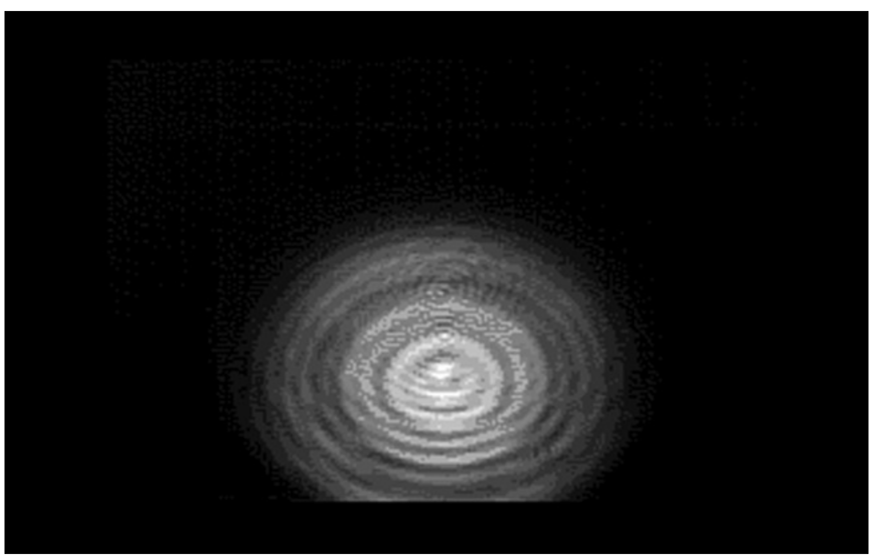


Figure A-6. Image with beam horizontally translated through rod 1

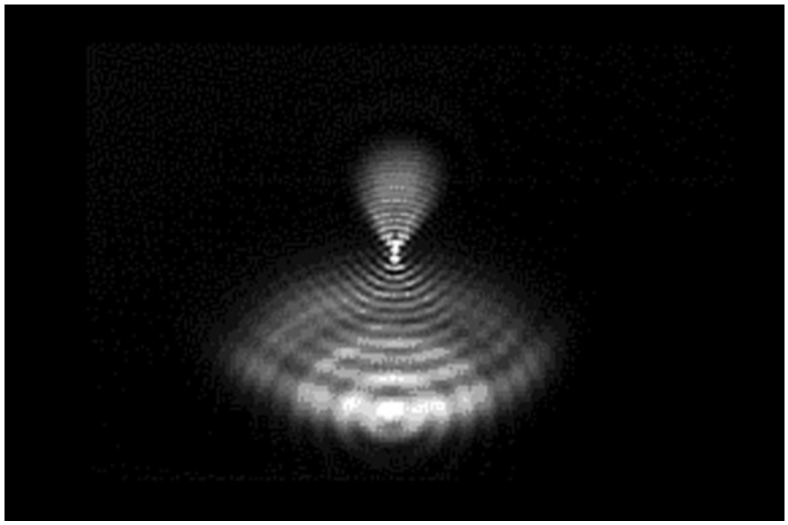


Figure A-7. Image with beam pointing down and centered horizontally

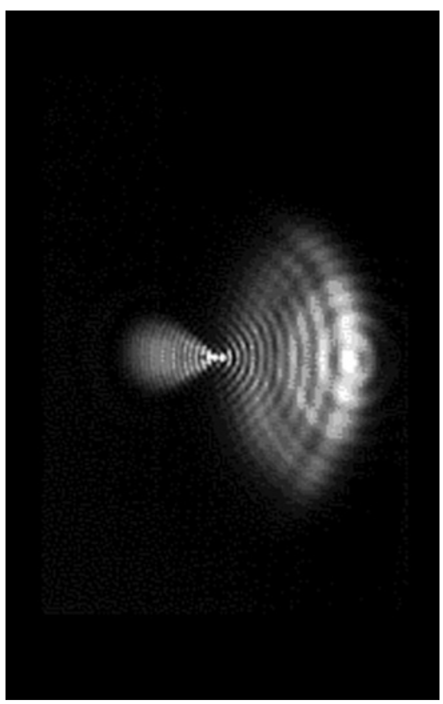


Figure A-8. Image with beam pointing to the right and centered vertically

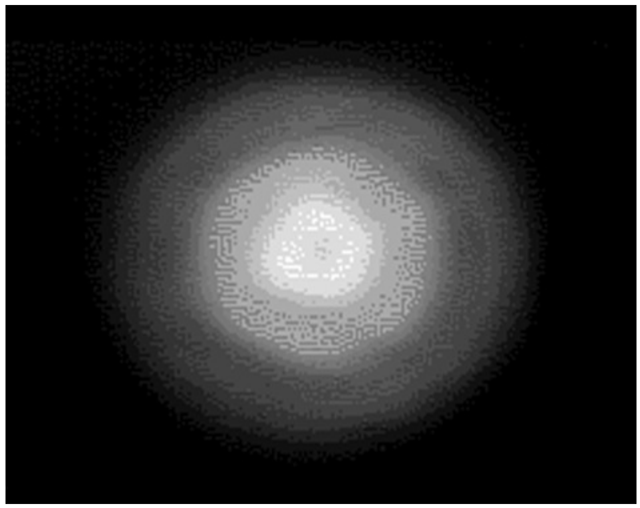


Figure A-9. Image with beam centered through rod 1 and the spatial filter inserted

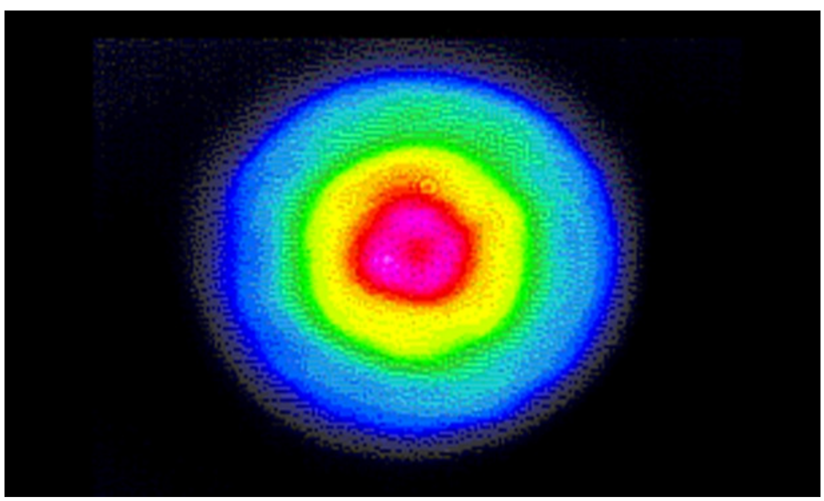


Figure A-10. Image with beam centered through rod 1 and the spatial filter inserted

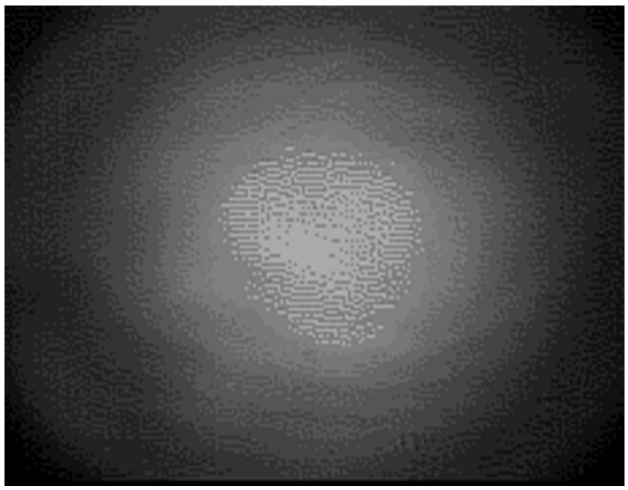


Figure A-11. Image after installation of relay vacuum cell

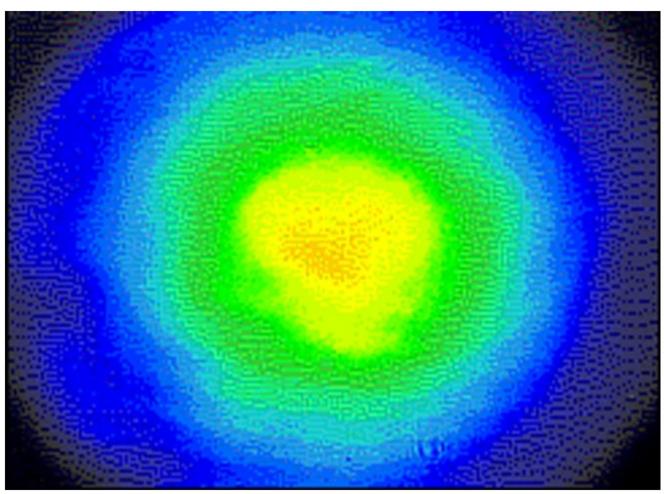


Figure A-12. Image after installation of relay vacuum cell

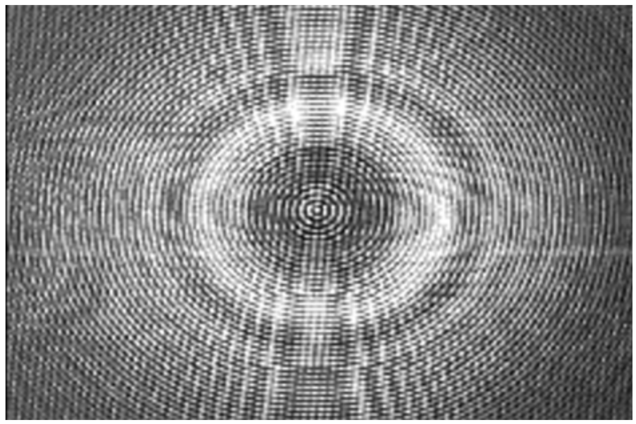


Figure A-13. Image with beam centered through rod 2 without relay lens 2 inserted

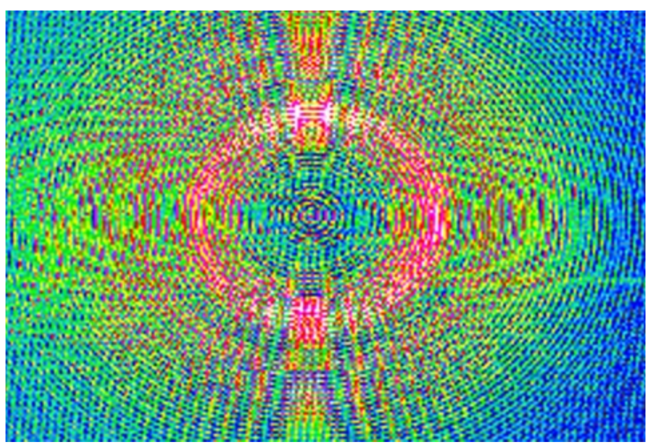


Figure A-14. Image with beam centered through rod 2 without relay lens 2 inserted

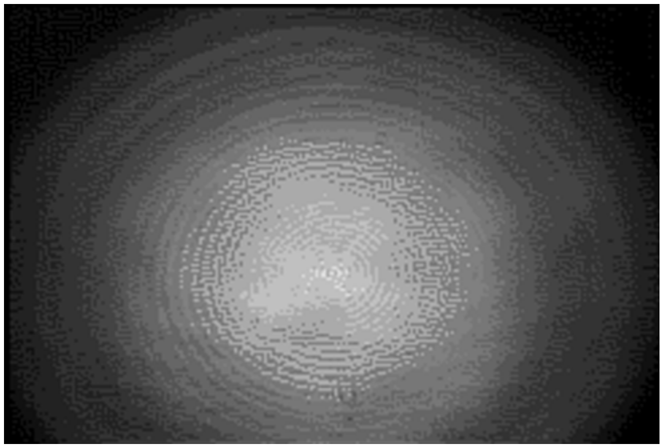


Figure A-15. Image with beam centered through relay lens 2

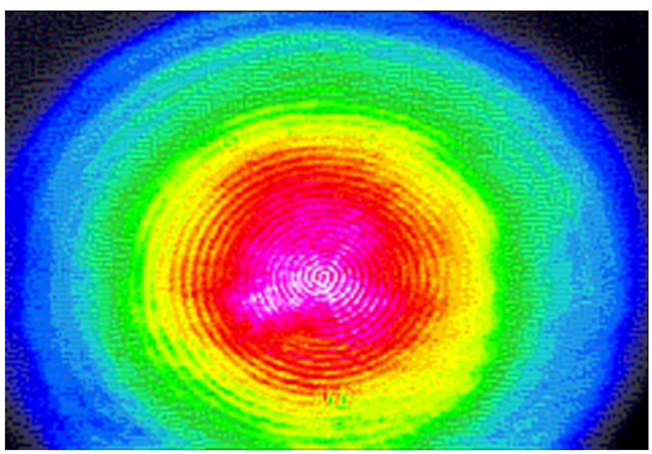


Figure A-16. Image with beam centered through relay lens 2

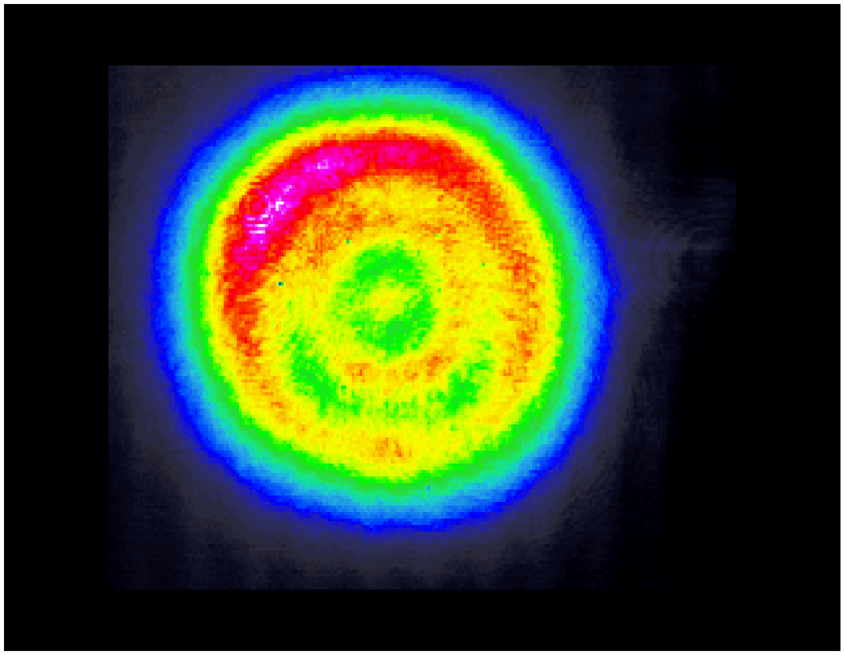


Figure A-17. Mode in the Intermediate Field when running the system at 10 Hz and 500 mJ/pulse

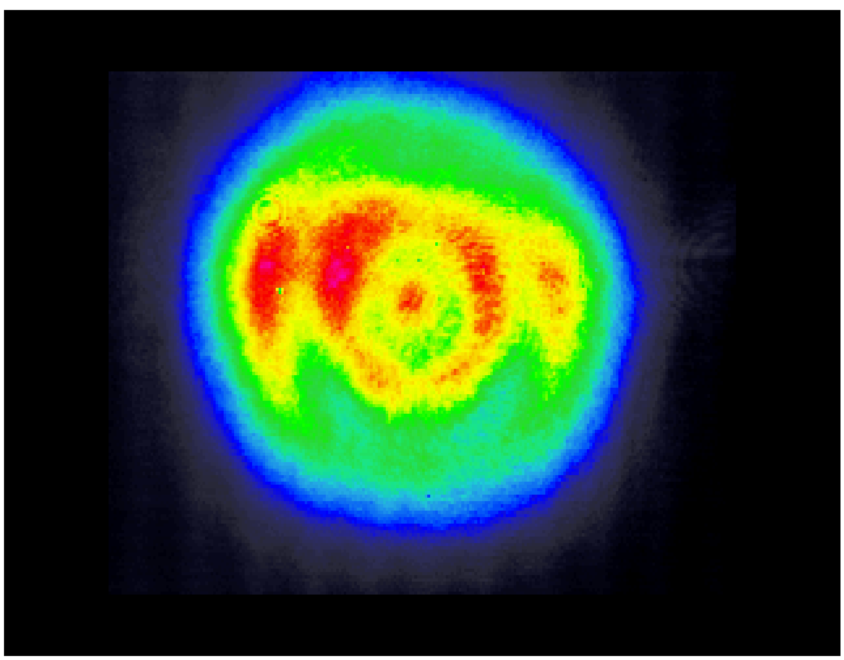


Figure A-18. Mode in the Intermediate Field when running the system at 50 Hz and 500 mJ/pulse

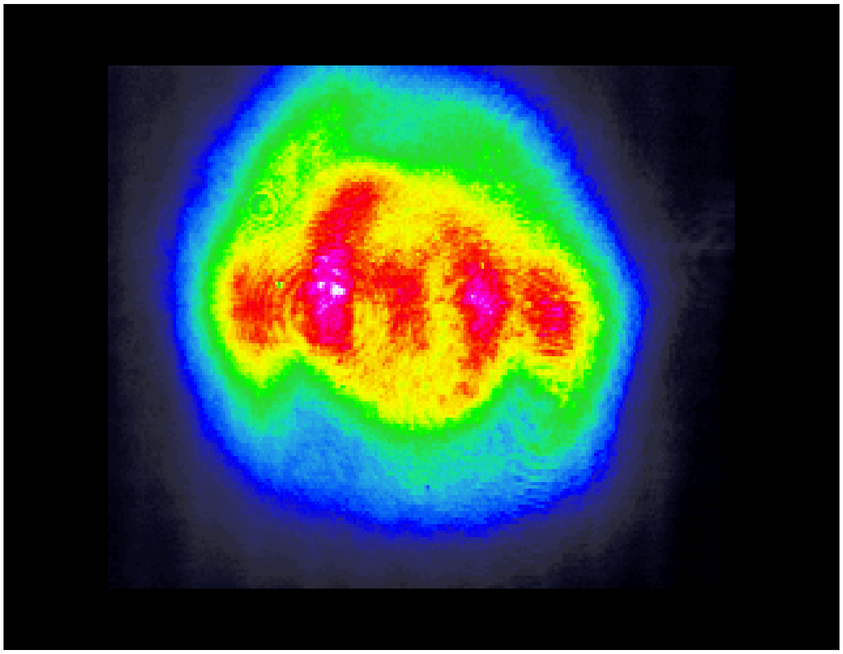


Figure A-19. Mode in the Intermediate Field when running the system at 100 Hz and 400 mJ/pulse

Introduction

This chapter is divided into several sections. Section 3.1, *Electronic Troubleshooting*, focuses on faults which are detected by the Infinity power supply CPU and then displayed to the end user by the Infinity software. Sections 3.2 through 3.6 address the issues which would prevent the Infinity laser system from performing up to the end users expectations as well as problems with system power up and computer communications.

Electronic Troubleshooting

Voltage out of range

+12VB out of range

+2VB out of range

+4VB out of range

+5V ref out of range

-12VA out of range

-12VB out of range

Definition: The CPU checks each of these voltages periodically and reports an error if the measured value falls outside a specified range. The pin numbers, voltage ranges, and sample rates are given in Table 3.1-1.

Table 3.1-1. Power Supply Voltage Errors

ERROR ("... OUT OF RANGE")	IC / PIN#	NOMINAL (V)	RANGE (V)	SAMPLE INTERVAL (S)
+ 12 VA	U102/9	2.39	2.21 – 2.82	60
+12VB	U111/12	2.39	1.99 – 3.17	10
+2VB	U111/10	2.0	1.67 – 2.65	60
+4VB	U111/9	4.0	3.33 – 5.0	60
+5V ref	U102/5	5.0	4.74 – 5.0	10
-12VA	U102/10	1.61	1.37 – 1.85	10
-12VB	U111/11	1.61	1.34 – 2.13	60

Action: If one or more of the fault messages above appears check the following:

1. Use the A/D converter available in the Service window to check the voltages. The 2 V and 4 V signals are read directly, the others are first divided down to fit into the converters 0 – 5 V operating range. Average over 50 readings. For the analog signals the standard deviation should be less than 1%. For the B- referenced signals the standard deviation should be about 2%. If the deviation is within its range, the diagnostic itself may have failed. Clear the message and see if it reappears.
2. If the software states that the voltage is out of range, remove the cover on the power supply and check the voltage at it's test point on the Control board. If the voltage is OK at the test point, the problem is in the digital electronics of the Control board. Replace the Control board.
3. If the voltage is out of range at the test point, check the supply voltages from the Condor power supplies on connectors J30 (floating ground) and J27 (B- referenced). Also verify that the Condor's are receiving their supply voltage through J25.
4. If the low voltage from the Condor's is there, the problem may be due to a load being placed on the system by another component. Attempt to isolate that component by trying the following:

   Disconnect all electrical connections to the laser head except the capacitor bank connector.

   Disconnect the booster boards from the control board.

Note: If the power supply responds to commands the + 5 V supply is working since the CPU runs off this voltage.

AC line voltage low

Definition: Pin 6 of U70 is high (LOWL_FAULT). The hardware shuts off the main contactor without CPU intervention. The CPU takes no other action.

Action: Check all phases on the main line against ground and that the three fuses are OK.

Wrong AC phase rotation

Definition: Pin 8 of U70 (ROT_FAULT) is high. The hardware prevents the contactor from engaging; the CPU takes no other action.

Action: Interchange any two of the incoming phases.

Contactor fan not turning

Definition: Pin 4 of U72 (FAN_SENSE_A) is either always high or always low.

Boost board fan not turning

Definition: Pin 5 of U72 (FAN_SENSE_B) is either always high or always low.

Transformer fan not turning

Definition: Pin 8 of U73 (FAN_SENSE_C) is either always high or always low.

For all three faults, when the fan is turning, this bit is supposed to cycle up and down. The CPU checks this bit every 0.1 s and reports a fault if the bit value remains unchanged 200 times in a row (approximately 20 s).

The CPU does not take any action when this fault occurs; it is reported to the user for information only.

Action:

1. Remove the power supply cover and inspect the fan. Check the polarity of the wiring if the fan is turning. The wrong polarity will cause the phase sense signal to be unreadable by the CPU.
2. Check for low voltages. +12 V ISO on J33 pin 1, pin 3 GND, for the Contactor fan. -12 V ISO on pin 1 and +5 V ISO on pin 5, J34 pin 3 or 4 GND, for the Booster board fan. -12 V ISO on pin 1 and +5 V ISO on pin 5, J35 pin 3 or 4 GND, for the Transformer fan.
3. If the wiring and are voltages are correct, check to see if a 2 - 4 VAC signal is present. For the Contactor fan this signal will be between pin 2(+) and pin 3(-), J33. For the Booster board fan check between pin 2(+) and pin 3 or 4(-), J34. For the Transformer fan pin 2(+) and pin 3 or 4(-), J35. If the AC signal is not there the fan has failed.

Cap bank not charged

Definition: This error occurs when two conditions are true: (1) Pin 6 of U72 (WAS_READY) is low during a lamp flash, and (2) the cap bank set point has not changed during the previous four shots. At any given instant, WAS_READY reflects the state of the charging circuit at the rising edge of the last lamp pulse: if the charging circuit was in its "maintenance charge" mode, WAS_READY will be high, otherwise it will be low.

When the cap bank set point is raised by a large amount, such as when the user increases the output power, it may take as many as four laser shots before the desired set point is actually achieved. During these shots WAS_READY will stay low. This is normal and is not reported as a fault.

The CPU does not take any action when this fault occurs; it is reported to the user for information only.

Action:

1. If the control board or the booster boards have been changed, recalibrate the control board.
2. Use a differential probe, connected between B ++ (upper right hand corner when key switch on right hand side) and LOAD – (the inner lead of capacitor C 44, upper left hand corner) to monitor the booster board charging cycle. An example of the charge cycle at 100 Hz is shown in Figure 3.1-1. The black solid line shows the increase in voltage across the capacitor bank as it is being charged by the booster boards. The grey diffuse area around it is due to the voltage spikes caused by the switching of the FETs. If the ramp up is much slower than shown in Figure 3-1 one of the booster boards might be defect.
3. The booster boards can be checked one at a time by running the system with only one board connected. Disconnect one of the boards by removing the AC input connectors coming from the EMI filter and connected to the rectifier mounted to the heat sink, see Figure 3.1-2. Note that you probably need to first remove the board from the power supply, disconnect it and then put it back into the power supply again.
4. This fault can also occur if the lamp pulse width has been set to long. Check on an oscilloscope, using inductive pick-off coupling from the cathode lamp lead, that the lamp current pulse is 250 µs long.

5. If any part of the capacitor/lamp circuitry is shorted to ground this fault condition may appear. The most likely cause, that has been seen in the field, is a short in the energy modulator board. Test if this is the case by, after having flipped the laser head on its side, bypass the energy modulator board (i. e., connect the flash lamp cable directly to the switching transistor [BFT]). Then turn on the system and try to flash the lamp. If the system works replace the energy modulator board. The system can be safely run without this board but it will run with a higher shot-to-shot intensity noise since Flash Track will be disabled.

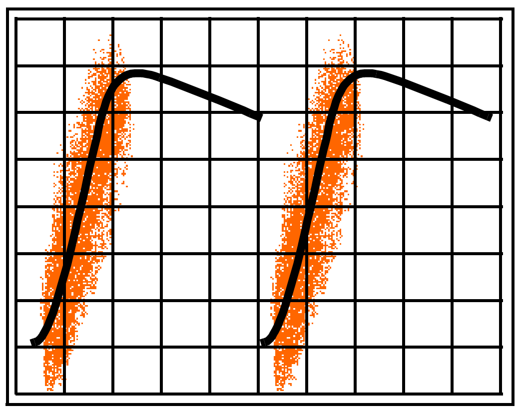


Figure 3.1-1. Booster Board Charge Cycle as Seen on an Oscilloscope, 2 ms/div.

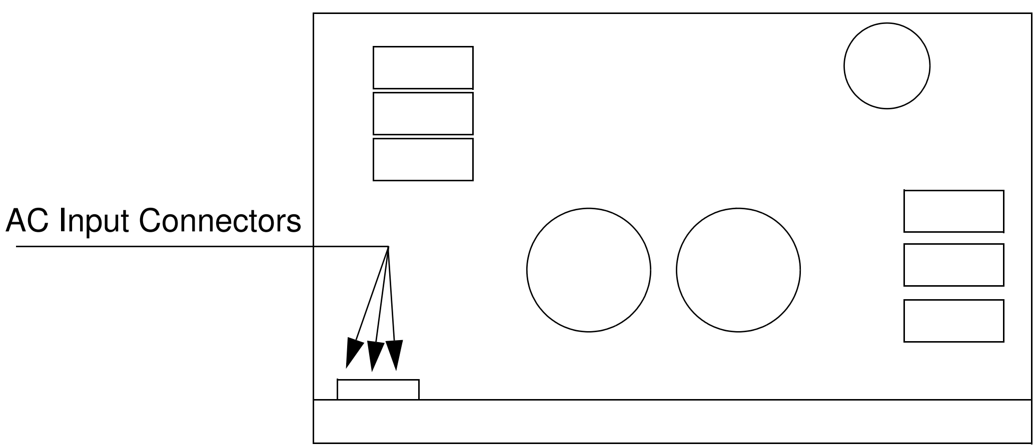


Figure 3.1-2. Position of AC input on Booster Boards

Control system failure

Definition: The contents of the control register, U3, have unexpectedly changed. The CPU reads the bits of this register through the read-back latch U75. When this fault occurs, the CPU attempts to shut down the entire laser system by turning off all CPU-controlled components; owing to the nature of this fault there is no guarantee that this will be successful. When this error occurs the message log contains an entry describing the actual and expected contents of the register: “Control register had ___, expected ___”. The numbers will be in decimal (base 10). By converting them to binary and referring to the schematic (sheet 10) it is possible to determine which control lines changed state.

Action:

1. Remove any test equipment that is connected to the laser.
2. Check that there are no strong emitters of EMI in the vicinity of the laser.
3. Check the registered bits in the Infinity control program up to the point of failure.

Depolarization too high

Definition: Two conditions can cause this fault: (1) A single laser shot had a depolarization greater than 10.0 mJ; (2) ten consecutive laser shots had a depolarization greater than 8.0 mJ. The depolarization is measured through the system A/D converter on pin 5 of U101. When the CPU detects this fault condition it disables the Q-switch trigger.

Action:

1. Make sure that the system is properly aligned. Check mode burns at different repetition rates in the near field and in the intermediate field (outside the cover).
2. Use the procedure described in section 2.9 of Chapter 2 to minimize the depolarized energy.
3. If the depolarized energy is still high, verify the energy using an external power meter. Recalibrate the internal meter if necessary.
4. If the depolarized energy is still high there may be an alignment problem with the POT assembly. Verify alignment with a CCD camera.

5. If the alignment and mode are OK, one of the POT assembly rods may have rotated during cleaning or lamp replacement. Return the assembly for realignment.

DPSS electronic fault

Definition: Three conditions must be true for this fault to be reported: (1) The DPSS is turned on through pin 18 of U3 (DPSS_ON), (2) The lamp trigger is enabled, (3) Pin 3 of U72 (DPSS_OK) must be low for one second. The CPU does not take any action when this fault occurs; it is reported to the user for information only.

Action:

1. Check that the DPSS cover is on and that the system has had time to warm up.
2. Check the supply voltage to the DPSS. This should be greater than or equal to 4.8 V with the DPSS turned on. If the value is lower, adjust the voltage up using the trim pot on the "floating ground" Condor board. Verify that the 5 V on the Control board in the power supply does not exceed 5.3 V. Voltages greater than this can damage the boards digital electronics.
3. Check to see if one of the DPSS internal interlocks has been tripped.

   Laser diode current fault, CR 5.

   Laser diode current off, CR 6

   Interlock fault, CR 11

   Laser diode temperature fault, CR 12

   A low on the cathode (black stripe) end of any of these diodes indicates a fault condition.
4. See "Unstable DPSS laser" in the next section of this chapter for additional information.

EEPROM hardware fault

Definition: The EEPROM chip (U6) is not responding to commands. This chip communicates with the CPU through a two-wire bus (SCL and SDA) controlled from the EPLD (U2). The fault can occur on either read or write and indicates that U6 failed to send a properly timed acknowledge signal. The CPU does not take any action when this fault occurs; it is reported to the user for information only.

Action: Replace the EPLD and/or the EEPROM

External (CDRH) interlock open

Definition: Pin 3 of U70 (EXT_INTLK_FAULT) is high. The hardware shuts off the main contactor without CPU intervention. The CPU takes no other action.

Action: Make sure that a RCA plus is inserted all the way into the External interlock connector on the power supply.

FlashTrack at limit

Definition: This error indicates that the FlashTrack centering servo is unable to center the FlashTrack error signal voltage, read on pin 7 of U101 (FT_ERR_SIG). Under normal conditions, the centering servo adjusts the FlashTrack coarse and vernier offset to maintain this signal at a level of 2.5 V. If the servo reaches the end of its available range this fault is reported. The servo will continue to monitor the voltage and keep trying to center it; the CPU takes no other action.

Action:

1. Check the A/D value from the photo diode. The value should be either 0 or 5 V (since it is at the limit). Block the light and see if there is any signal or noise.
2. Check the connector to the photo diode housing.
3. Disconnect the cable and check if the ± 12 V supply voltages to the heater circuit are present. +12 V on pin 1 (black wire), -12 V on pin 3 (white wire). Pin 2 (red wire) is common.
4. If low voltages are not there verify their presence on the Signal Distribution board. Pin 2 for +12 V, pin 3 for -12 V, and pin 1 common.

5. Connect an oscilloscope probe inductively to the cathode flash lamp lead. Terminate 1 MOhm. Check if Flash Track is present in the trace. Is it moving (i.e., trying to adjust the lamp current)? If it is, see if you can move it by changing the gain and coarse parameters in the servo window.

   If you do not see it, disable Autocentering and set the coarse side bar to zero and decrease the Gain in steps of 5 until zero is reached. The Flash Track spikes should appear on the oscilloscope at some point.

If you suspect that the photo-diode board is bad, test it in the IR power meter position. The boards are the same, but be aware that you will have to recalibrate the IR power meter after the original board has been put back. If the board works and Flash Track is present on the oscilloscope but doesn't regulate, the problem is on the signal distribution board. The only signal that comes from the outside is a flash lamp trigger from head board #1. If there are no signs of Flash Track whatsoever this trigger might be missing. In order to locate the faulty component/board the signal distribution board needs to accessed.

6. Flip the laser head on its side. Remove the bottom cover and check that the photo diode is correctly connected to the signal distribution board.
7. Using an isolation probe and an oscilloscope, check TP 4(+) and TP 10(-) on the signal distribution board. A signal with the same repetition rate as the flash lamp should be present. See section 4.8 of Chapter 4, Circuits, for more information.

High water temperature

Definition: This fault occurs when the signal on pin 12 of U102 (W_TEMP) exceeds 4.5 V, indicating a water temperature greater than 45°C. The CPU will immediately shut down all subsystems, including the heat exchanger.

Action:

1. Make sure that the primary cooling water is on and that the regulating valve inside the power supply allows a sufficient amount of water to flow through it.
2. Check the temperature of the internal water.
3. Check the connections to the temperature sensor.
4. Verify that the pump is spinning in the right direction (clockwise).

Interlock open

Definition: Pin 2 of U70 (INTLK_FAULT) is high. The hardware shuts off the main contactor without CPU intervention. The CPU takes no other action.

Action: Check all interlocks, wiring and connectors.

Lamp not simmering

Definition: This fault requires two conditions to be true: (1) The starter has been on for at least two seconds, or else it was on at some time in the past and was turned off because of a successful lamp start; in other words, the lamp should be lit. The starter is controlled by pin 14 of U3 (SIM/IGN_ENABLE). Note that when the CPU detects a high on pin 7 of U72 (SIMMER_ON), it assumes that the lamp has been successfully lit and it turns off SIM/IGN_ENABLE. (2) Pin 7 of U72 (SIMMER_ON) is low. The CPU takes no special action when this fault occurs.

Action: Components in either the lamp ignite or in the simmer circuit may have failed if the lamp does not simmer.

See sections 4.4 and 4.5 of Chapter 4 for a detailed overview of the ignition and simmer circuits.

Low capacitor bank voltage

Definition: Pin 9 of U70 (LOWV_FAULT) is high. The hardware shuts off the main contactor without CPU intervention. The CPU takes no other action.

This fault indicates that the capacitor bank voltage dropped below a critical level. In order for this fault to appear, the CPU must first enable the fault by setting pin 16 of U8 (LVF_ENABLE). The CPU sets LVF_ENABLE high when the power module boards are first enabled, and resets LVF_ENABLE low when the contactor is opened.

Action:

1. Measure the charging voltage on the booster boards, between B++ and LOAD –.
2. Check that the clock is working by verifying that the 10 MHz signal is present at its BNC on the power supply.
3. Check the Lamp Flash Out signal, from the trigger board, on the Control board using an isolation probe, TP 203(+) and TP 908(-). The observed TTL should be clean and have the same frequency as the software set repetition rate.

Low water flow

Definition: Pin 5 of U70 (WFLO_FAULT) is high. The hardware shuts off the main contactor. The CPU shuts off the heat exchanger to minimize the amount of water being squirted around the room, quite possibly at innocent bystanders.

Action:

1. Check that the water pump is spinning in the right direction (clockwise) and that there are no leaks in the system
2. Check the signal from the flow sensor, J29 pin 7(+) and pin 8 (-), with the heat exchanger on and off. There should be between a 4 and 5 volt difference between the two states.

Power module A over current

Power module B over current

Power module A over temperature

Power module A voltage fault

Power module B over temperature

Power module B voltage fault

Definition: The Power module faults result from logic low signals appearing on the associated signal lines (see control board schematics sheet 2). The relationship between signal names and fault names is straight forward. The fault lines are latched by U110 and U205, then multiplexed by U118. The OR gate U117 produces an output that is the logical OR of all 6 faults; this is also multiplexed by U118. The CPU sets the address bits on pins 11, 12, and 13 of U118 to select which of the fault signals is multiplexed out on pin 14. This output is transmitted across the isolation boundary by U156 (pin 5) and arrives at pin 9 of U72 (ALERT). Because of this complex scheme, these fault bits cannot be read directly by the CPU and are not available in the Peek window. All of these faults are latched. Like other system faults, they are cleared by the CPU when it receives the Clear Faults message from the PC.

It is often the case that all of these faults appear at once, typically preceded by a deafening auditory signal and a bright flash of light emanating from the power supply.

Action: This fault messaged usually indicates that one or both of the booster boards have failed and needs to be replaced.

Remote buffer overflow (characters lost)

Definition: The PC sent a message to the power supply before the previous message was acknowledged by the power supply. Since the message exchange is under software control, this error would normally indicate a software defect, although it could possibly be caused by a noisy serial line or defective serial port.

Remote framing error

Definition: The UART on the CPU reported a framing error, which typically results from a baud rate mismatch between the serial sender and receiver. Since the baud rate settings are under software control, this error would normally indicate a software defect, although it could possibly be caused by a noisy serial line or defective serial port.

Remote parity error

Definition: The UART on the CPU reported a parity error, indicating noise on the serial line.

Action for above 3 faults:

1. Check that the serial ports and the cable is OK.
2. Eliminate any sources of EMI.
3. If possible, try another computer.

Remote UART overrun (character lost)

Definition: The UART on the CPU reported a character overrun, a condition in which a character was received while the previous character was still read by the CPU. This indicates a software defect.

Action: Reinstall the software.

SCR failure

Definition: Two conditions must be true for this fault to occur: (1) The contactor must have been closed for at least 10 s; and (2) Pin 7 of U70 (SCR_ON) must be low. This pin is supposed to go high when the soft-start SCR engages. The CPU will not turn on the power modules under these conditions. It takes no other action.

Action:

1. Measure the voltage between B ++ and LOAD – on the booster board.
2. Measure the incoming line voltages and the output from the EMI filter.

Servo error: no lasing

Definition: Two conditions must be true for this fault to occur: (1) The lamp voltage must be above 460 V (highest non-lasing voltage), and (2) The signal on the laser power photocell must indicate less than 10.0 mJ (lowest meaningful energy). This does not disable the servo; the CPU will continually try to reach the power set point. The error has no real effect on laser operation.

Both the lamp voltage and the pulse energy are subject to calibration. During the startup sequence the PC sends messages to the laser that define what numbers to use for the highest non-lasing voltage and the lowest meaningful energy. The former is in D/A counts on a scale where 4095 is 5 V (the D/A converter chip is U207); the latter is A/D counts on a scale where 4095 is 5 V (the A/D converter channel is pin 6 of U101, IR_PC).

Action:

1. Make sure that there are no obstructions in the beam path.
2. Check the A/D signal from the photo diode in the Service window. Verify that it reads 0 V when the photo diode is blocked.
3. Check that the photo diode connector is pushed all the way into its socket.
4. Check the Q-switched output of the DPSS for large intensity drop outs, or missing shots, with a fast photodiode.
5. Verify that the signal goes to the Signal Distribution board. If the signal goes into the Signal Distribution board but does not appear on pin 6 of U101 the problem is probably located on the signal distribution board.

Water flow low

Definition: Pin 5 of U70 (WFLO_FAULT) is high. The hardware shuts off the main contactor without CPU intervention. The CPU shuts off the heat exchanger.

Action:

1. Make sure that the umbilical is not crimped
2. Check that the pump is spinning in the right direction (clockwise)
3. Check the flow sensor and its connections

Water level low

Definition: Pin 4 of U70 (WLVL_FAULT) is high. The hardware shuts off the main contactor without CPU intervention. The CPU shuts off the heat exchanger.

Action:

1. Check the water level in the reservoir
2. Check the level switch in the reservoir (floater) and its connections

Water resistivity low

Definition: Pin 11 of U102 (W_RES) is greater than 2.86V (2342 counts), indicating a water resistivity less than 0.5 MOhm-cm. (Resistivity is computed by dividing the A/D counts into 1171.) The CPU shuts off the main contactor but leaves the heat exchanger running. The contactor is shut off for reasons of electrical safety, which do not apply in the case of water resistivity too high.

Action: Increase the flow through the deionization filter by turning the valve located between the two filters counter clockwise, e.g. half a turn. Wait until the value has stabilized.

Water resistivity high

Definition: Pin 11 of U102 (W_RES) is less than 0.71V (585 counts), indicating a water resistivity greater than 2.0 MOhm-cm. (Resistivity is computed by dividing the A/D counts into 1171.) The CPU takes no action.

Action:

1. Decrease the flow through the deionization filter by turning the valve located between the two filters clockwise, (e.g., half a turn). Wait until the value has stabilized.
2. Verify that the connector P29, on the Control board, is firmly connected and that all contacts are secure (pins 1(+) and 2(-) are connected to the resistivity sensor).
3. If the resistivity does not decrease stop the flow through the deionization filter by turning the valve fully clockwise. Wait approximately 4 hours, if the resistivity does not decrease replace the sensor located in the water tank.

System Troubleshooting

Low IR Power

It is only meaningful to talk about low IR energy relative to the lamp voltage. A well aligned system with a new flash lamp (and a well working DPSS laser) should be able to produce 500 mJ/pulse with less than 515 V on the lamp at 10 Hz.

Table 3.2-1. Low Output Power Trouble Shooting Guide

A	CAUSE	DETAILS & REMEDY
1	Old flash lamp	Normally the lamp voltage should not increase by more than 40 V/100 Mshots. A typical voltage increase is 15–20 V for the first 30 Mshots on the lamp. **Remedy:** Replace the flash lamp.
4	IR photo diode calibration off	Use an external power meter to verify the calibration. The photo-cell is temperature stabilized at ~60 C. Check that the photo-cell board is hot. If not, check that the cable is connected at both ends. **Remedy:** Recalibrate the detector. Replace the photo-cell board.
5	Beam partially blocked	Check mode burns and beam path. Check that the flash lamp lead teflon wing protectors are not obstructing the beam.
6	Burned optics	Inspect all optics with a flash light at different angles. Remove the pot and inspect the rod surfaces. **Remedy:** Replace damaged optics. If rod surfaces are damaged return the assembly to CLG for refurbishment.
7	Uncoated expanding lens	Make sure that the expanding lens used is coated for 1064 nm.
8	System misaligned	Check mode burns in near field and outside cover. High depolarized energy and/or high noise can also be indications of poor alignment. **Remedy:** Realign the system.

Table 3.2-1. Low Output Power Trouble Shooting Guide (Continued)

9	Water contamination	Contaminations in the water can accumulate on the rod, decreasing the absorption of lamp light. Check for signs of contamination in the pot manifold. **Remedy:** Follow the maintenance procedure given in section 6.2 of Chapter 6.
10	Capacitor bank calibration	The capacitor bank voltage, as read out through the system software, should agree with the voltage measured with a DVM across the capacitor bank to within ± 5 %. Let the system warm up for 30 minutes before checking. **Remedy:** Use the automatic calibration procedure, available in the software under the calibrations menu in the Service window. If there is still a problem with the Capacitor bank voltage, follow the calibration routine given in section 6.1 of Chapter 6.

Poor IR Mode Quality

Table 3.2-2. Poor Mode Quality Trouble Shooting Guide

B	CAUSE	DETAILS & REMEDY
1	Poor beam alignment through the amplifier	Possible signs of poor alignment are: • asymmetric mode burns, especially if in the near field. • scattered light from the pot side of the pinhole. • high level of depolarized light. • excessive IR noise. • high lamp voltage. **Remedy:** Check the mode on the CCD camera while running the system at 10 Hz and 500 mJ/pulse. Use two wedges to split off a reflex that can be checked on the camera. If the mode can't be improved by adjusting the SBS lens realign the system using a CCD camera.
2	Poor DPSS laser mode	Use a CCD camera to check the DPSS mode. **Remedy:** If the mode from the DPSS laser is bad it needs to be replaced.
3	Burned optics	Inspect all optics with a flash light at different angles. Remove the pot and inspect the rod surfaces. Use a CCD camera if available. **Remedy:** Install new optics. If rod ends are burnt, return the POT to CLG for refurbishment.
4	Beam clipping	Check mode burns and beam path. Check that the flash lamp lead teflon wing protectors are not obstructing the beam.
5	Break down in SBS cell	DPSS laser mode hopping or contaminated liquid. Problem with Q-switch/Lamp timing. **Remedy:** Verify the Q-switch/lamp timing. Change the DPSS temperature 5 counts in either direction. Replace the SBS cell.
6	Position of SBS lens not optimized	**Remedy:** Align the lens using the procedure described in section 2.10 of Chapter 2.

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High IR Noise

The Infinity noise specifications are only valid for a system that has a new flash lamp, is operated with the covers on, at maximum specified energy and, is in thermal equilibrium.

Table 3.3-1. High Noise Trouble Shooting Guide

C	CAUSE	DETAILS & REMEDY
1	Q-switch/lamp timing not optimized	Check the settings in the lamp timing window. **Remedy:** See Operator's Manual for procedure on setting timing values.
2	Old flash lamp	As the lamp ages the shot-to-shot fluctuations will increase. The systems specification are only guaranteed with a new flash lamp. **Remedy:** Replace the flash lamp.
3	Poor alignment	Poor beam alignment can cause excessive noise. Check the mode using burn paper. **Remedy:** Realign the system using a CCD camera.
4	Unstable DPSS laser	Monitor DPSS laser output on photo diode. Peak to peak fluctuations should be less than 5 % on the oscilloscope. **Remedy:** Adjust the prelase level on pot R4, located on the main DPSS control board, while monitoring the output on a photo diode.
5	Poor control board calibration	**Remedy:** Use the calibration procedure in the Service Manual, section 6.1 Chapter 6.
6	Noisy A/D converter or detection circuit	Check A/D converter in Service window. The 2.0 V reference voltage should have a standard deviation of less than 10 counts when averaged over 50 cycles. **Remedy:** Check the cable path from the IR photo diode to the signal distribution board. It is important that the cable goes close to the base plate (ground plane) and as far away from other boards as possible.
7	Low power/bad IR calibration	The noise specifications are only valid when the system is run at its maximum energy. Check that the output energy is 400 mJ at 100 Hz or 500 mJ at 10 Hz. **Remedy:** Operate the laser at maximum energy.

Table 3.3-1. High Noise Trouble Shooting Guide (Continued)

C	CAUSE	DETAILS & REMEDY
8	FlashTrack not optimized	Verify that FlashTrack is working using the same type of inductive pick off signal as when setting the lamp/Q-switch timing. **Remedy:** Engage autocentering in the Servo window. Optimize the gain setting while monitoring the noise in the monitor window.
9	Intensity too low on photo diode	At 500 mJ/pulse the voltage on the IR photo diode should be approximately 3.5 V. At 400 mJ/pulse the voltage should be approximately 3.0 V. Check these voltages through the A/D converter in the Service window. **Remedy:** Increase the light intensity on the photo diode and recalibrate.
10	Energy modulator board broken	Connect an oscilloscope probe inductively to the cathode flash lamp lead. Besides the on and off spike in the trace there should also be a spike from the turn-on of the FlashTrack signal in the middle. If FlashTrack is active this spike should be moving back and forth. It should also respond to changes in the gain setting in the Servo window. **Remedy:** Replace the Energy Modulator board.
11	Noisy environment	Both the DPSS cover and the laser head cover should be on for optimum performance. Removing the DPSS cover increases noise due to EMI disturbance from the flash lamp and opening the laser head cover increases noise due to air turbulence. **Remedy:** Move laser or eliminate sources of interference.
12	Missing phase on booster board	If one of the incoming lines on the booster boards is missing or disconnected the noise at 100 Hz will be high. **Remedy:** Verify the 3-phases at Contactor 2, the EMI filter, and at the main connector on each of the Booster boards.

Unstable DPSS Laser

Manifestations of an unstable DPSS laser can be:

- high intensity noise
- higher than normal lamp voltage and
- Large timing jitter.

Verify that the DPSS laser is unstable with a photo-diode. The DPSS cover should be on and the flash lamp off.

Table 3.4-1. Unstable DPSS Laser Trouble Shooting Guide

D	CAUSE	DETAILS & REMEDY
1	Pre-lase too low or too high	**Remedy:** Minimize DPSS jitter by adjusting prelase level (pot R4) on DPSS Control BD (monitor DPSS Q-switch output with a fast photodiode). Jitter should be <700 ps. Counterclockwise increases prelase. Monitor RF voltage between gnd (TP 3) and inner most leg (with respect to DPSS output window) of transistor Q3 for stability.
2	DPSS cover off / other EMI interference	**Remedy:** Put on the DPSS cover to eliminate EMI interference or move laser.
3	Supply voltage too low	**Remedy:** Check voltage between TP 14(+) and TP 3(-) on DPSS Control BD. Should be between 4.7 asnd 4.9 V. If lower than 4.7 adjust on analog housekeeper board. Care should be taken that +5 V does not go over 5.3 V on the P/S Control BD.

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Laser Does Not Turn On

When the key switch is turned "ON", the power supply fans should come on and the display above the BNCs should display a zero.

Table 3.5-1. System Will Not Power Up Trouble Shooting Guide

E	CAUSE	DETAILS & REMEDY
1	AC line phase missing	Check the incoming AC lines.
2	Connectors not fully in sockets	Reinsert all connectors. Go through the connectors on the boards.
3	Line fuse blown	Check the fuses.
4	House keeper board failed	Check supply voltage to housekeeper boards. Check low voltages on control board and on connectors.

No Response To Computer Commands

If the computer works and the power supply comes on and displays all expected life signs but does not respond to commands from the computer check the possible causes in Table 3.5-2.

Table 3.5-2. Laser Not responding to Computer Commands

F	CAUSE	DETAILS & REMEDY
1	Serial cable not connected	Connect cable or use new cable.
2	Program has crashed	Reboot computer.
3	Cable connected to the wrong port on the computer	Verify that the cable is connected to the port set in the software. Default is COM 1 for the power supply and COM 2 for the mouse. Use the 9 to 25 pin adapter for the mouse.
4	RS 232 card in computer broken	Insert new card or try another computer.

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Low SHG Efficiency

Table 3.6-1. Low SHG Efficiency Trouble Shooting Guide

G	CAUSE	DETAILS & REMEDY
1	SHG crystal not optimally phase matched	Rotate the SHG crystal slowly around the horizontal axis in small steps. Note that there are several weaker side maxima. Make sure you are on the absolute maxima.
2	IR power low	Make sure that the internal energy meter is correctly calibrated. Check the IR power with an external power meter.
3	Poor mode quality	Check the IR mode in the near field and outside the cover. There should be no signs of beam clipping or pronounced asymmetries.
4	Crystal/Optics damaged	Remove the crystal and inspect the optical surfaces of the crystal and the input and output window. Check the dichroics and the window on the head cover for damage.
5	Wrong coating on expanding lens	Make sure that the coating on the dichroics and on the expanding lens are for the right wavelength.

Low THG/FHG Efficiency

Table 3.6-2. Low THG/FHG Efficiency Trouble Shooting Guide

H	CAUSE	DETAILS & REMEDY
1	(THG only) SHG crystal not optimally phase matched for maximum THG energy.	The position of the SHG crystal that produces the most energy at 532 nm might not be exactly the same as the optimum position of the SHG crystal position for THG energy. After the SHG crystal has been optimized for maximum energy at 532 nm and the THG crystal angel optimized for maximum energy at 355 nm, go back and tweak on the SHG phase matching angel again, maximizing the energy at 355 nm.
3	THG/FHG crystal not optimally phase matched	Rotate the THG/FHG crystal slowly around the vertical axis in small steps. Note that there are several weaker side maxima. Make sure you are on the absolute maxima.
4	(THG only) Waveplate position not optimized	Rotate the wave plate, positioned between the SHG and THG crystal, so that the energy at 355 nm is maximum.
5	IR power low	Make sure that the internal energy meter is correctly calibrated. Check the IR power with an external power meter.
6	Poor mode quality	Check the IR mode in the near field and outside the cover. There should be no signs of beam clipping or pronounced asymmetries.
7	Crystal/Optics damaged	Remove the crystal and inspect the optical surfaces of the crystal and the input and output window. Check that the proper optics are installed and that they are not damaged.
9	Head cover window installed	If there is a window in the head cover it should be removed.

Circuit Descriptions

The following chapter contains descriptions and simplified diagrams for each of the principle electronic functional groups of the Infinity laser system. For a complete representation of the Infinity electronics, the user is directed to the schematics at the end of this manual.

Supply Voltages

The current path of the incoming AC line is illustrated in Figure 4.1-1. Each phase first goes through a line fuse, mounted in a fuse box that also serves as a breaker. Inside the power supply line voltage is applied to three locations as soon as the main wall breaker is closed, to contactor 1, contactor 2 and to connection J25 on the control board. Due to the fact that the water pump is phase dependent, it is necessary to monitor the line phase order to insure proper operation of the pump. This is the reason for the direct connection, J25, on the P/S control board. One of the lines (red) going into the control board goes out and to the key switch. It comes out as orange from the key switch, is split into two cables, each one going as supply voltage to one house keeper board. Incoming black is also taken out from the control board on two cables, each going to one of the house keeper boards. When the key switch is closed line voltage is supplied to the house keeper boards which in turn supply all low voltage electronics in the system. When contactor 1 is closed from the software the pump starts. When contactor 2 closes line voltage is applied to the rectifiers on the booster boards through the EMI filter. As soon as the second contactor is closed the voltage across the capacitor bank is equal to rectified line voltage, 290 V.

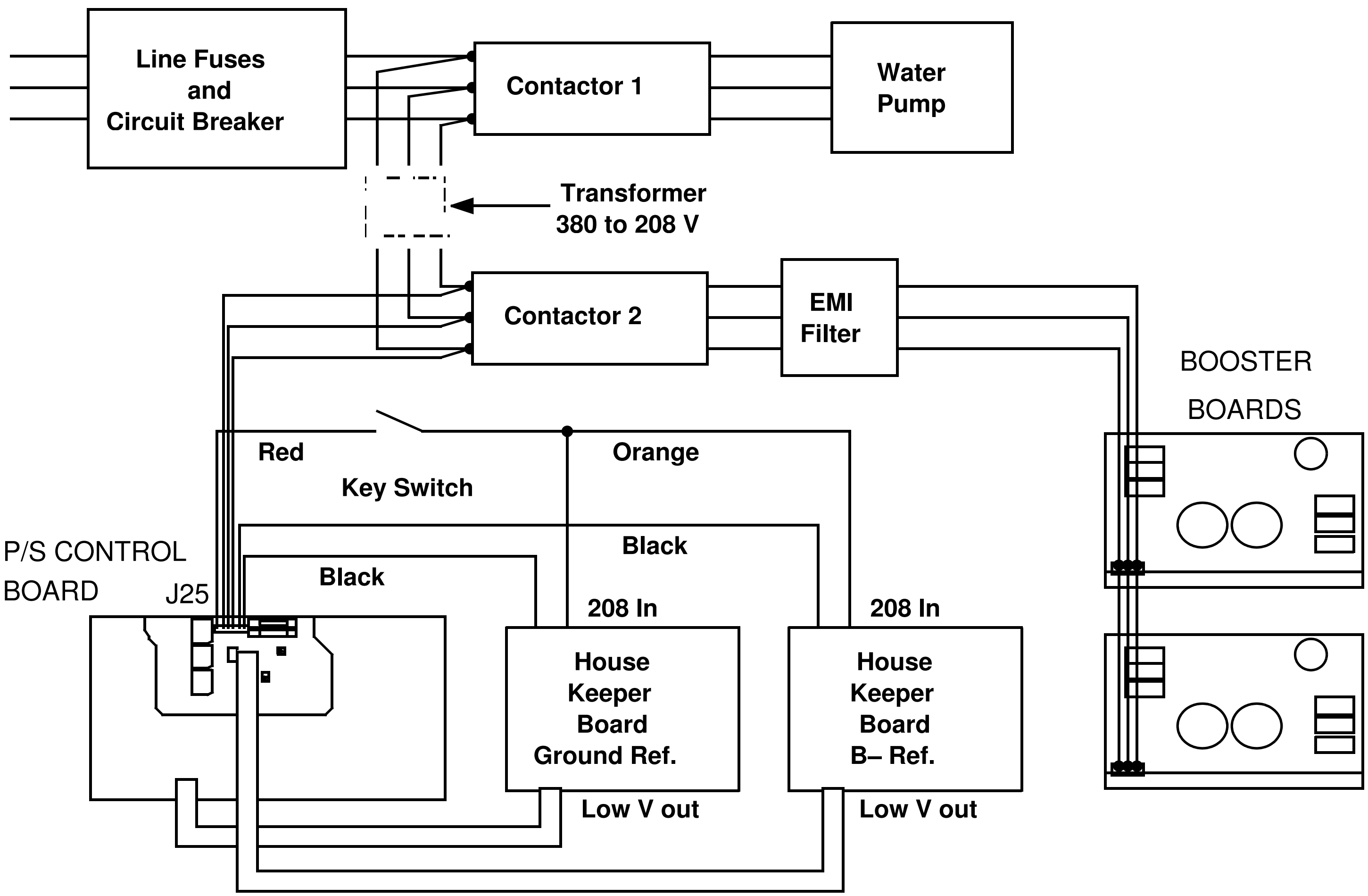

Figure 4.1-1. Block Diagram of Supply Voltage Distribution

DPSS Prelase Regulation

An overview of the prelase/autolock circuit is given in Figure 4.2-1. The prelase level of the Infinity DPMO is set by adjusting trim pot R4. As the effective resistance of R4 is increased, clockwise rotation, the prelase level of the DPMO increases. A typical adjustment range for the RF drive voltage is 8 to 13 V. A typical hold off voltage (i.e., the voltage where all prelase is suppressed) is 18 V. For voltages ≤ 5 V, the prelase output is a maximum. The RF drive voltage can be measured on the inner most leg of Q3 on the DPSS Control Board (part of 5x RF drive voltage circuit, see DPSS control board schematic sheet 7).

Located inside the DPSS housing is a photodetector. The diffracted beam from the AO cell is directed into this photodetector. As the RF drive voltage is increased the diffraction efficiency of the AO cell increases in an “almost” exponential manner. However, since the prelase level of the DPSS decreases with increasing voltage the amount of diffracted energy also decreases.

The prelase trim pot and the photodetector form an active stabilization loop for the DPMO. During normal operation the voltage at the junction point, and thus the input to the inverting input of U6, should be approximately zero (see Figure 4.2-1). As a result the output from U6 will be a small negative voltage. This voltage is inverted by U11 before being sampled on pin 3 of U3. The output from U3, pin 5, is sent to a 5x RF drive voltage amplification stage and to comparator U10.

The purpose of the comparator is to determine if the output from the sample and hold, U3, is within a 0.250 to -0.250 V window. If the voltage from U3 is outside this window the gate of Q3 will go high, causing the FET to be forward biased. This will pull J3 pin 10 low (to ground) giving a DPSS electronic fault.

Additionally, the output voltage from amplifier stage is monitored by the electronics on the “Prelase Autolock” board. The RF drive voltage is sent through a 2:1 voltage divider to the noninverting terminal on U1. If this voltage is less than 2.5 V the output from U1 will go low, forward biasing Q1. This will provide an additional 5 V to the output of the sample and hold, U3, on the DPSS control board. The end result is a RF drive voltage greater than the prelase hold off voltage.

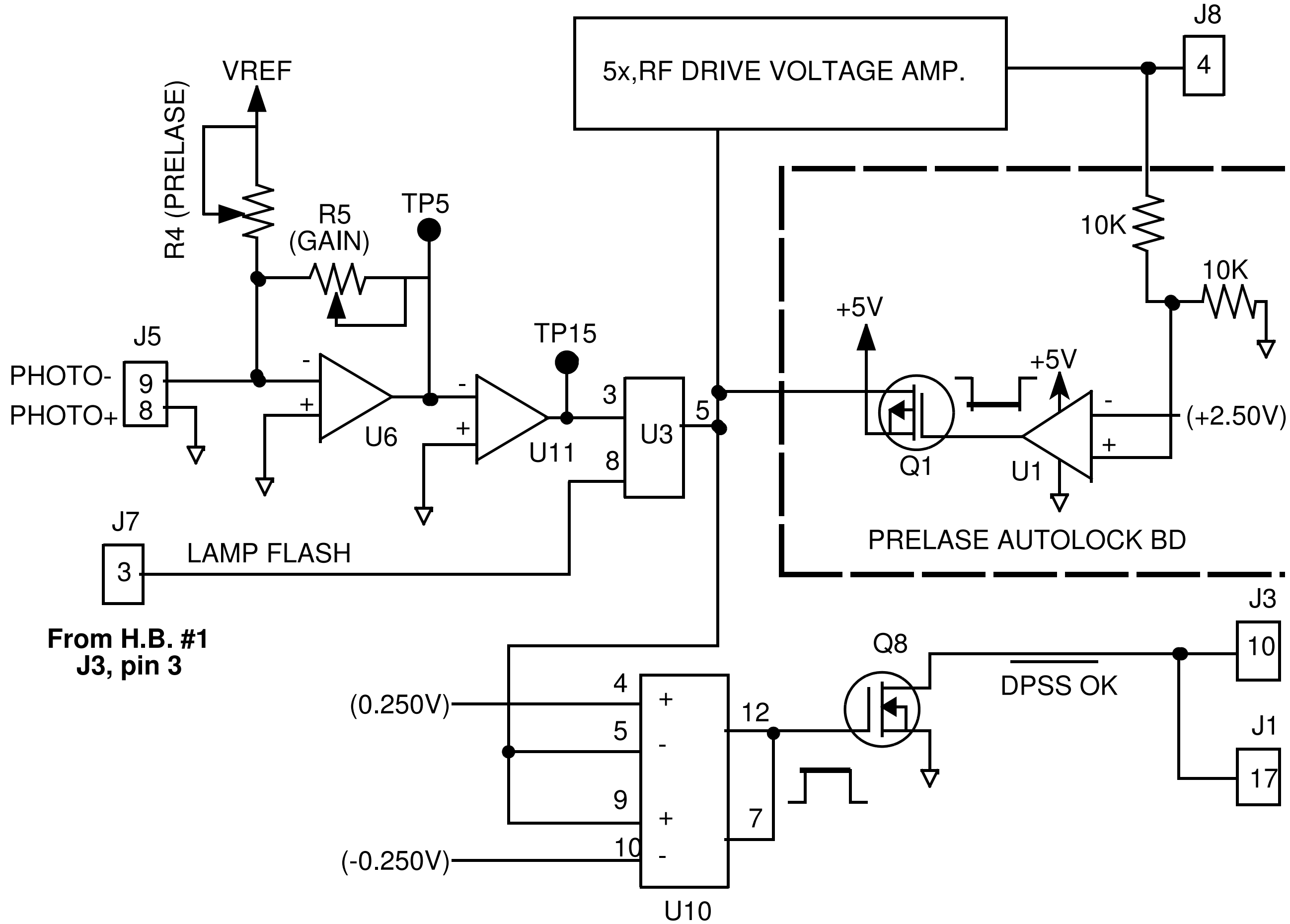

Figure 4.2-1. DPSS Prelase/Autolock Circuit

If there are no problems with the DPSS the drive voltage will relax back to the set point established by R4. If there is a problem with the internal alignment of the DPSS, or one of the internal optical surfaces, the DPSS will continue to ramp.

In addition to the voltage input on pin 3 of the Sample and Hold (U3), there is a second input on pin 8. This is the lamp flash trigger from the Trigger board in the power supply (via head board #1). The purpose of this input is to lock the last state sampled during the flashing of the lamp.

DPSS Fault I/O Logic Circuits

There are four fault conditions monitored, and latched, by the DPSS. These faults are:

- 5 volt power supply, CR6
- Laser diode temperature, CR12
- Interlock, CR11
- Laser diode current, CR5

It is important to note that these individual faults are not reported to the CPU. To determine which fault has been set the output voltage from the appropriate flip-flop logic gate must be measured. This is most easily accomplished by measuring the voltage at the cathode end of the diodes given in the above list. In most cases a DPSS electronic fault will be displayed on the computer monitor when one of these faults occurs.

Once a fault condition is achieved it is latched by the circuit. To clear a fault, with the exception of the “interlock” fault, the entire laser system will need to be cycled off and on. The interlock fault can be cleared by simply cycling the DPSS off and on. These points are discussed in detail in the sections that follow and are illustrated in Figure 4.2-2. Note that the diode temperature and power supply fault circuits are similar to the diode current. For these latter circuits the reader is referred to the schematics. Table 4.2-1 gives an overview of the fault logic flow (i.e., expected voltage states) for these circuits.

Interlock Fault

If there is an interlock fault a "low" should be measured on the cathode end of CR11. Since there is an inverter between this diode and the signal source the combined output of the two NAND gates (making up the latching flip-flop) must be high for the fault to be true (i.e., to get a low voltage out).

At initial start up, if there are no open interlocks, pin 12 of U9-C will be high. When the system is first turned on pin 11 of U9-C is pulled to ground through Q5, which is forward biased for 1.5 seconds. The two NAND gate pins which are not externally connected are low at start up. At this point the output of the lower NAND gate will be high. This will cause the output of the upper NAND gate to go low giving U9-C a high output. When Q5 closes the lower NAND gate will have a low on its upper terminal (the output of the upper logic gate) and a high on the lower terminal (pin 11) giving a high output. This gives two highs on the inputs of the upper NAND gate producing a low out put (no fault condition). Due to the inverter after the output of the upper logic gate the output of U9-C will be high when no fault condition exists.

If an interlock is opened the base of Q4 will go high forward biasing the FET. This will pull pin 12 of U9-C to ground, causing the output of the upper NAND gate to go high. This will give two highs on the lower NAND gate which will latch with a low output giving two lows on the inputs of the upper NAND gate. The final output of the flip-flop will be high and converted to a low after passing through the inverter (fault condition).

To clear this fault, once the interlock problem is found, the DPSS needs to be cycled off and on. Cycling the DPSS off will cause the base of Q7 to go high, forward biasing the FET. This will pull pin 11 of U9-C to ground causing the output of the lower NAND gate to go high (since one terminal will be high and the other will be low). This will give two highs on the input terminals of the upper NAND gate giving an overall low output. When Q7 closes pin 11 will go high and the upper pin will now be low. The condition required for normal operation.

Diode Current Fault

The circuit for monitoring the laser diode current fault state works in exactly the same manner as the interlock fault monitor. However, the only way to clear the fault is to cycle the entire system on and off (this includes the power supply).

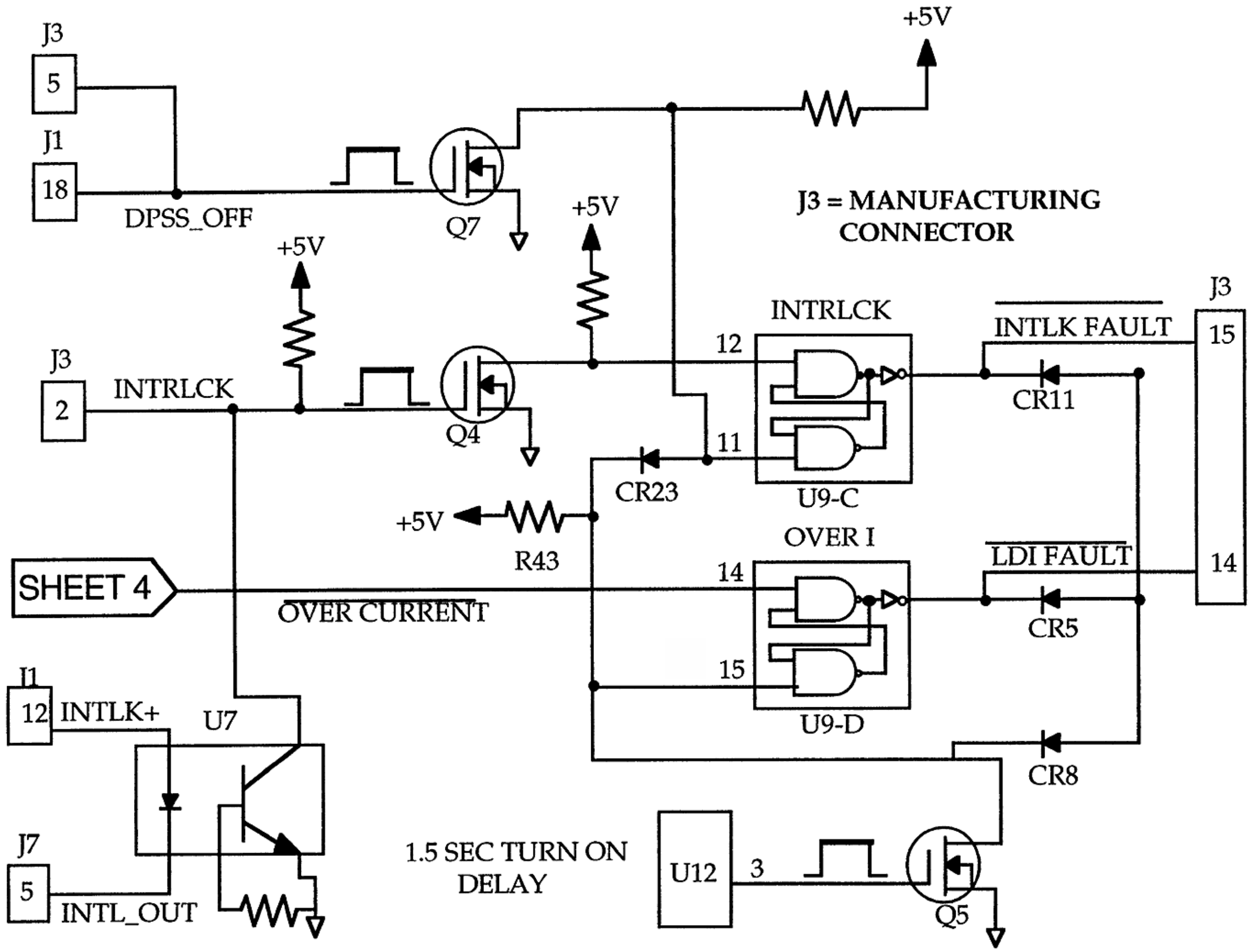

Figure 4.2-2. DPSS Fault I/O Logic

When the system is first turned on pin 15 of the lower NAND gate of U9-D is pulled to ground via Q5. This will cause the output of the logic gate to go high. At this point both inputs to the upper NAND gate will be high giving the desired low output before the inverter. When Q5 closes pin 15 of the lower NAND gate will go high. Since the upper input of this gate is low at this point, the output of the gate will latch high causing both inputs of the upper NAND gate to be high. This gives the required low output, before the inverter, for a "no" fault condition.

If the diode current goes 16% over the set value, pin 14 of U9-D will go low causing the output of the upper NAND gate to go high. This will cause the output of the lower NAND gate to latch low, since both inputs will be high at this point. This is the fault condition.

I/O Logic States

There are four possible combinations of input logic states for each of the flip-flop fault monitors on the DPSS control board. These are given in Table 4.2-1, along with the resulting output logic state. In the Table 4.2-1 a "1" indicates 5 volts and a "0" indicates 0 volts. This applies to the inputs and the output. In determining the output logic state, it has been taken into account that there is an inverter between the flip-flop and the output pin.

Looking at the Interlock flip-flop given in Figure 4.2-2, the header "Upper Pin" in Table 4.2-1 is referring to pin 12 and the header "Lower Pin" is referring to pin 11. The header "Output State" is the logic state that would be expected on the cathode of CR11.

Table 4.2-1. Logic Table for DPSS Interlock Monitoring Circuits

	UPPER PIN	LOWER PIN	OUTPUT STATE
1	1	1	1
2	1	0	1
3	0	1	0
4	0	0	0

DPSS Trouble shooting

There are three reasons for a FSE to suspect that there is a problem with the Infinity master oscillator. These are:

- System is reporting a DPSS electronic fault.
- System is reporting a servo error, system not lasing.
- System is reporting that the long term shot-to-shot noise is out of specification.

Based on these indicators, it can be expected that the DPSS will be in one of two states:

- Emitting light.
- Not emitting light.

To aide in the troubleshooting of the DPMO a flow chart is provided. Since it can be expected that the DPSS will be in one of two states, this is where the flow chart begins. Figure 4.2-3 gives the first part of the flow chart, and focuses on troubleshooting measures for the case where the DPSS is emitting light. Figure 4.2-4 gives the second part of the flow chart for the case where no light is being emitted by the DPMO.

The numbers given in Figures 4.2-3 and 4.2-4, next to the flow chart elements, indicate that additional information for that step is provided in the text of this section.

When troubleshooting the DPMO, the Infinity laser system should always be controlled from the "Service" window.

Supplemental information for Figure 4.2-3.

1. To determine if the DPMO output is stable, the CW output should be viewed with a photodiode. The CW power output of the DPMO can be determined using a 210 or 212 power meter. A typical output power is 10 mW. The RF drive voltage can be measured on the inner most leg of Q3. A typical drive voltage range is 11 to 14V. The prelase level, and thus the RF drive voltage, are adjusted via R4 on the DPSS control board.
2. The jitter in the DPMO output is determined with the Q-switch enabled. The repetition rate should be set to 100 Hz. To minimize the jitter in this output, the prelase level is adjusted (R4). A typical range for the Q-switched output of the DPMO is 28 to 35 µJ.

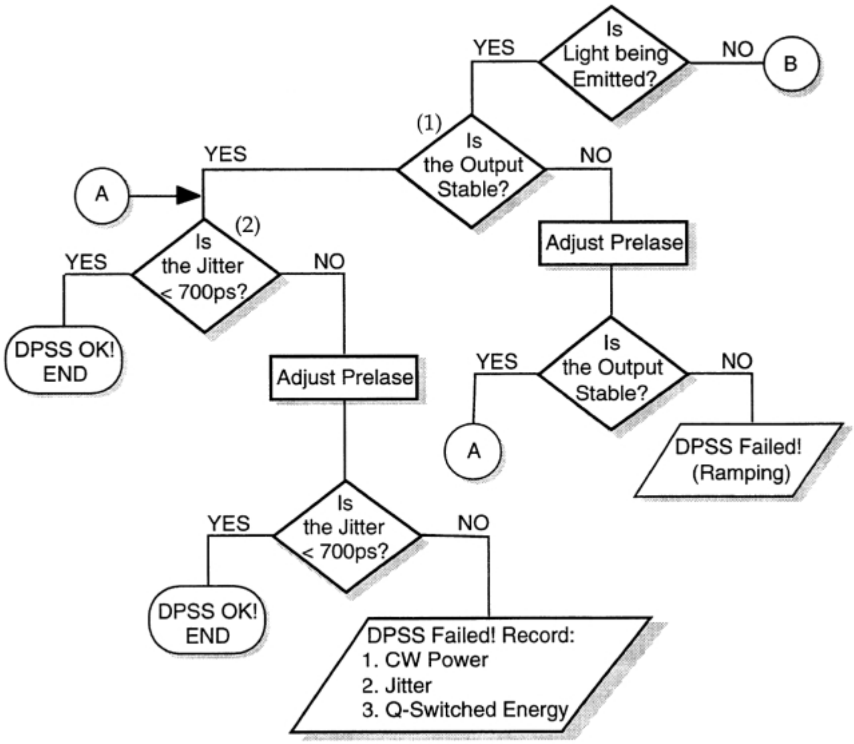

Figure 4.2-3. DPSS Troubleshooting Flow Chart, DPSS Lasing

Supplemental information for Figure 4.2-4.

1. If a DPSS electronic fault is encountered, the first thing which should be checked is the +5 volts supplied to the DPSS from the analog housekeeper board. This voltage should be adjusted so that it is at least 4.7 volts when the DPSS is turned on.

When adjusting the +5 volts on the DPSS control board it is important to check that the voltage does not increase beyond 5.3 volts on the power supply control board. If the voltage on the P/S control board exceeds 5.3 volts the digital electronics on this board may be damaged.

In addition to the +5 volts, the -5 volts (TP 11) and the +24 volts (TP 16) on the DPSS control board should be verified. Both of these voltages are generated by the DPSS electronics.

2. If the power supply voltage is OK, then one of the four DPSS faults are latched. Measure the voltage on the diodes indicated in the “DPSS Fault I/O Logic Circuits” section above.

   Once the fault has been identified, verify that there is indeed a problem. If there is a latched fault, but no “real” fault condition exists, there may be a problem with the flip-flop for that fault. If this is the case than the DPSS will need to be replaced. Contact Santa Clara with your findings.

   If it is found that one of the faults is latched, and the fault condition is true (with the exception of the interlock fault), than the DPSS will need to be replaced. Contact Santa Clara with your findings.

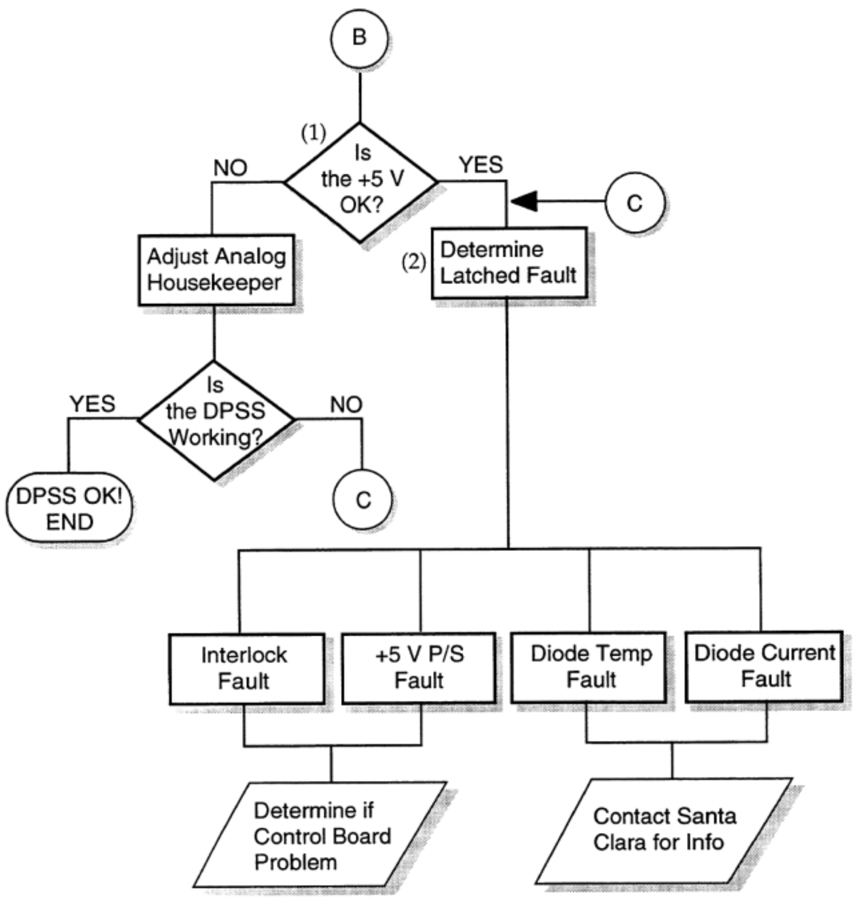

Figure 4.2-4. DPSS Troubleshooting Flow Chart, DPSS not Lasing

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Lamp Ignition

The lamp ignition process is initiated by the SIM/IGN_ENABLE going low, see Figure 4.3-1. As a result pin 8, INH, on U301 goes low. This activates the FET drive, output 11 from U301. The frequency of the FET base drive signal is ~40 kHz, determined by a RC circuit connected to pins 5 and 6 on U 301.

Every time the FET, Q300, is turned on current is pulled through the transformer T301. As a consequence the voltage on the line going to pin 10 on U 301 starts to rise. When it reaches a preset value the FET is turned off. When the FET is turned off it creates a reverse voltage spike in the transformer T301 (200:1 current reduction). The spike in the secondary side then creates a current flow through the diode, CR 302, that charges the capacitor, C390. The U 301 circuit will continue to drive the FET, at a switching frequency determined by the RC network and a duty cycle determined by the feedback coming in on line 10, until the voltage across the capacitor is large enough to break down the spark gaps, SG1 and SG2.

While the FET is charging the capacitor, both sides of spark gap SG1 will be at the same potential since current can flow through the five 2 MΩ resistors across it. Consequently spark gap 2 will break down first. When this happens the potential on the side of spark gap 1 connected to spark gap 2 will fall like a rock, thus creating a large potential across it. On the time scale that this takes place no current will have time to flow through the resistors and SG2 will also spark. With both spark gaps conducting current can flow freely from the capacitor out to the ignition transformer. An inductor is placed after the last spark gap to slow down the current flow.

The voltage spike (~1 kV) travels into the ignition transformer. Since the primary side with its single winding has a much lower impedance than the secondary side it will receive most of the current. A 15 kV voltage spike is induced and sent to the flash lamp where it ionizes the gas.

If the ignition attempt was successful the simmer current comes on and current will start to flow through SIM A and SIM B. This will cause pin 8, INH, on U 301 to go high which inhibits any further charging of the capacitor (i.e., no more ignition pulses will be sent to the lamp).

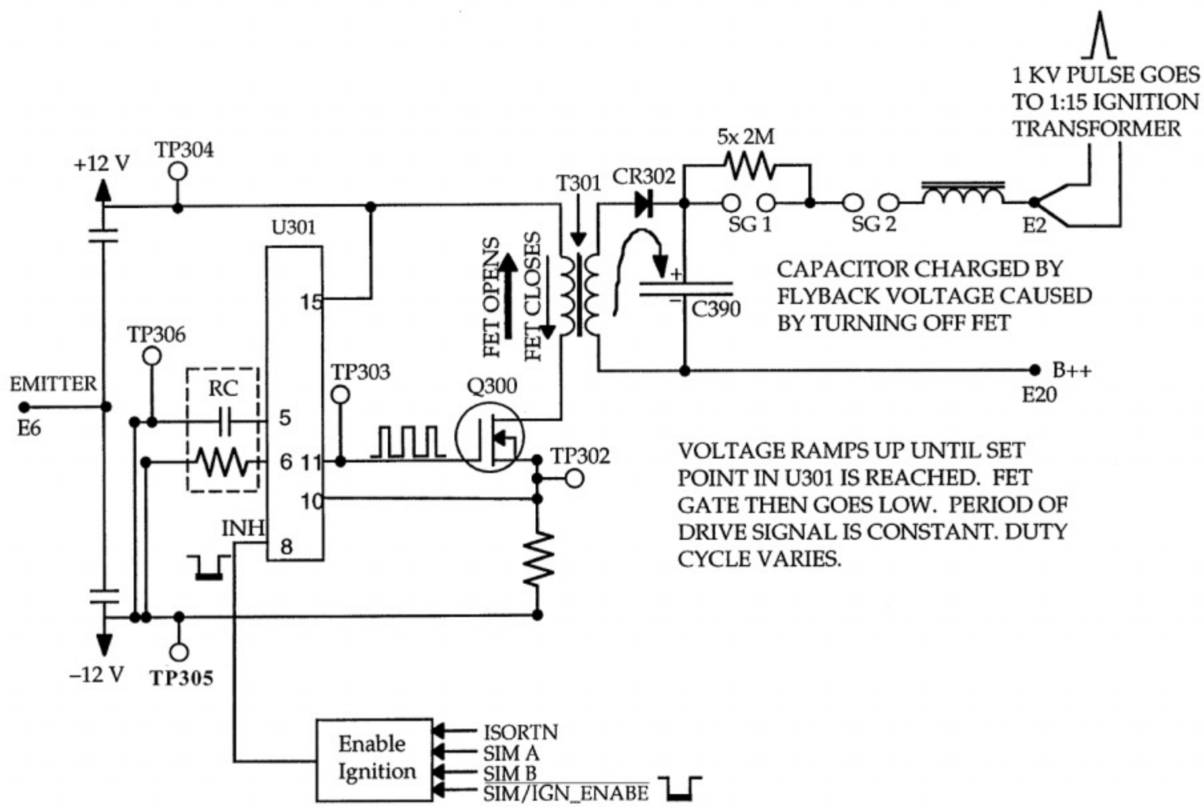

Figure 4.3-1. Block Diagram of Ignition Circuit, Head Board #1

If the ignition attempt was unsuccessful, U 301 will simply continue to charge the capacitor by driving the FET. If the lamp hasn't ignited within 2 seconds the CPU will set SIM/IGN_ENABLE high preventing any further ignition attempts to be made. An error message is displayed stating: "Lamp not simmering".

All circuits participating in the ignition process are referenced to B–. This means that an isolation probe must be used when measuring voltages using an oscilloscope. The only exception is the main trigger signal coming from the CPU, SIM/IGN_ENABLE. This signal is TTL and is isolated from the ignition circuit by an opto isolator, U302. Referencing the circuits to B– is accomplished by connecting the BFT emitter capacitively to ± 12 V (C303 and C301).

Troubleshooting

If the lamp fails to ignite:

1. Check that the flash lamp leads are connected and that the lamp is OK
2. Check the house keeper voltages, ± 5 VA and ± 12 V (both analog and B- referenced) on the control board in the power supply.
3. Check that there is 24 V between TP 304 and TP305 on head board #1.
4. Check that the main trigger, SIM/IGN_ENABLE, is present on J2 pin 14 (pin 2 of U302).
5. Check the voltage in time between pin 8 on U 301 and Tp 305. It should go high, up ~6 V, when the laser attempts to ignite the lamp (i.e., if either On or Standby is pressed in the user interface). Use a DVM with min/max sensor.
6. Check the FET base drive, TP 303. It should swing between + and - ~6 V.
7. Check the voltage across transformer, T 301. Pins 1 and 2 on the primary side, 4 and 5 on the secondary side.
8. Check the spark gaps. Replace the spark gaps if they do not fire.

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Lamp Simmer

Simmering the lamp means that a DC current, capable of sustaining the plasma created by the ignition pulse, flows through the lamp. This means that every time the lamp flashes, the current pulse from the capacitor bank can travel through the ionized corridor maintained by the simmer current instead of having to first break down the gas. The advantages of simmering the lamp are: higher pumping efficiency, increased lamp life, improved pulse to pulse reproducibility of lamp output and reduced time jitter.

The simmer electronics is located on Head Boards #1 and #2. A simplified circuit diagram showing the main components in the path of the simmer current and the corresponding control electronics is shown in Figure 4.4-1.

Circuit Description

Ionizing the gas in the flash lamp opens a conductive path from B++ to Cap-. The simmer current, ~1.5 Amp, can then go from the capacitor bank through the ignition transformer, the lamp, the energy modulator and into the simmer circuit on head board 2, continue into head board 1 and back to the B– side of the capacitor bank.

The primary regulating component of the simmer current is a FET located on head board 1, Q8. The gate of this FET is driven by the amplified output from a free running IC, U1. Free running means that there are no trigger or synch signals connected to it. As soon as the laser is turned on it will start to try and regulate the (initially non-existent) current. The switching frequency, ~60 kHz, is determined by the RC network, R8 and C4, connected to pin 4 of U1.

The duty cycle of the FET drive, which determines the current through the simmer circuit, is set by U1 based on feed back from two current sensing components, LA25-NP (LEM) and a pick-off transformer, T1. Both are located on head board 2.

When the FET is closed (conducting) current starts to flow through the inductors, L6 and L7. As this happens a magnetic field is created, through inductance, that chokes the current flow and it continues to rise linearly until the FET is opened. Mathematically the inductors behavior is described by the relationship: $V = -L\,di/dt$, where V is the voltage across the inductor, L its inductance and i the current.

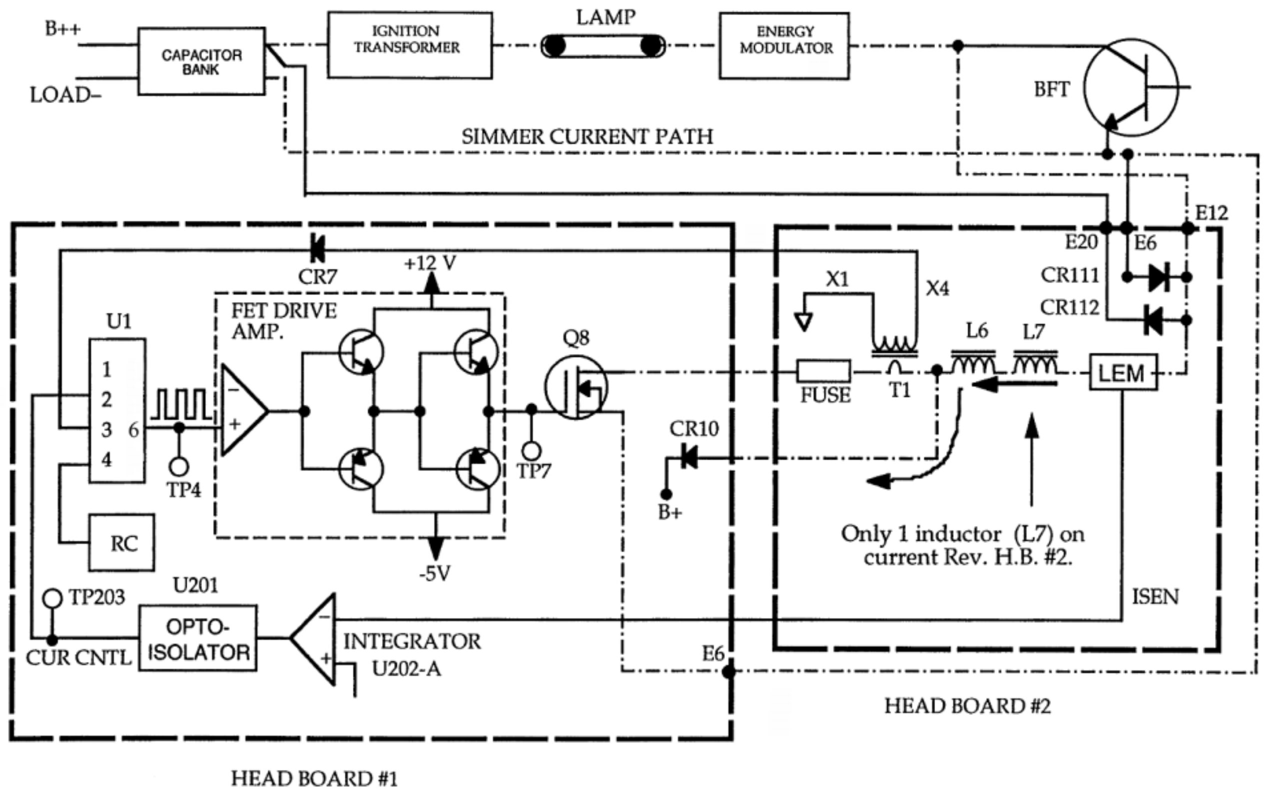


Figure 4.4-1. Block Diagram of Simmer Circuit

When the FET is opened (non-conducting) the flyback voltage in the inductors causes the voltage at point "A" of Figure 4.4-2 to increase with time. During this event current flows out of the inductors, through CR10, to B+.

The simmer current consequently goes either through the FET or through the diode. The function of the inductors is to provide a current source while the FET is non-conducting. The simmer current oscillates slightly around its average value of 1.5 Amp, as shown in Figure 4.2-3.

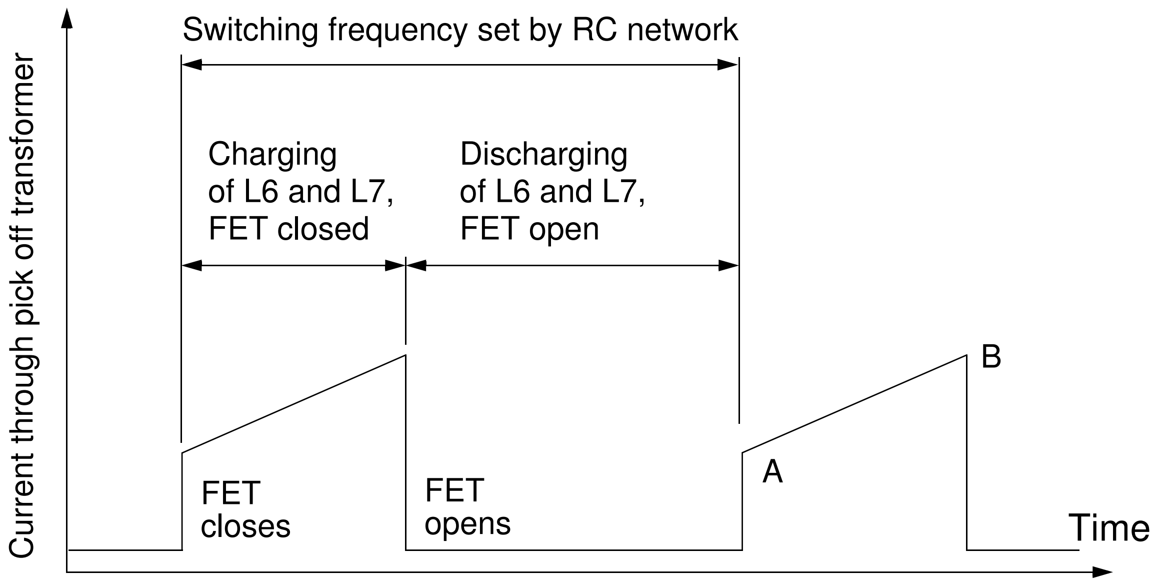


Figure 4.4-2. Current through the Pick off Transformer as a Function of Time

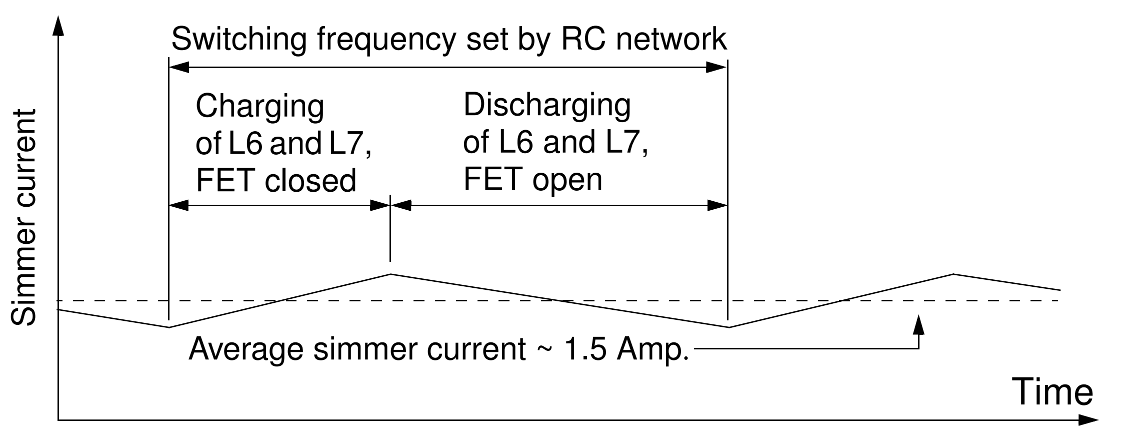


Figure 4.4-3. Current through Inductors, L 6 and L 7

Current Regulation

A signal proportional to the simmer current is generated by the LEM. This signal, ISEN, goes through an integrator, described in a later subsection, and an opto-isolator before it is connected to pin 3 on U1. U1 increases the time that the FET is conducting if the simmer current goes down and decreases it if the simmer current goes up. When the FET stays closed a longer time the current through the inductors can ramp up to a higher value, allowing the simmer current to increase. Since the LEM connected to the integrator only measures the average value of the simmer current, it does not provide U1 with any information about when to turn off the FET. This information is instead supplied by the pick off transformer measuring the current through the FET only.

Based on the feedback from the LEM, U1 determines how much extra current is needed. The signal from the pick off transformer then tells U1 when this goal has been reached.

LEM Feedback

The integrating circuit converts the output from the LEM, ISEN to the U1 input, CUR CNTL. If there is no simmer current, no current flows into the LEM. The voltage at TP201 is then 2.5 volts, the same as at TP202, since there is no voltage drop across R203 (see head board #1 schematics, simmer 2). The inverting input to U202-A is then higher than the noninverting input causing the op-amp output to go low, TP204. This reduces the diode current in the opto isolator, U201, reducing the current through the transistor which lowers the potential at TP203, CUR CNTL. This causes U1 to maximize the duty cycle of the FET in an effort to increase the simmer current.

When the simmer is on, current is drawn into the LEM and the potential at TP201 falls as the voltage drop across R203 increases. When the voltage drops below 1.25 V, the sign of the output from U202-A swings positive and the current through the diode in U201 increases. When this happens the current through the transistor increases and the potential of CUR CNTL increases, TP203. This decreases the duty cycle off the FET drive from U1. The capacitors across the op-amp integrate the input signal thereby increasing the time constant of the feedback.

Diode Clamp

Diode clamps are mounted on Head Board #2 where the simmer current enters. They ensure that the incoming voltage can’t go higher than B++ and not lower than Cap –. If the potential in the simmer circuit rises above B++ current will flow through diode CR 112 and if it falls below Cap- current will flow through diode CR 111. These diodes protect the simmer circuitry when the lamp is ignited.

FET Base Drive

The output from U1, pin 6, driving the base of the FET is amplified in several steps. First by an op-amp, U2, and thereafter by the transistors Q1, Q4, Q5 and Q7. The transistors are arranged in a push pull configuration so that when Q4 and Q1 are conducting (FET conducting) Q7 and Q5 are non-conducting (FET non-conducting).

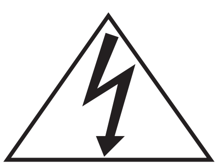
Soft Start

If the -5 V connected to the transistor emitters in the FET drive was to disappear, they would not be able to switch off the FET. This would lead to catastrophic failure since it would short-circuit B++ to Cap-. To prevent this from happening a soft start circuit is implemented. This consists in principle of the FET Q6 and the diode CR1. If the -5 V were to disappear, Q6 will pull pin 6 on U3 low. The output from U3 would then go high, TP1, turning off Q3 (no voltage difference between base and collector on Q3). With Q3 off the voltage at TP3 can't swing.

If the -5 V is present, the gate of Q6 becomes negative pinching off the current path through it. Pin 6 on U3 will then be close to 6 V (voltage division of 12 V between R1 and R2) bringing TP1 to ground. Q3 is turned on bringing its emitter to 12 V. TP3 can then swing freely against the 12 V through R18.

Troubleshooting

If it has been established that the ignition circuitry works and there still is no simmer current, go through the list below. Presence of a simmer current is manifested by intense DC emission from the flash lamp.



All circuits described in this section are referenced to B-. Therefore a differential isolated oscilloscope probe must be used when performing time resolved measurements.

1. Check that the lamp leads are properly connected and that the flash lamp is OK.
2. Check the fuse on Head Board #2, Figure 4.4-2.
3. Check the house keeper voltages, ±5 V and ± 12 V on the control board.
4. Check the amplified FET gate drive by connecting an isolation probe to TP7 (TP13 or the BFT emitter lead can be used for ground). The signal should swing between + 10 V and - 4 V.

5. If there is no signal on TP7, check for the non-amplified FET gate drive on TP3. If there is a signal on TP3, but not on TP7, one of the FET drive transistors is defective (Q1, Q4, Q5 or Q7).

6. If the FET drive measured on TP7 is OK, the FET itself (Q8) or the diode (CR 10) may be defective. Measure the resistance through the diode both ways using a DVM.

Lamp Flash

The sequence of events that ultimately leads to the lamp flashing starts with the LAMP_FLASH_OUT signal being set high by the trigger board. After having gone through a NAND gate and a Schmitt trigger (U702-C) this trigger is split along two paths, one going to head board 1 and the other to the booster boards.

Once on head board 1 the trigger is conditioned and amplified before it is sent as the base drive to the BFT. The trigger that goes to the booster boards inhibits charging of the capacitor bank while it is discharging into the flash lamp. The paths of these signals and their logical states at various points are illustrated in Figure 4.5-2.

The output from the NAND gate, located on the P/S control board, will always be high if the LOWV_FAULT line is low (active, fault present). Consequently the flash lamp will not be fired as long as this condition exists.

The function of the Schmitt trigger is to reduce sensitivity to noise and to enhance the switching speed. The principle of operation is shown in Figure 4.5-1. The input signal will not change the output state until it has risen above a high threshold value on its way up or before it has fallen below a low threshold value on its way down. The voltage difference between the two thresholds (dashed horizontal lines) is called hysteresis. Hysteresis means that the output state not only depends on the input voltage but also on the history of the input signal.

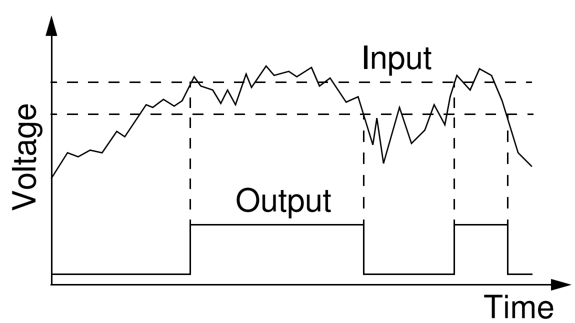


Figure 4.5-1. Principle of Schmitt Trigger

The repetition rate of the system is set by the LAMP_FLASH-OUT signal and the restrictions imposed by the A and B inhibit signals. (An external lamp trigger replaces the LAMP_FLASH_OUT trigger.)

Base Drive Signal

Before the trigger is sent through the umbilical it is divided into two separate signals by U288, one going up when the other goes down. The advantage of this is greater immunity against noise. Both signals are referenced against ground as shown in Figure 4.5-2. When the signals arrive at head board 1 they are summed (U101) and then sent to an opto-isolator. When the output from U101 goes low current is drawn through the diode, light is emitted and the transistor in U100 is turned on. A comparator, U102-A, adjusts the voltage level and speeds up the transition. Four transistors (Q105, Q106, Q151 and Q108) par wise arranged in a push-pull configuration amplify the signal. The main current path goes through R120 and R121. A fraction is picked of and sent through a red LED, Q102. Every time the BFT is closed the LED flashes.

The summed signal sent to the opto isolator is also used as a trigger for the DPSS circuit and the signal distribution board.

Charge Inhibiting Signal

This signal, after having gone through an inverter and an opto isolator, is divided again (see B- referenced section of P/S control board, Figure 4.5-2). One signal goes into a NAND gate where it is checked against A_ENABLE. The other goes through a delay line consisting of a NAND gate and an inverter before it also goes into a NAND gate where it is checked against B_ENABLE. Only if both signals are high will the charging be inhibited.

Since the circuity on the booster boards are referenced to B- the inhibit signals are sent through two isolating switches, OG211.

On the booster board the inhibit signal goes into a comparator, U5-D. If the inhibit signal from the P/S control board is low, the optical switch connecting this signal to the booster board will be open. As a result pin 10 of U5-D will be at 12 V. Since there is + 5 V on pin 11, the output of U5-D will go low, TP27, causing U4 to stop driving the FETs.

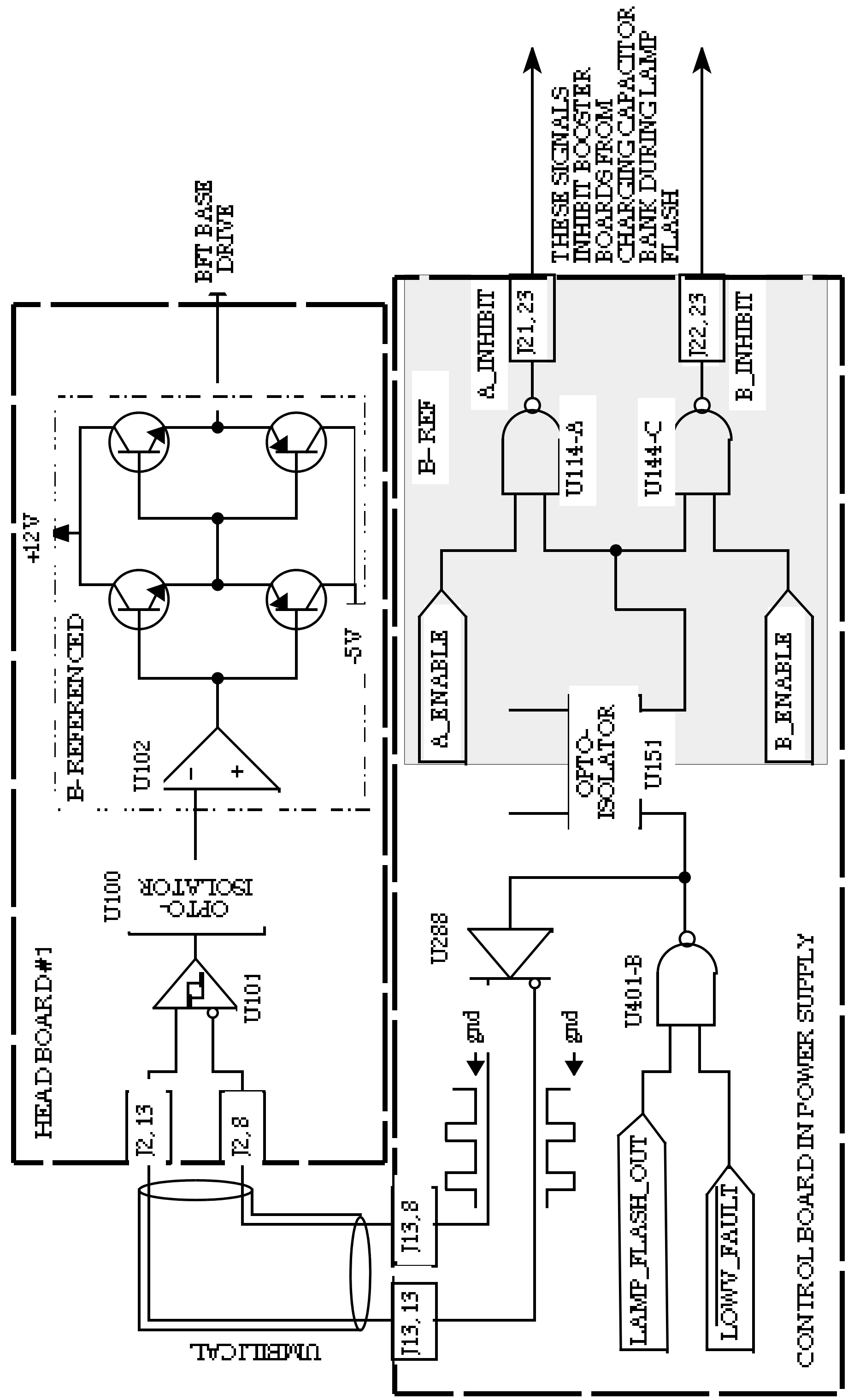


Figure 4.5-2. Block Diagram of Lamp Flash Circuit

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Booster Board

The IR output power from Infinity is continuously adjustable. This is made possible by a power supply design that allows a continuously adjustable amount of energy to be stored in the capacitor bank.

The energy stored in a capacitor with a capacitance C depends on the voltage, V, across it as:

$$Energy = \frac{1}{2}CV^2$$

Since the total capacitance is constant the voltage must be continuously variable. This function is performed by the booster boards.

When the second contactor is closed, 290 V is applied to the capacitor bank. Consequently this is the minimum voltage that can be applied to the capacitors. Through a voltage booster circuit, essentially consisting of an inductor and a FET switching the current through it, the voltage can be raised above the 290 V by a chosen amount. How this is done is explained in detail below. A simplified circuit diagram showing the principal components is shown in Figure 4.6-2.

Principle of Operation

The current enters the board at terminal TB2, Figure 4.6-2. When the FETs are conducting, the current goes through two large inductors, L1 and L2, and then through three parallel water-cooled FETs to LOAD–. The voltages across the inductors are V1 and V2, respectively.

The gates of the three FETs are switched at a frequency of approximately 25 kHz, set by a RC network connected to the IC controlling the FETs, U4. Every time the FETs stop conducting the voltage across the large inductors (L1 and L2) is reversed according to the relationship:

$$V = -L\frac{dI}{dT}$$

where L is the inductance and I the current through them. If the FET is switched off fast the derivative dI/dt will be large which means that the flyback voltage, V, will be large as illustrated in Figure 4.6-1. The flyback voltage pushes current through the two diodes (D1 and D14) and charges the capacitor bank.

The voltage across the capacitor bank is measured by electronics on the P/S control board. A comparator, U303-C, generates an output signal CAP VOLTAGE proportional to the voltage difference between B++ and LOAD–. This signal is available on TP307. A second comparator, U303-D, generates an output signal proportional to the difference between the actual capacitor bank voltage and the voltage set in the computer (TP308). The exact relationship between this signal, and the signal that goes back to the booster boards, (ICNTRL) is set in a calibration procedure where the pots R332 and R314 are adjusted. (See Chapter 6 section 6.1 for the calibration procedure.)

The feedback signal ISENSE on the booster boards is used by the IC (U4) to determine when to switch off the FETs.

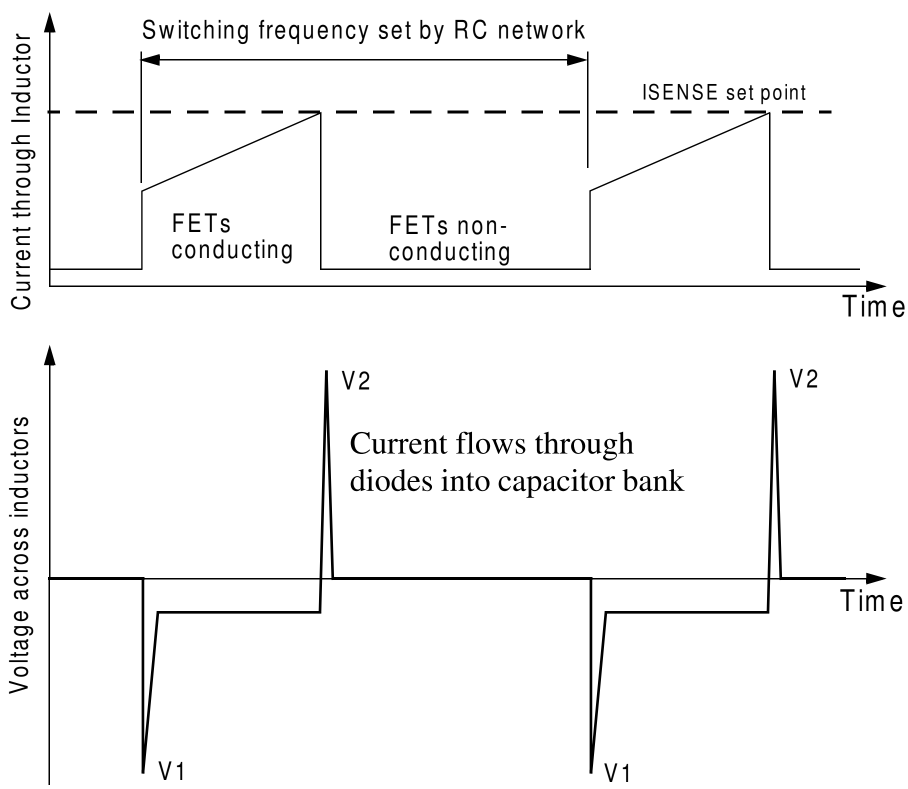

Figure 4.6-1. Principle of Operation of Booster Boards

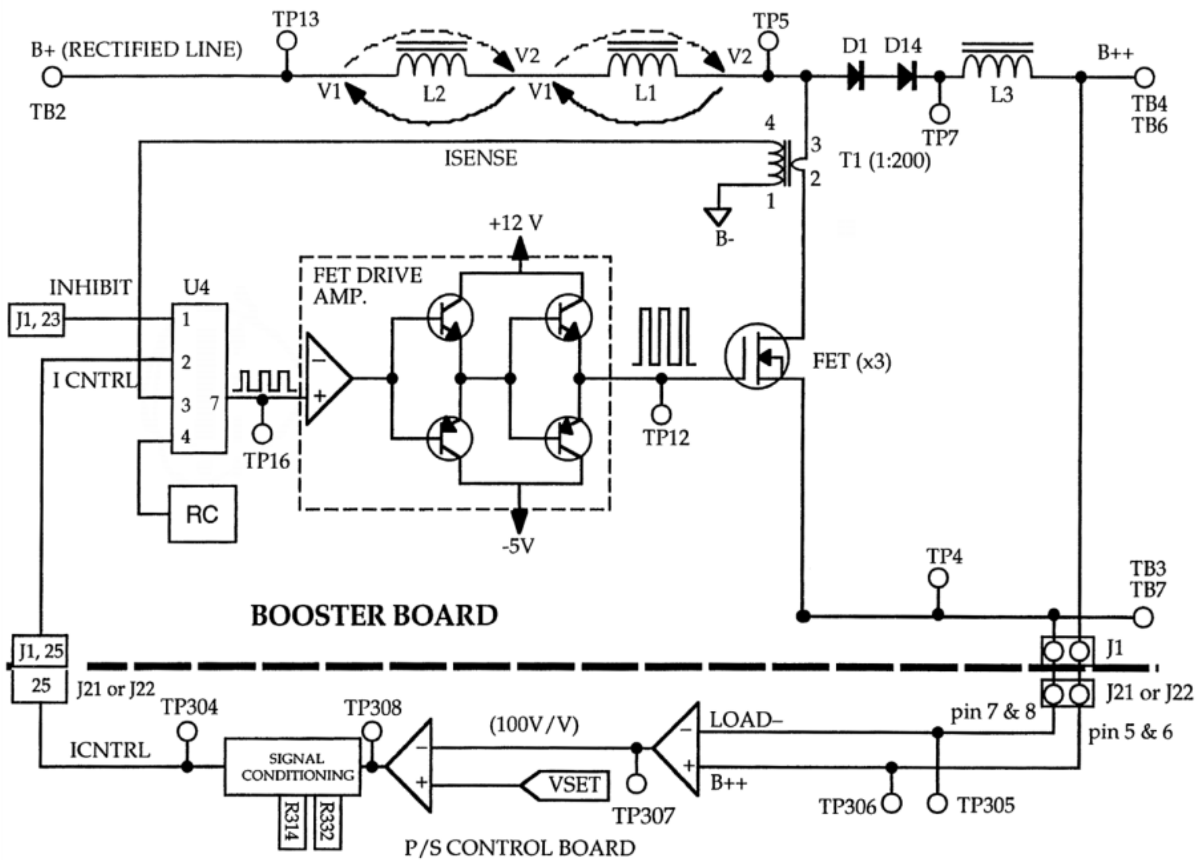

Figure 4.6-2. Block Diagram of Booster Board

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FlashTrack

FlashTrack is an control system that reduces shot-to-shot intensity fluctuations. A photo-diode monitors the emission intensity of the flash lamp during the first half of the lamp pulse. The on-board microprocessor uses this information to make a small correction to the lamp current during the second half so that the total amplification remains constant over time. The principle of operation is illustrated in Figure 4.7-1. A photo-diode connected to an integrator monitors the flash lamp emission. After half of the lamp width time has elapsed a sample and hold circuit reads out the integrator and converts it to a voltage. This voltage is summed with V_{Coarse} and $V_{Vernier}$ (both inverted) and then amplified. When auto-centering is chosen the microprocessor sets $V_{Verinier}$ and V_{Coarse} so that the amplified output is at 2.5 V which is the center of the A/D converters operating range. The amplified signal is converted to a pulse that has a length proportional to the voltage. This pulse is used to drive a transistor switch regulating the current going through the lamp. If the first half of a lamp pulse is brighter than average the amplifier output goes down (signal applied to inverting input of amplifier). This yields a shorter drive pulse for the transistor switch which decreases the amount of current going though the lamp during the second half of the lamp pulse.

Troubleshooting

1. Using a isolation probe, set for 200:1 attenuation, connect to the signal distribution board [TP15(+) and TP10(-)]. At 10 Hz, with 500 V applied to the lamp, a typical signal voltage for the Flash Track detector is 25 mV. At 10 Hz and 400 V this signal should decrease to approximately 15.5 mV (probe set to the same attenuation as above).

   On a system where the detector and circuit is working properly, the Flash Track signal will decrease if the lamp voltage is held constant and the repetition rate is increased.

2. The Flash Track signal can also be averaged from the A/D monitor window. For an average of 50 cycles, at 100 Hz and 400 mJ/pulse IR energy, a value of approximately 2.4 V with a standard deviation of 8% should be measured. At 50 Hz and 500 mJ/pulse a value of 2.3 V with a standard deviation of 11% is typical.

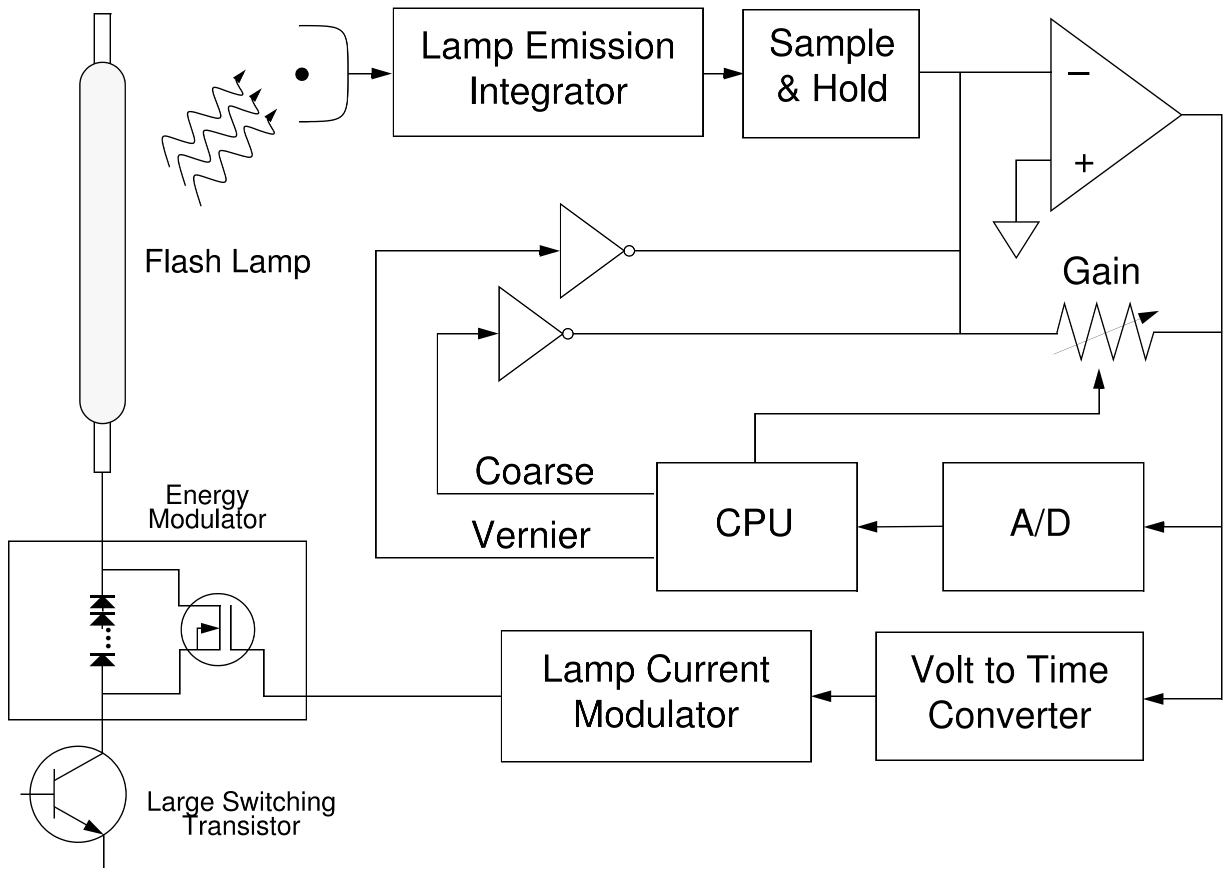

Figure 4.7-1. Block Diagram of FlashTrack

For additional information on the operation of the FlashTrack circuit see Chapter 5, Section 5.2.

IR and Depol. Detectors

An overview of the circuit path for the IR and Depolarized energy detector signals is given in Figure 4.8-1. The current output from the photodiodes, of these two detectors, goes onto the Signal Distribution board where it is integrated by three parallel capacitors. To clear the integrators the Q-switch trigger (from the Trigger board located in the power supply) is employed. This TTL signal is inverted and stretched before being applied to the gate of a FET (Q3 for the depolarized energy meter and Q2 for the IR energy meter). When the FET is forward biased, after the Q-switch is fired, it grounds the capacitors thus dissipating the accumulated signal. From the signal distribution board the signal from these two detectors is routed to Head board #1 and then to U101 (multiplexer) on the P/S Control board where the signals are monitored by the CPU.

Troubleshooting

1. Verify the low voltages on the photodiode boards: TP6 +12 V and TP5 -12 V.
2. Verify that the +5 V reference, TP11, and the +5 V, TP1, are OK. Both of these voltages are generated on the signal distribution board from the +12 V, TP10 can be used for ground.
3. Try to reposition the photodetectors to increase their response signal. The housing holding the board should be approximately centered on its mount. A typical voltage reading for the depolarized energy detector is 0.7 V at 2 mJ/pulse depolarized energy. A typical 10 Hz voltage reading for the IR detector is 4.5 V at 500 mJ/pulse (for the new 40-100 or the 25-50 this reading should be 4.5 V at 600 mJ/pulse).

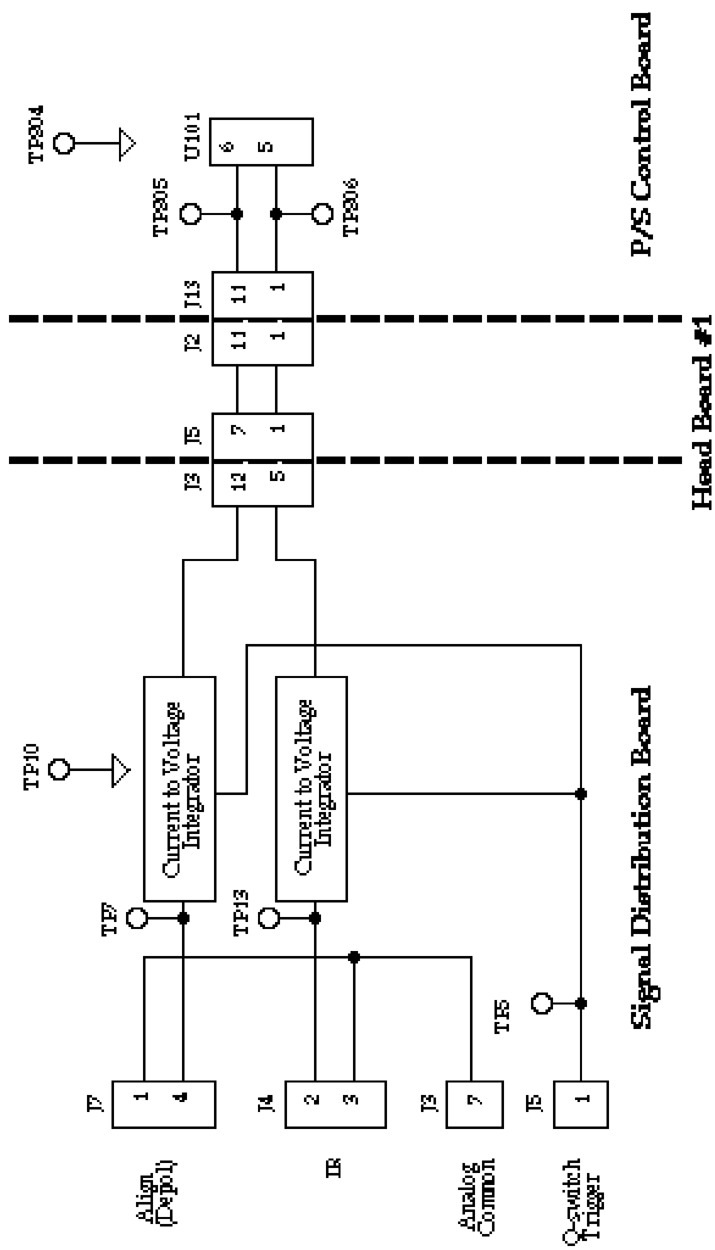

Figure 4.8-1. Signal Paths for the IR and Depolarized Energy Meters

Harmonic Heater Circuit

The heating and regulation of the harmonic mounts is software servo controlled. The “Infinity crystal Temperature” window consists of two bar graphs, one slide bar, a push button harmonic selector, and a Servo mode indicator. The slide bar is labeled “Set Point”; the two bar graphs are labeled “Drive Signal” and “Temperature”. The Drive signal is generated based on the difference between the Set point and the Temperature. Five possible messages can be displayed by the Servo mode indicator: Seek, Seek2, Not Installed, Roundout and Hold. The messages “Seek” and “Seek2” will be displayed while the software tries to stabilize a mount temperature. The message “Roundout” will be displayed when the set and actual temperatures are within a few hundred counts of each other. The message “Hold” will be displayed after the temperature is stabilized. The message “Not Installed” will be displayed after 15 minutes if the software is unable to regulate the temperature of a harmonic mount. To reactivate the software and heater circuit, both the laser system and the software program must be cycled off and on.

The physical heating, as well as the temperature regulation, of the harmonic mounts is accomplished by controlling the amount of current flowing through a heating resistor secured to the top of each mount. The temperature of the mount is determined by a thermistor embedded in the side of the mount. Figure 4.9-1 gives an overview of these signal paths. It should be noted that the circuits for both heaters are identical. Therefore, for clarity, the electronics discussion will focus on the SHG circuit.

To control the amount of current, flowing through the heating resistor, a switching voltage regulator is employed. The drive signal for this regulator is the SHG_SET signal. This signal consists of a series of paired pulses. Using the switching regulator signal as the scope trigger, the first pulse determines the frequency (25 µsec) of the square wave output and the second pulse determines the duty cycle. The duty cycle is based on the difference in magnitude between the SHG_TEMP+ signal (returned by the thermistor located in the harmonic mount) and the set point established in the software. The output of the switching regulator is either +12V or 0V (ground). When the regulator receives the first pulse its output goes to +12 V shutting off the current flow through the heating resistor. The second pulse causes the output pin of the switching resistor to be tied to ground allowing the maximum amount of current to flow through the heating resistor. The path of this current flow is from SHG/THG Heat+ to SHG HEAT-.

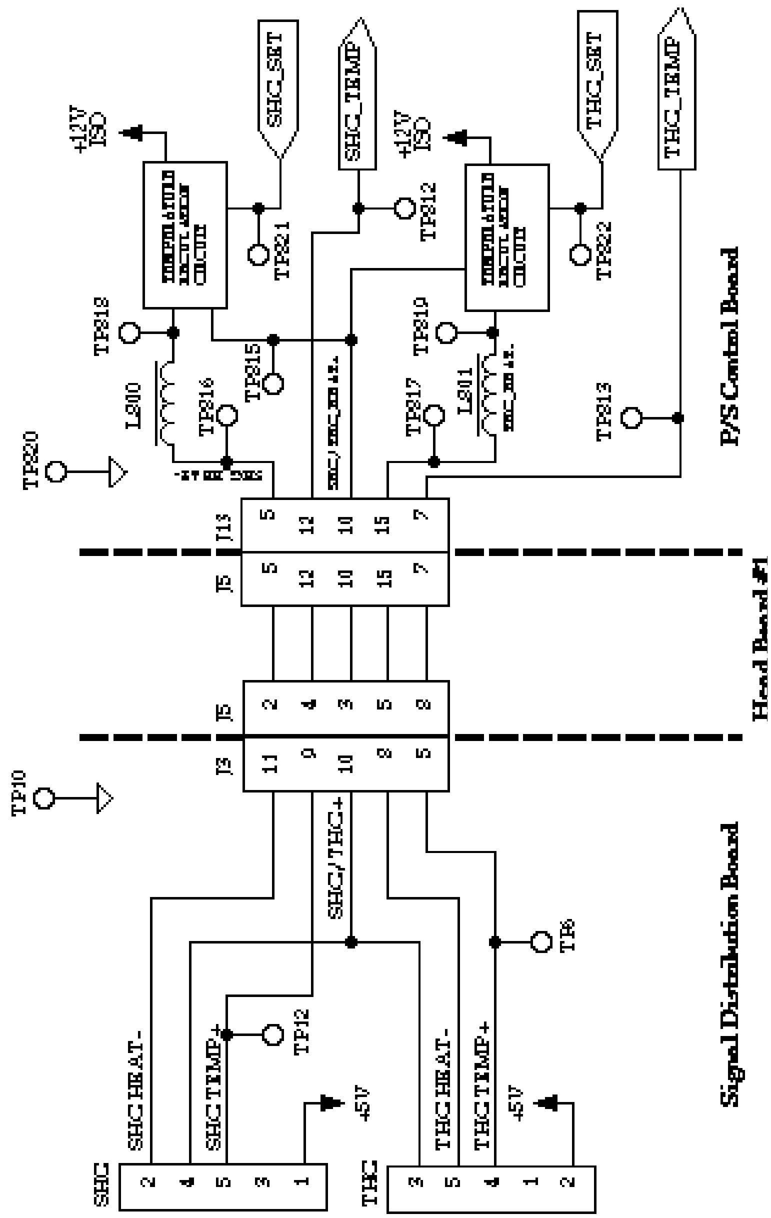

Figure 4.9-1. Signal Path for Harmonic Heater Circuit

Troubleshooting

1. Determine if the problem is on the mount or in the heater circuit by switching the plugs used for the THG and SHG mounts. If the problem does not follow the mount it is in the electronics for that plug.

2. Using an isolation probe look at the "set" signal for the harmonic, TP821 for SHG and TP822 for THG, on an oscilloscope. The pulses should be between 2 and 3 volts in intensity. Note that there will be a significant amount of ringing on this signal.

Hook an isolation probe to the output of the switching regulator, TP818 SHG and TP819 THG. Adjust the software "set point" to its maximum value. The duty cycle of the square wave should go to its minimum value after approximately 1.3 µsec. Decrease the set point to its minimum value, the duty cycle should go to its maximum value after approximately 24 µsec. Note that it will take a few seconds for the circuit to stabilize after a set point change is made.

If there is a break in the signal path between SHG/THG Heat+ and SHG/THG Heat-, the switching regulator signal will be unadjustable. The circuit will need to be traced to determine/find the break in the heating resistor current path.

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Interlock Circuit

An overview of the interlock circuit path is given in Figure 4.10-1. The interlock voltage is provided by transformer T500. The primary side of this transformer is connected to phase A and C of the line voltage. The output from the secondary side is rectified to provide 24 V, the interlock circuit voltage. This voltage, INTLK+, leaves the P/S control on J12 pin 14 and goes to the DPSS control board. Here the interlock voltage is used to drive an opto-isolator. The function of this IC is to pull the input of the DPSS interlock logic circuit to ground (see above DPSS discussion). If the interlock chain is broken, this connection to ground is broken causing the DPSS to shut down with an interlock fault.

From the DPSS the interlock voltage goes onto head board #1 and then directly onto head board #2. Here the interlock voltage is used to drive two opto-isolators, U150 and U151. When these two ICs are “on” (i.e., the interlock chain is complete) they connect the gate of FET Q150 and Q151 to CAP-. Since the source of each of these FETs is also at CAP-, the FETs will not conduct (the drain of each FET is connected to B++). If the interlock chain is broken the FETs will be forward biased allowing the capacitor bank to discharge.

From head board #2 the interlock voltage goes back to head board #1. This voltage is now labeled INTLK in the schematics. On head board #1 the interlock voltage goes through four switches. Three of these switches are for the POT assembly, one for the cover and two for the clips which hold the assembly to the head. The fourth interlock switch is for the head cover.

From head board #1 the interlock voltage is routed back to the P/S control board where it goes through the key switch and is then used as the supply voltage for the +12 V regulator U601. The +12 V is used to drive a relay, U604, which pulls the interlock fault line to ground. If the interlock chain is broken the relay will open causing the interlock fault line to go high which is the fault condition. From the relay the interlock voltage goes back to the transformer, T500, as INTLK-.

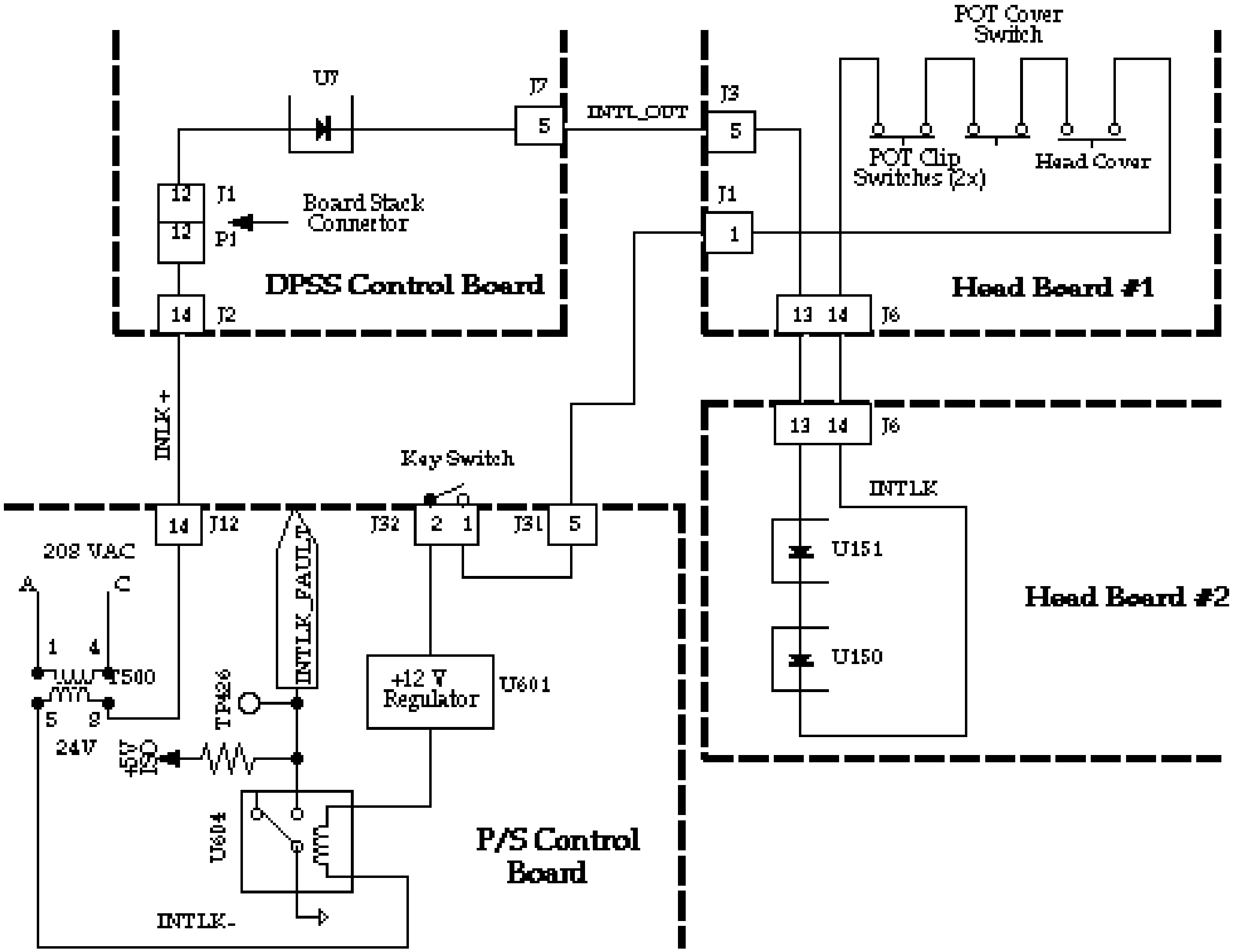

Figure 4.10-1. Interlock Block Diagram

Software Theory

The Infinity software comprises two separate programs:

1. The Control Program, which is stored permanently on an EPROM inside the power supply and runs on the power supply CPU, an NEC V40;
2. The User Interface Program, INFINITY.EXE, runs on a standard IBM-style personal computer (PC) equipped with the Windows operating system.

The Infinity software was originally released for the OS/2 operating system. There may still be several of these systems in the field, and all of the information presented here applies equally to those systems unless otherwise indicated.

The two programs communicate by RS-232 over a serial cable. One of the PC's serial ports must be dedicated to this communications link; otherwise there are no special requirements for the PC. It does not matter which program is started first.

The User Interface program manages several data files that save the state of the program and hold data regarding the laser system. Normally the user does not have to concern himself with these files as their creation and modification is automatic.

1. LAMP.IN contains data describing the system lamp change history.
2. CONFIG.IN contains information about the configuration of the system serial ports.
3. CALCONST.IN contains calibration constants.
4. SHOTS.IN contains the last system shot count. A backup copy of this information is kept on board the laser power supply in case this file is lost.
5. STATE.IN contains the last settings of all the adjustable laser parameters, such as repetition rate, power and timer values.

To insure the most up to calibration and system information, the Infinity software data files (i.e., files with an "in" extension) should always be backed up onto a floppy disk after a system is serviced. This data disk should be stored in the system maintenance kit.

The OS/2 User Interface Program also requires five "dynamic link library" files (identified by the.DLL extension). Four of these five files are third-party products; the fifth, INFINITY.DLL, contains icons and other resources needed by the program.

.DLL files required by the OS/2 Infinity program:

1. VPMOFL20.DLL
2. VPMVM20.DLL
3. VPMRES.DLL
4. VPMBAS20.DLL
5. INFINITY.DLL

The total disk space occupied by all of the Infinity files is about 2.5 MB for the OS/2 program, 4.2 MB for the Windows program.

The Control Program runs on the NEC V40 microprocessor (an Intel 8086 compatible) found on the Infinity CPU board. The program code is physically located in an EPROM on the CPU board. The program is running whenever the power supply keyswitch is on. A small amount of critical data is stored in an EEPROM on the system control board, including:

1. The system serial number
2. The laser shot count (updated every one million shots)
3. The factory-set lamp and Q-switch timing.

Division of software functionality

The overall software functionality is distributed between the Control Program and the User Interface.

- The Control Program is responsible for the management of all low-level hardware signals, fault detection, pulse energy monitoring and servo systems. It is important to realize that all direct access to the hardware is done by the Control Program. If the communications fail between the power supply and the PC, the laser remains in a safe operating state since the Control Program knows how to "take care of itself."
- The User Interface program accepts input from the user in order to change the laser's state. It also displays diagnostic and status information about the laser. Its primary purpose is to interact with the user; it has no ability to exercise direct control over the laser hardware.

The two programs communicate over an RS-232 link. To optimize performance, commands are sent and received in a proprietary format that is highly compressed; many commands are as short as a single byte. Some of this communication can be viewed by opening the "Message window" (launched from the Service Window menu bar), which provides a real-time log of selected messages. In order to make the log entries understandable to humans, the log contains expanded versions of what is actually transmitted. Furthermore, some of the repetitive messages that are exchanged continuously (such as the messages that update the pulse energy bar graph) are omitted.

As more of the User Interface windows are opened, there is an increase in the number of messages that must be exchanged in order to keep the screen up to date. On some computers this will have a noticeable effect on the rate at which the various bar graphs are updated. It is a good practice to close windows when they are no longer needed.

Multitasking

Both parts of the Infinity software (the Control Program and the User Interface) are multitasking. This means that each program is subdivided into a set of small procedures or “tasks” that operate independently of one another. Since there is only one CPU for each program, the CPU divides its time between the tasks, allocating slices of time alternately to each task. To the user it often appears that all of the tasks are running simultaneously.

It is not very important to know that the Control Program is multi-tasking, but for the User Interface Program it is critical. There are actually two levels of multi-tasking at work in the UI program: one within the program itself, and a higher one provided by the operating system. Only the second of these is of interest to the user.

Both OS/2 and Windows can multi-task entire applications, making it possible for the Infinity program to share the computer with other programs. This sharing is entirely under the user's control; the Infinity program does not “know” how many other programs are running or what they are doing. This is a useful and powerful feature, but it can be abused. It is possible to load so many programs at one time that the CPU practically grinds to a halt, spending most of its time swapping chunks of data back and forth between memory and disk. Use common sense when multi-tasking. There is only one processor, and its computing power is limited.

Servo Control

The Infinity pulse energy is regulated by means of three interacting control systems: (1) FlashTrack, (2) the FlashTrack Centering Servo and (3) the Pulse Energy Servo. Of these systems, FlashTrack is implemented entirely in hardware; the other two are entirely in software.

FlashTrack

FlashTrack will be first discussed in simplified, but slightly incorrect terms. The details will then be filled in, in order to illustrate how FlashTrack and the software interact.

A photocell in the laser head receives a fraction of the flash lamp light during each shot. During the first half of each lamp flash (about 100 microseconds), a circuit integrates the signal from this photocell. The result of this integration is a voltage proportional to the total lamp energy (V_L) produced during the first half of the flash. Because of fluctuations inside the lamp, this signal will vary from one shot to the next; such variations are a major source of laser output noise.

During the second half of the flash, the FlashTrack circuit applies an extra burst of current to the lamp. FlashTrack is designed to vary the duration of this extra burst in order to correct for fluctuations in V_L. To do this, a circuit compares V_L with a reference voltage V_R, and sets the duration of the extra current burst (T_B) so that it is proportional to V_R-V_L (the “error signal”). Thus a larger V_L results in a shorter burst; a smaller V_L results in a longer burst, see Figure 5.2-1. The net effect is to reduce shot-to-shot fluctuations in the total lamp energy and as a consequence also in the laser output.

There are some simplifications in this description. To be correct, the relationship between error signal and current burst must include both a constant term and two gain factors. An accurate mathematical expression for the current burst duration is:

$$T_B = G_0(2.5 + G(V_R - V_L(+)))$$

where T_B is the duration of the current burst, G_0 is a fixed gain factor, G is a variable gain factor, 2.5 is a constant voltage offset, V_R is the reference voltage and V_L is the integrated photocell voltage. V_L varies from shot to shot. G and V_R are under software control and must be adjusted manually for optimum laser performance.

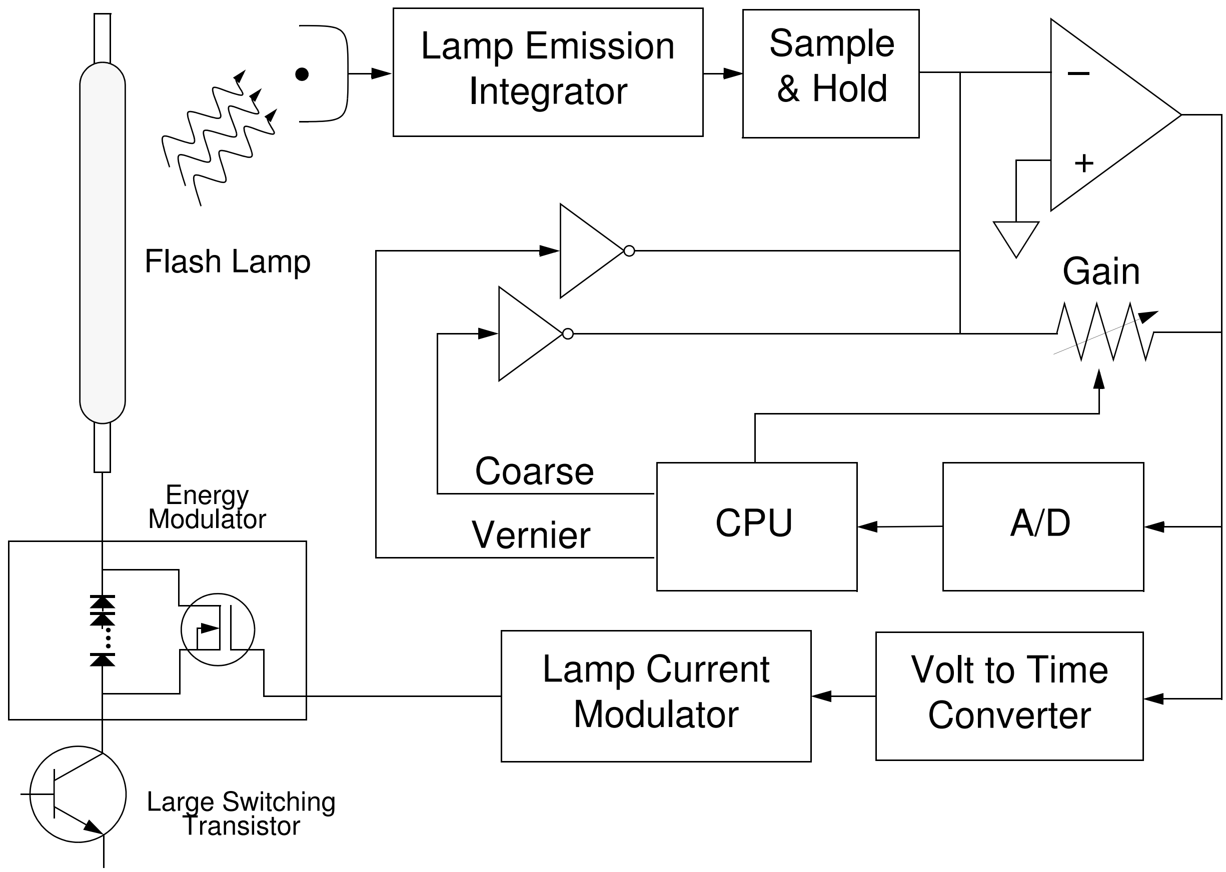

Figure 5.2-1. Principle of Operation of FlashTrack

The adjustment of the gain is straightforward: a control is provided in the user interface that allows the operator to experiment with different gain settings. By trial and error the user selects the one that gives the lowest laser output noise. In practice we have found that the optimum gain is independent of power level and repetition rate, therefore we can set the gain at the factory and the customer should not need to readjust it.

The reference voltage, V_R, is more complicated. Its optimum value is a function of lamp voltage, repetition rate and lamp age. Fortunately its control can be automated; this is the job of the “FlashTrack Centering Servo.” The rule for this servo is easy to state: V_R should be kept equal to the average value of V_L. In order to do this, the quantity:

$$2.5 + G(V_R - V_L)$$

is sent to an A/D converter so it can be read by the CPU on each lamp flash. The last value can be read through the Service window by clicking on “Photocell FlashTrack” in the A/D list. When the servo is properly centered, these readings will fluctuate around an average of 2.5 V. The average of the last 16 values can be seen by looking at the bar graph named “Error(V)” in the Servo window. In this display, 2.5 V have been subtracted so that the readings will fluctuate around zero (the center of the bar).

During normal operation the Error(V) reading will typically be within 0.1 V of center. Excursions of more than about 0.2 V are abnormal and indicate either excessive noise on the photocell signal or too much gain.

The centering servo is engaged when the box in the Servo window is checked. In practical operation there is no reason to disengage the servo, but the software provides this in order to verify that FlashTrack is indeed reducing the system noise. Use the noise readings in the Monitor window to check this.

Note that un-checking the box does not disable FlashTrack, it simply disables the centering servo. To turn off the action of FlashTrack, disengage the servo and run the Coarse slider all the way to one side. The Error(V) readings should go to either +2.5 V or -2.5 V, indicating that the FlashTrack A/D converter is saturated. It doesn't matter whether +2.5 V or -2.5 V is chosen; the point is to run FlashTrack up against one of its rails. In this state you will see the inherent noise of the laser when it operates without the noise-reducing effects of FlashTrack.

If you change the laser pulse energy while the centering servo engaged, you will notice that the Error(V) reading immediately departs from zero. The centering servo will then bring it back to zero by changing the coarse and vernier. You can watch this happening in the servo window. The laser will not deliver its best noise performance unless Error(V) is sitting near zero, so when the pulse energy is first changed the laser will operate for a while with increased noise. How long it takes to settle depends on the repetition rate, since the servo control logic can take steps only as fast as it receives data from the laser.

Pulse Energy Control

The second servo on the system is the pulse energy control. The basic idea here is to control the lamp voltage in order to make the laser deliver the pulse energy desired by the user. This feature makes Infinity easy to operate, since the customer sets the laser power in units that he cares about. The servo also holds the power steady as the lamp ages, allowing us to extract the maximum possible lamp life. Finally, using the servo protects the laser and the customer's optics by preventing pulse energies in excess of specification. Therefore we have made it the default mode of operation and we expect customers to use it all the time. The more direct method of control - setting the lamp voltage - is available only through the Service window. It is necessary for certain kinds of manufacturing and service procedures, but beyond that it should not be used.

The pulse energy servo itself is rather complicated since it interacts with the FlashTrack centering servo: each time the voltage changes significantly, the centering servo must be re-optimized. Furthermore, traditional servos have a tendency to overshoot their set point. Since this would be disastrous in a laser of this kind, the pulse energy servo is designed to have only a very small overshoot. This implies approaching the set point rather slowly and carefully. In order to combine this “conservative” design with a fast response to the user, the pulse energy servo is programmed to remember its past voltage settings. This memory is lost when you turn the keyswitch off, so each time you turn the laser power off the servo must “re-learn” the power versus voltage curve. You will notice when you operate the system that it takes a few seconds (even at 100 Hz) the first time you ask it to establish a new power value. The next time you return to that setting it will snap there much more quickly.

The pulse energy servo has two operational modes which are called “seek” and “hold.” Seek mode takes effect when the user changes the power by a large amount (>25 mJ). Here the servo bases each voltage change on a single laser shots in an effort to reach the new set point as quickly as possible. Once the power is close to the set point (within 25 mJ), the mode changes to “hold.” In this mode the servo works by averaging groups of 80 laser shots before deciding how much voltage change is necessary. The difference in gain between seek and hold is therefore this factor of 80. You can watch this process in action by observing what happens in the lower section of the Servo window as you change the pulse energy.

Calibration

There are three calibration procedures in the software: two for the photodectectors (energy and depolarization) and one for the lamp voltage. The latter is straightforward because it is completely automatic. The purpose of this procedure is to correct for the gain of the opto-isolator on the control board, that delivers the cap bank command voltage across the high-voltage isolation boundary. Therefore this procedure must be done whenever the control board and/or the booster boards are changed.

The photocell procedures correct for the sensitivity of the photocells themselves, and the filter stack in front of each photocell. These procedures are crucial to correct laser operation since they are the basis for the servo systems.

All three procedures use simple linear least squares to extract calibration constants from the raw data. The constants are computed and stored to disk after each procedure. It is a really, really good idea to convince the customer to keep a backup file of the latest constants. A new backup should be made every time a re-calibration is performed. In the User Interface program (V1.018 or later) the constants can be viewed (but not modified) from a text window accessible from the Service window menu bar.

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Special Error Windows

Besides the Fault window, the program contains a number of other windows that pop up when needed to let the user know about some special condition. These are listed below in Table 5.3-1. For corrective actions see Chapter 3, Troubleshooting.

Table 5.3-1. Infinity Fault Messages

FAULT MESSAGE	EXPLANATION
Syntax error: [text]	There is a syntax error in a command text read from the clipboard.
No file [path] STATE.IN was found. All system parameters will be initialized to their default values. Check timer settings before operating laser.	The indicated file was not on the disk. It should always be in the same directory as the program file INFINITY.EXE.
Configuration file [path] CONFIG.IN is missing -- the default configuration will be used.	
The system sent an indecipherable message (opcode [a number]). There is a possible problem with noisy communications. Processing will continue.	
COM Port [a number] could not be opened, either because it is not present on the system or is in use by another process.	Make sure that the power supply is connected to COM port 1.
The non-volatile memory (EEPROM) in the laser is blank. To initialize it use Init Eeprom in the Special menu of the Service Window	This should not arise in the field unless the EEPROM or the hardware that controls it has failed.
Calibration problem: at photocell saturation the IR energy is [a number] mJ. For proper operation it should be between 550.0 and 720.0 mJ. Adjust the light level and recalibrate.	The photo diode receives to much or to little light. Optimize the signal by translation of the photo diode housing and/or adjust the slit in front of the photo diode. In order to adjust the slit the housing needs to be opened and the board removed.
Calibration problem: at photocell saturation the depolarized energy is [a number] mJ. For proper operation it should be between 11.0 and 40.0 mJ. Adjust the light level and recalibrate.	The photo diode receives to much or to little light. Optimize the signal by translation of the photo diode housing.
Unable to locate file [path] CALCONST.IN. This file contains calibration data. The program will run with default calibration values. Do not operate the pulse energy servo before calibrating!	The software file containing calibration data for the photo diodes is missing or corrupted.
No lamp change data is available. File named [path] LAMP.IN is missing	

Table 5.3-1. Infinity Fault Messages (Continued)

FAULT MESSAGE	EXPLANATION
The system sent an unknown asynchronous message number [a number]. There is a possible problem with noisy communications.	
The serial number text contains a non-digit. Please re-enter it.	
No text in clipboard	This appears when the user selects the clipboard menu item from the Infinity Application window and the clipboard is either (1) empty or (2) loaded with a graphic.

Control Board Calibration

If the control board or one of the booster boards have been replaced, the following calibration procedure must be carried out. All voltages specified in the text below are volt meter readings. Do not use the values available in the computer interface.

1. Connect a DVM to B++ (TP 17) and Load– (e.g. inner lead of C44) on one of the booster boards. The DVM now monitors the voltage across the capacitor bank.
2. Turn on the system and go to standby mode (at this point the lamp might not ignite, i.e. no simmer current goes through it). Set the lamp voltage in the Service window so that the DVM reads 400 V. If necessary turn R314 c.c.w until the voltage becomes adjustable.
3. Adjust R314, Figure 6.1-1, clock wise to find the roll over point. Once roll over point is found turn R314 a half a turn counter clock wise. The minimum voltage has now been set.
4. Set the lamp voltage in the Service window so that the DVM reads 530 V. If necessary turn R314 c.c.w and increase Lamp Max in the Control window.
5. Adjust R314 clock wise to find the roll over point. Once roll over point is found turn R314 a half a turn counter clock wise. The maximum voltage has now been set.
6. Open the “Service...” window and select “Calibration” from the menu. From the drop down menu select “Cap bank voltage...” to run the capacitor bank calibration routine.
7. Connect the differential oscilloscope probe so that it measures the voltage across the capacitor bank (use the same points on the booster board as for the DVM).

   Set the probe to 1:200 attenuation. Set the oscilloscope to AC coupled and at 5 mV/div sensitivity. Insert a 3 kHz lowpass filter between the oscilloscope and the differential probe. A circuit diagram of a 3 kHz filter is shown in Figure 6.1-2. Connect the input side to the booster board.

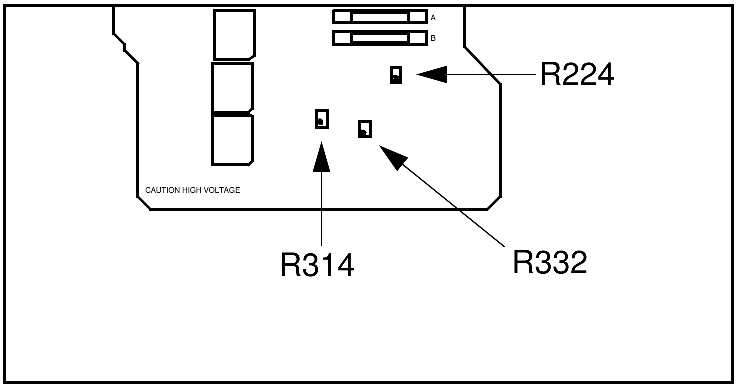

Figure 6.1-1. Location of Trim Pots on Control Board

8. Set the lamp voltage to 480 V in the Service window (DVM meter reading)
9. For 40-100 60 Hz AC systems set the repetition rate to 50 Hz.

   For 40-100 50 Hz AC systems set the repetition rate to 60 Hz.

   For 25-50 50 or 60 Hz AC systems set the repetition rate to 25 Hz.

   For 15-30 50 or 60 Hz AC systems set the repetition rate to 7.5 Hz.
10. Enable the lamp trigger in the Service window. The lamp should now be flashing.
11. Using the oscilloscope adjust potentiometer R224 for minimum Vcap voltage fluctuations. It should be possible to get 3 mV or less, $\Delta V < 3$ mV in Figure 6.1-3. Otherwise readjust R314 but not by more than 1/4 turn.
12. Adjust R332 for minimum voltage over shoot voltage at the maximum repetition rate of the system, see Figure 6.1-4.
13. Go back and adjust R224, step 12, again.

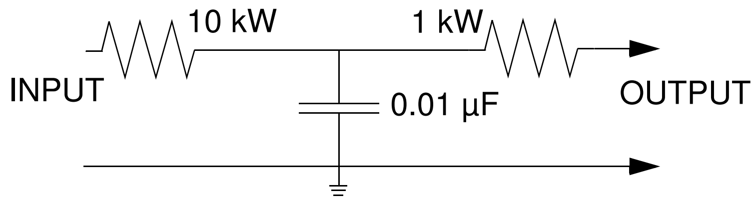

Figure 6.1-2. Schematic of 3 kHz filter

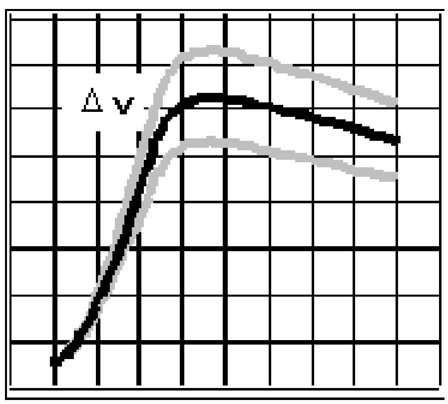

Figure 6.1-3. Adjust Potentiometer R224 for Minimum Voltage Variation, ΔV

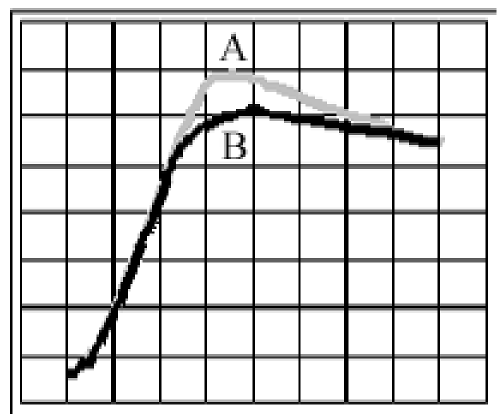

Figure 6.1-3. Adjust Potentiometer R332 for Minimum Overshoot. Trace B is Better than Trace A.

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Cooling System Maintenance

The following procedure should be followed when performing cleaning maintenance on the cooling system of the Infinity laser system.

Indications that cooling system cleaning may be required are:

1. Increase in threshold lamp voltage.
2. Increase in operation lamp voltage.
3. Increase in shot-to-shot noise.

Note that most of these symptoms are the same as those for a failing flash lamp! Therefore, if there is no clear indication of a contamination problem such as an extra dirty water filter, always request that the user replace the lamp before going in to perform this procedure.

Tools

To prevent structural stress in the Nd:YAG rods, it is highly recommended that a small torque wrench or torque driver capable of holding “Allen bits” be employed when working on the POT assembly. The torque specification for the screws (3/32) securing the 2 inner endplates are the same as that for the screws (7/64) securing the outer endplate, 8 lb/in.

Preliminary Data

Before beginning this maintenance procedure the following data should be recorded:

1. Threshold lamp voltage.
2. Lamp voltage at 100 Hz, 400 mJ IR output.

   Lamp voltage at 50 Hz, 500 mJ IR output.

   Lamp voltage at 10 Hz, 500 mJ IR output.
3. Noise at 100 Hz, 400 mJ IR output.

   Noise at 50 Hz, 500 mJ IR output.

   Noise at 10 Hz, 500 mJ IR output.
4. Depolarized energy at 100 Hz, 400 mJ IR output.

   Depolarized energy at 50 Hz, 500 mJ IR output.

   Depolarized energy at 10 Hz, 500 mJ IR output.

The end surfaces of the Nd:YAG rods are cut and polished at an angle. It is therefore very important to insure that the rods are not rotated when they are being removed from the POT housing and when they are being cleaned.

Cleaning Procedure

1. Drain the cooling water from the heat exchanger side of the power supply.
2. Remove the POT assembly from the laser head taking care not to drip water on to the optical components.
3. Following the outline below, referring to Figure 7-10 of the Service Manual and Chapter 5 of the Operator's Manual to dismantle the POT assembly, to clean the POT assembly and booster board cooling channels.

When handling the POT assembly special care should be taken not to bump or touch the ends of the Nd:YAG rods. Even the smallest blemish on a rod end will have devastating effects on the performance of the laser system.

a. Remove the lamp from the POT assembly (See Chapter 5 of the Operator's Manual). Do not reinstall the two outer most endplates.

b. Remove the screws holding the second endplate on the cathode end of the POT. Remove the screws holding the third endplate from the POT housing. Reattach the second endplate.

Make sure that the second endplate is tightened securely. The pressure of the second endplate, against the third endplate, will be all that holds the rods in position during the cleaning process.

c. Remove the second and third endplates from the anode side of the POT assembly.

d. Slide the rods out of the POT body from the cathode side.

e. Clean the rods using steam distilled water and cotton swabs by lightly rubbing a wet swab on the rods. The contamination on the rods will appear as a light gray/black film on the cotton swabs.

Continue cleaning the rods until no more contamination can be seen on the cotton swabs.

The screws securing the inner and outer endplates should always be tightened in a "star" pattern. This will reduce the likelihood of "Stress Induced Birefringence".

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f. Reassemble the POT (i.e., slide the rods back into the assembly housing). Attach the third endplate to the anode side of the POT assembly body.

g. Attach the second endplate to the anode side of the POT assembly body.

h. Remove the second endplate from the cathode side of the POT assembly and reattach the third endplate to the POT assembly body.

i. Reattach the second endplate to the cathode side of the POT assembly. Inspect the ends of the rods for water marks and clean if necessary with methanol.

j. Reinstall the lamp as outlined in Chapter 5 of the Operator's Manual.

k. Using paper towels, absorb as much water as possible out of the cooling water channels in the POT assembly base mount located in the laser head.

l. Using cotton swabs, clean the cooling water channels in the POT assembly base mount.

m. Install the POT assembly into the laser head and secure.

Clean each of the Booster boards following the procedure outlined below.

a. Remove the Booster board from the power supply.

b. Using paper towels, remove as much water as possible from the cooling channel in the Booster board mounting plate, located in the power supply.

c. Using cotton swabs and steam distilled water, clean the bottom of the Booster board and the inside of the cooling channel.

d. Reinstall the Booster board.

The cooling system maintenance procedure is now complete. Replace the filter cartridge (if the customer has not already done so) and the deionization cartridge. Fill the heat exchanger with water and turn the water pump on for a few seconds. Inspect the POT assembly and the heat exchanger for leaks.

Turn the system on, top off the water reservoir, and allow the system to run for at least 30 minutes. Verify that the operating temperature is the same as that on the packing/performance list shipped with the system. If this document is not available, set the temperature of the system to 26 - 30°C.

Verify all system specifications and threshold voltage.

Trouble Shooting

If the lamp voltages are still high, and the system is noisy, it could be an indication that there is a problem with the lamp or the alignment of the system.

1. If the Lamp has more than 50 million shots on it, try replacing the lamp.
2. If replacing the lamp does not help try to adjust the relay imaging system for the POT.
3. If making small adjustments to the imaging system does not help the system may need a full alignment.

   A good indication that there is a fundamental alignment problem is black residue inside the spacial filter on the high energy side.
4. Inspect the system for damaged or dirty optics.
5. Verify that the output from the DPSS is stable and in specification (See the DPSS Trouble Shooting section in Chapter 3).

If the depolarized energy has increased, even at low repetition rates, it could be an indication that there is "Stress Induced Birefringence".

1. Remove the POT assembly from the laser head. Loosen and then retighten the screws securing the outer endplates. Always tighten the screws using a "star" pattern.
2. Inspect the ends of the rods for dirt or water marks.
3. Inspect the system for damaged or dirty optics.

If the depolarized energy can not be reduced, and the alignment of the system and the performance of the DPSS has been verified, the POT assembly should be returned to Coherent Santa Clara for evaluation.

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Brightness

The brightness of a laser beam is defined as the number of photons it delivers per area and solid angle [W/cm^2 sr]. It is an important characteristic of a system because it cannot be increased using passive optical components, e.g. lenses or mirrors. If the beam diameter is decreased so that the same number of photons are transported through a smaller area the intensity [W/cm^2] will increase but so will the beams divergence.

The brightness of a laser system is determined by how close to its diffraction limit it can be operated. For a given output energy a laser achieves its maximum brightness when it runs in TEM_{00} mode. This corresponds to the situation where the beam can be focused to its smallest spot size and it follows the rules of geometrical optics. Another advantage of high brightness is that it increases the efficiency of non-linear processes such as frequency doubling and mixing.

Diffraction Limit

One of the most important parameters for a laser beam is its divergence. It is desirable that it is as small as possible since this translates to higher brightness and a smaller beam waist when focused. Higher brightness means higher conversion efficiency in non-linear optical processes. The divergence is also closely related to the quality of the spatial mode since a laser will exhibit its smallest divergence when it runs in TEM^{00} mode. The brightness can easily be increased using optics but not decreased using only passive components. The ultimate limit for a beams divergence is set by diffraction, the wave nature of light. Sometimes the divergence is therefore expressed in terms of closeness to diffraction limit. A convenient way of quantifying the quality of a laser beam is to use a parameter called M^2. This compares the divergence of the real beam to that of a true Gaussian having the same diameter at the focal point. A M^2 value of 1 thus means that the beam behaves like a Gaussian beam.

Faraday Isolator

Already in the nineteenth century it was found that the polarization state of light could be rotated by passing it through a medium immersed in a magnetic field. Empirically it was found that the degree of rotation could be written as

$$b = g\, B\, l$$

where g is a constant specific for the material used, B is the strength of the magnetic field and l the traversed distance. A special feature of this effect, named after its discoverer Michael Faraday, is that the direction in which the polarization is rotated is independent of the direction in which the light enters the system. This can be used to build an optical diode, which will transmit light in only one direction. Such a device is illustrated in Figure 7.1-1. Light entering from the left is first polarized and then this polarization is rotated 45 degrees clockwise before it is transmitted through a second polarizer which is aligned to have maximum transmission for this state. For light entering from the other direction, however, the faith is very different.

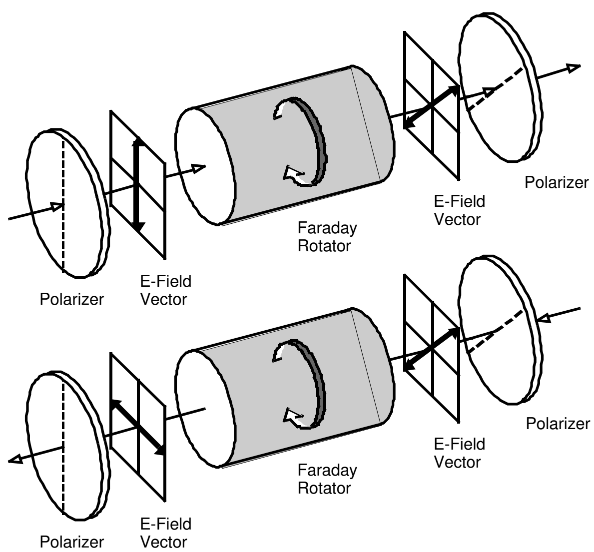


Figure 7.1-1. Polarization States of Light Passing through a Faraday Isolator from the left (top) and from the right (bottom)

It is first linearly polarized and then its polarization is rotated in the same direction and by the same amount as the light entering from the other direction. If the rotation angle is 45 degrees, the light entering from the right will be perpendicularly polarized to the second polarizer (analyzer). Using high quality optical components, rejection ratios in excess of 40 dB can be achieved.

Far Field

The term far field refers to the spatial regime in which the diameter of a laser beam increases linearly with distance. This also corresponds to the regime where Fraunhofer rather than Fresnel diffraction theory is applicable. A convenient way of getting access to the intensity distribution in the far field is to focus the beam. Doing this brings the intensity distribution at infinity into the focal plane of the lens.

Far Field Mode, Near Field Mode

The intensity distribution in the output from a Nd:YAG lasers is often specified for both the near and far field. The reason for this is that they can be dramatically different. Since Nd:YAG lasers, for efficiency reasons, are operated with large rod fill factors the beam will exhibit pronounced diffraction ripple in the intermediate field. In the far field most of these will have disappeared and the intensity profile will more closely resemble a Gaussian intensity distribution.

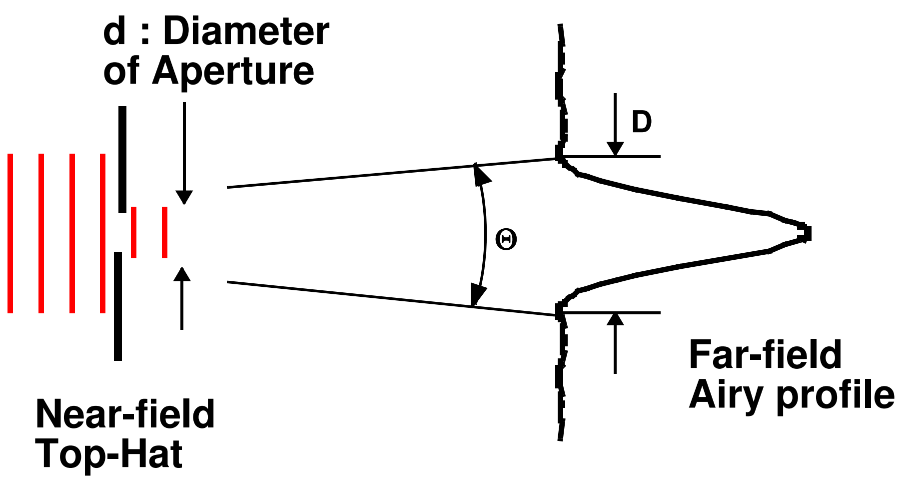
For the Infinity, the near field is defined as the position of the SHG crystal and the far field as the focal plane of a focusing lens positioned in the beam path. The function of the lens is to bring the mode at infinity to the focal plane where it can be examined. In the near field more than 95 % of the total energy must pass through an aperture with a diameter of 5.5 mm. In the far field more than 84 % of the energy must be contained within a cone having a divergence of 0.7 mrad full angle, Figure 7.1-2.

The specifications does thus not specify that the mode must be round or symmetrical. However, if the mode, especially if in the near field, exhibits asymmetries, this is a sign of the alignment not being optimized. If the system is operated under such conditions optics might be damaged.

Figure 7.1-3 shows how an Infinity like mode propagates through space according to theory. The z coordinate is the distance from the laser in meters. In the intermediate field the mode is to a large degree affected by diffraction from the aperture edges in the laser. Far away from the laser these have had time to diverge away from the mode and the result is a Gaussian like mode.

The calculated evolution shown in Figure 7.1-2 is for the IR output from the Infinity laser. Frequency doubling, tripling and quadrupling amplifies the intensity variations in the mode.

In Figure 7.1-2 d is the diameter of the aperture creating the top hat, λ the wavelength of the light, Θ the divergence angle and D the diameter between the first minimis of the Airy profile.



$$\Theta = 2.44 \frac{\lambda}{d}$$

Figure 7.1-2. Relationship Between Near Field Top Hat Profile and Far Field Airy Profile

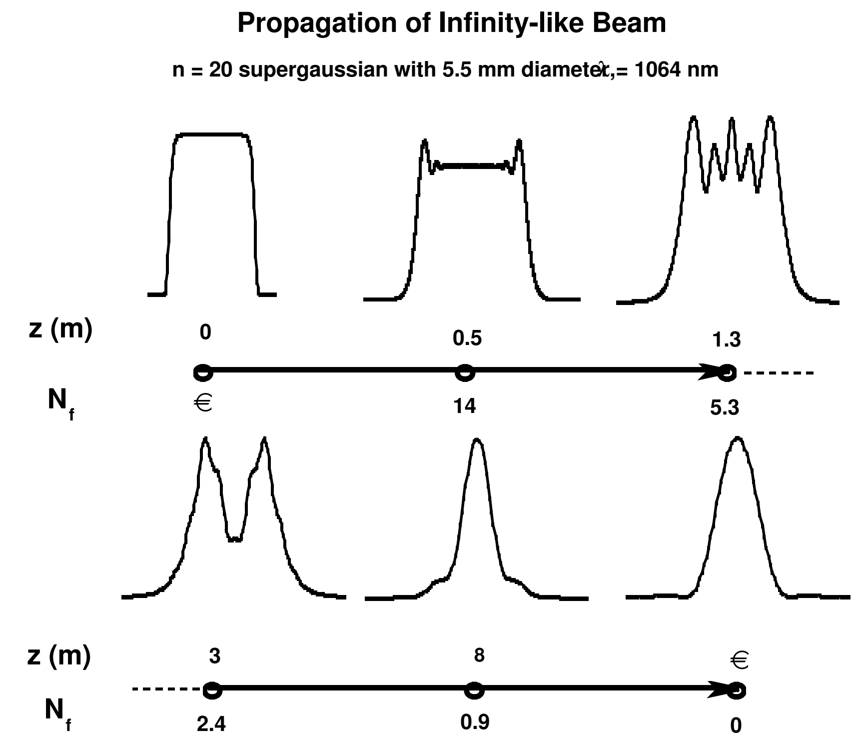


Figure 7.1-3. Evolution of Near Top Hat Mode Into an Airy Profile

Fault Handling

Within the multi-tasking structure of the Infinity control program, one task is solely responsible for all error detection and reporting. In the software nomenclature this is referred to as the “Fault task.” It runs in an infinite loop with a nominal 0.1 s delay between iterations. This implies that there will be a worst-case delay of about 0.1 s between the occurrence of a fault and its detection by the CPU.

Because of this slow response, certain critical faults are handled directly by the control board electronics. In these cases the CPU is merely reporting to the user on a system shutdown that has already happened. See the individual fault descriptions in the Trouble Shooting section of this manual.

The fault task also manages the flashing red LED on the CPU board. Each time through its loop, the fault task toggles the state of the LED, creating a blink rate of 5 Hz. It may be useful in some trouble-shooting situations to note whether the LED is flashing, especially if one suspects that the CPU might be dead. Under no circumstances should one attempt to run the laser when CPU's LED fails to flash.

When the Fault task detects a new fault, it posts an RS-232 message to the PC. It maintains a list of faults that have already been detected (the “active faults”) in order to avoid sending the same fault code repeatedly. The user interface program makes the user aware of a fault by opening the Fault window (if it is not already open) and sounding a warning tone on the PC speaker. The Fault window appears regardless of what the user is doing at the time and cannot be minimized.

The system CPU will only post a given fault once until the faults are “cleared”, which the user does by pressing the Clear button in the Fault window. Clearing faults actually has three effects: it closes the fault window, it causes the CPU in the power supply to clear the hardware latches that are applied to certain fault lines, and it clears the Fault task's list of active faults.

Each fault code received by the PC also causes an entry to be made in the message log. The log can be used to determine exactly when the fault occurred and (with some effort) the operating conditions at the time.

Imaging

The principle of imaging is illustrated in Figure 7.1-4 for three different cases. In this illustration f is the focal length of the lens, So the distance to the object and Si the distance to the image. The relationship between these quantities is:

$$\frac{1}{f} = \frac{1}{So} + \frac{1}{Si}$$

This means that if the object is two focal distances away from the lens then the image will be two focal distances away from the lens on the other side. If the object is one focal length away from the lens then the image will be in infinity, i.e. no image is formed. Assuming that the lens has a focal length of 50 cm and the distance between the lens and the object is 200 cm, the image will be formed at: 1/50–1/200=66.7 cm from the lens.

As can be seen in Figure 7.1-4, moving the object closer to the focal point of the lens moves the image further away from the lens. It also changes the size ratio between the object and the image. It can be shown that the size ratio is Si/So. Using the previous example where the object distance was 200 cm and distance to the image 66.7 cm the image would be 66.7/200=0.33 of the original size. From this it can be concluded that 1:1 imaging requires that the image and object

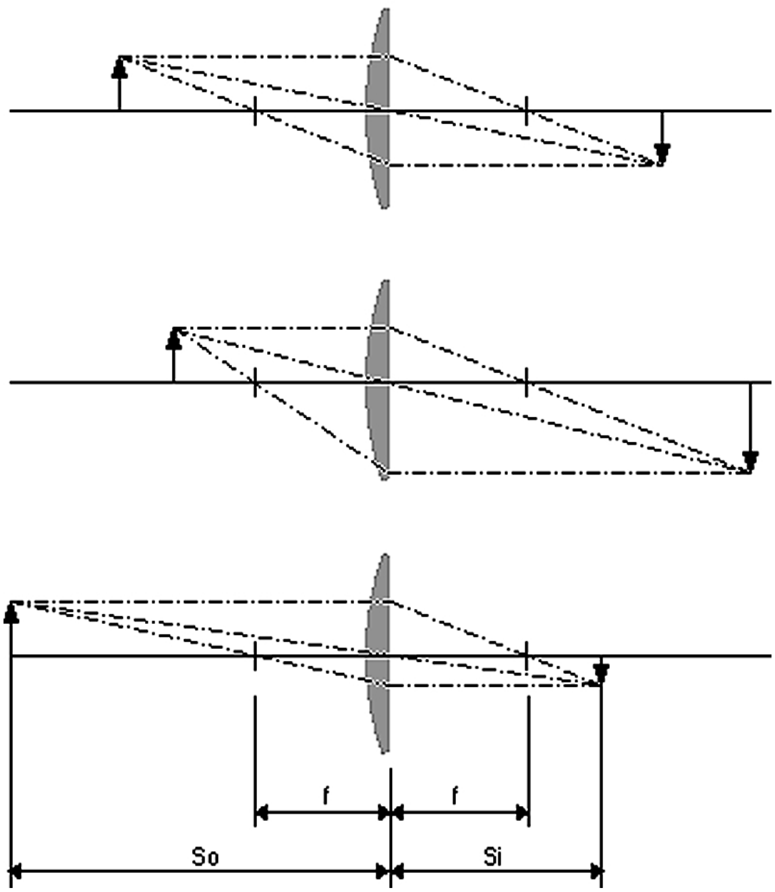


Figure 7.1-4. Imaging Using a Single Lens

distances are the same. This in turn means that they both must be equal to 2f. The shortest distance you can have between the object and the image is hence 4f.

Figure 7.1-6 shows an example of Relay imaging. The size and position of the arrows relative each other is preserved. This type of imaging is used in the Infinity system to image rod #1 into rod #2 and also to image the hard aperture on which the collimating lens is mounted into rod #1. The lenses in the spatial filter images the aperture into rod #1 and the two lenses and concave mirror in the relay imaging system images rod #1 into rod #2.

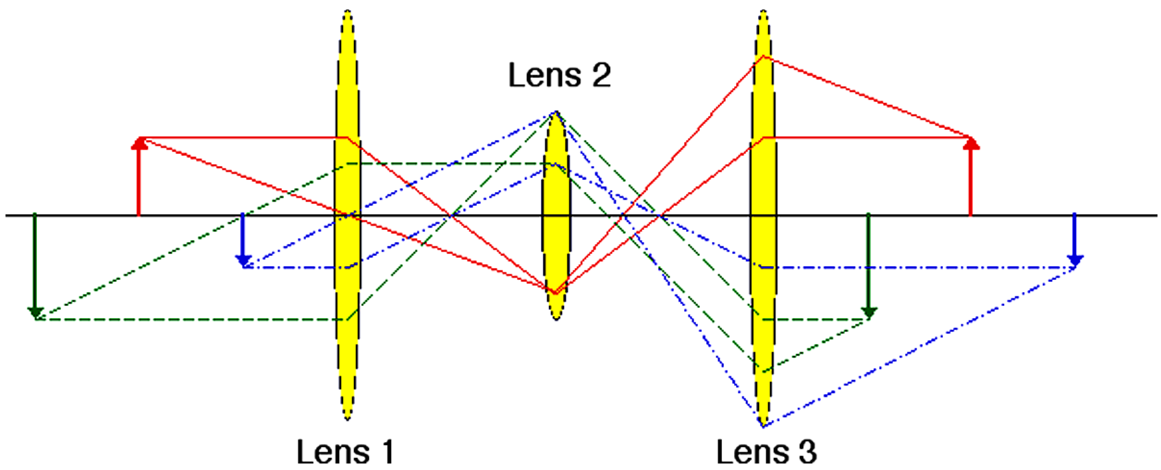


Figure 7.1-5. Relay Imaging Using 3 Lenses

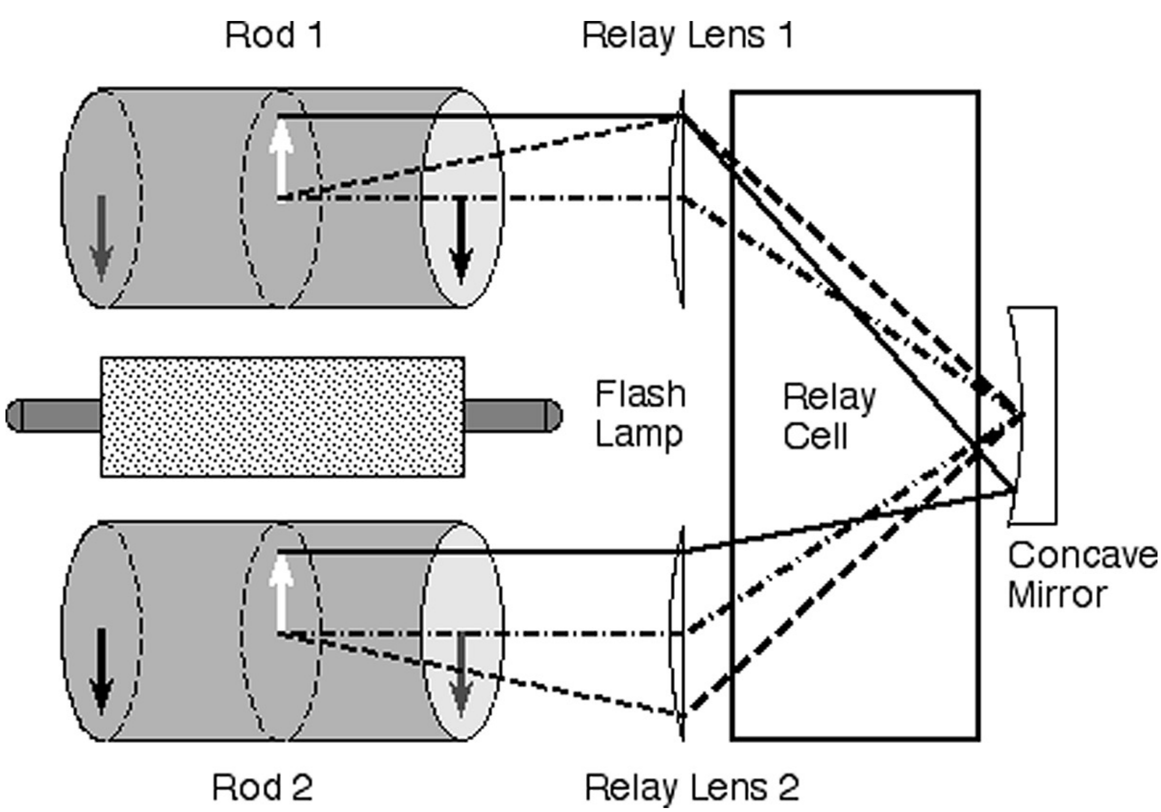


Figure 7.1-6. Principle of Relay Imaging in the Infinity System

Injection Seeding

An oscillator lasing in several longitudinal modes can be brought to lase in only one (single) longitudinal mode by injecting photons from an external narrow bandwidth source. When this is done, the laser process, instead of being building up from spontaneously emitted (noise) photons, builds up from the seeded photons. The result is a temporally smooth pulse without mode beating. The SLM pulse also occurs slightly before the unseeded pulse since orders of magnitude more photons are available during the initial build-up phase compared to what's provided by spontaneous emission. Because of the (Fourier) relationship between frequency and time, the SLM pulse exhibits a very narrow line width. A common line width for a non-seeded Nd:YAG oscillator is 1 cm^{-1} while an injection seeded system can be close to transform limited, 0.003 cm-1.

Invar

This material is an iron alloy which contains 31 % Nickel and 5 % Cobalt. It exhibits a very low thermal expansion coefficient which makes it attractive for application where thermal stability is important.

Master Oscillator Power Amplifier

This is a design concept in which the output from a low power well controlled master oscillator is amplified in one or more power amplifiers. Compared to the output from one big high power oscillator, a MOPA yields an attractive combination of high energy and excellent beam quality.

Non-Linear Crystals

Although all material can be made to behave in a non-linear way, assuming that the applied electrical field can be made large enough, the term non-linear crystals is usually reserved for materials that are birefringent as well. The reason for this is that the birefringence is a prerequisite for being able to utilize the non-linear property for frequency conversion. Important factors to consider when choosing a non-linear crystal are non-linear coefficient, damage threshold, transparency range, phase-matching range, available crystal size and chemical and mechanical stability.

The crystals most commonly used for frequency conversion in high power solid state lasers are tabulated below. There are other materials on the market that have higher non-linear coefficients but they do not have a damage threshold high enough to be of interest. The conversion efficiency of a crystal depends of course on the intensity since it is a non-linear process. Higher incident power leads to a higher conversion efficiency. There is, however, an upper limit on the laser intensity the crystal can be exposed to and hence also an upper limit on the conversion efficiency that can be achieved in practical devices.

KD*P is the crystal most commonly used because it can easily be manufactured in large sizes, has excellent optical properties and good stability. One disadvantage it suffers from is that it is hygroscopic, i.e. it is sensitive to water. Therefore it has to be mounted in a cell which can be kept dry. This is the material favored for frequency doubling of the Nd:YAG fundamental by most laser manufacturers. Its efficiency for this process is usually around 40 to 50%, sometimes up to 60% depending on mode quality and pulse intensity.

BBO is an excellent non-linear material. It is efficient, has a high damage threshold and it is transparent over a very wide range, from 190 nm to 2.5µm It is, however, unfortunately also very expensive.

KTP is a relatively new material and it has proven to have a very high efficiency for frequency doubling of the Nd:YAG fundamental, above 80% in some cases. It also has the advantage of being chemically inert. Another advantage of KTP is that it has a very wide angular tolerance, almost a factor of ten higher than that of KD*P. Unfortunately it suffers from photo induced darkening of the material under heavy power loadings over extended periods of time.

Table 7.1-1. Material Constants For Non-linear Crystals

SYMBOL	MATERIAL NAME	FORMULA	RELATIVE NON-LIN. COEFF.	DAMAGE THRESHOLD [MW/CM2]
BBO	Beta-barium borate	$ß\text{-}BaB_2O_4$	4.1	4600
KDP	Potassium dihydrogen phosphate	KH_2PO_4	1.0	400
KD*P	Potassium dideuterium phosphate	KD_2PO_4	1.06	400
KTP	Potassium titanyl phosphate	$KTiOPO_4$	29.2	250

Polarization

Light, or electromagnetic fields, are in physics described by Sinusoidal waves characterized by their amplitude, frequency, direction of propagation and the direction in which the electrical field oscillates. In Figure 7.1-7 the wave has an amplitude of E, a wavelength of λ, propagates along the Y-axis and its E-field oscillates along the Z-axis. This light is linearly polarized parallel to the Z-axis and unless it encounters a polarization changing optical element it will stay this way.

When calculating the effects of a birefringent element, e.g. a waveplate, on polarized light, using a coordinate system that has one of its principal axises parallel to the optical axis of the waveplate is convenient. Figure 7.1-8 shows a waveplate and an incoming wave that is linearly polarized in a plane tilted 45° with respect to the optical axis. It is easier to see the effect of the waveplate on the incoming wave if it is separated into its components, E_X and E_Z, along the X- and Z-axises, thin lines in Figure 7.1-8.

If the optical component in Figure 7.1-8 is a quarter wave plate the component parallel to the optical axis will travel faster than the component perpendicular to it. After having passed through the waveplate the components will be oriented as shown in Figure 7.1-9. The X-axis component is now quarter of a wave ahead of the Z-axis component. As can be seen in the figure the direction of the total E-field is no longer fixed in a plane but undergoes a rotating motion, it is circularly polarized.

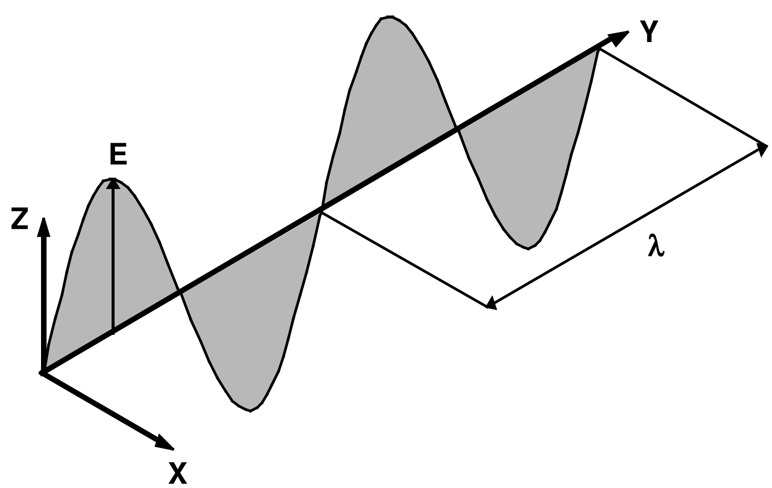


Figure 7.1-7. The E-field of Linearly Polarized Light

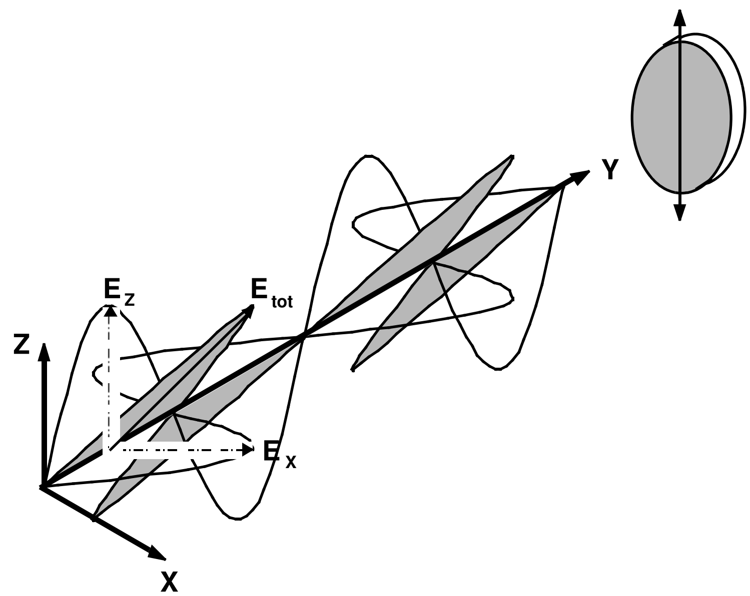


Figure 7.1-8. The E-field of Light Linearly Polarized in a Plane 45° to the X-axis

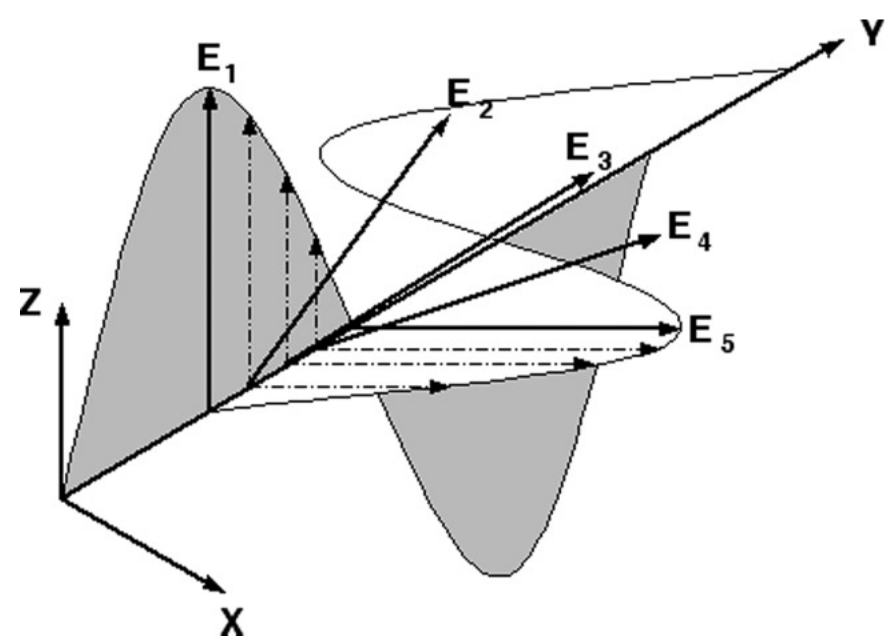


Figure 7.1-9. The E-field Components of Circularly Polarized light

POT

The amplifier assembly holding the rods and the flash lamp is called the pot, see Figure 7.1-10. The rods and the flash lamp are positioned inside a quartz flowtube that sits inside a ceramic diffuser. This geometry ensures that the rods and the lamp are efficiently cooled and at the same time that the light from the lamp is coupled into the rods uniformly.

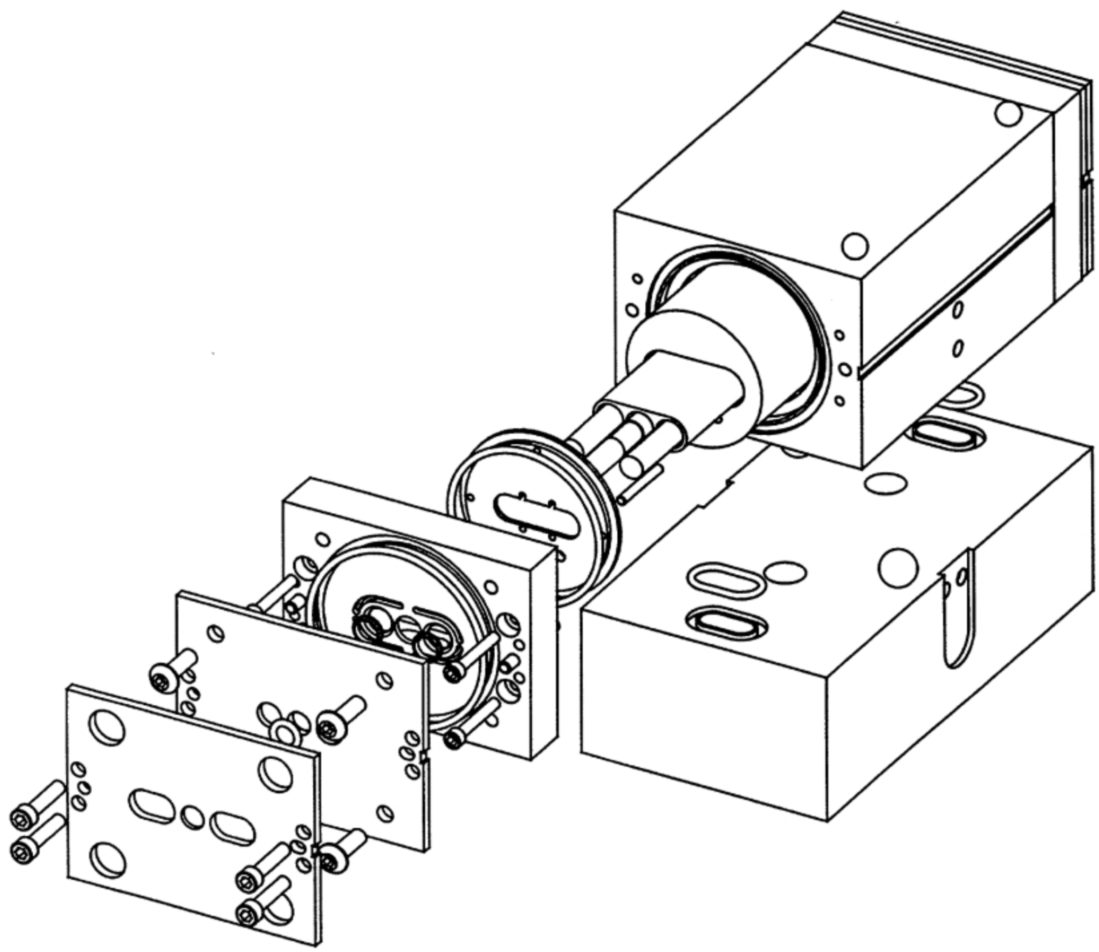


Figure 7.1-10. Exploded View of the POT Assembly and the POT Manifold

Q-Switch

A Q-switch is a device that can change the losses in a cavity. It is commonly used to achieve control over when the lasing starts. By using a QS the spiking behavior usually associated with non-QS systems can be avoided.

The Q-value is the ratio of stored energy to the losses pro cycle. The energy deposited into the cavity through the flash lamp is constant but the transmission is drastically changed by the QS.

Simmer Current

Before the lamp is flashed for the first time following start up, a short voltage spike is sent through the flash lamp to create an ionized corridor through the gas. This corridor then acts like a lightning rod for the main discharge, directing it through the gas in a well controlled manner.

Stress Induced Birefringence

When a Nd:YAG rod is heated a thermal gradient is created inside it. The temperature will reach a maximum value in the center of the rod and the surface temperature will equal that of the cooling medium. Due to the temperature increase the rod will expand. Since the temperature is different across the rod surface different parts of the rod want to expand to a different degree. When the hotter parts in the center of the rod wants to expand more than the cooler surrounding volumes stress (pressure) is created in the rod. The components of this stress is different in the radial and the tangential direction and also a function of radius, see Figure 7.1-11.

The stress in the material will give rise to a change in the refractive index. Since the stress is a function of radius and direction the refractive index will also be a function of radius and direction. This means that different incoming polarization states will see different refractive indexes, i.e. the material is birefringent. Figure 7.1-12 shows how the change in refractive index varies across the rod surface.

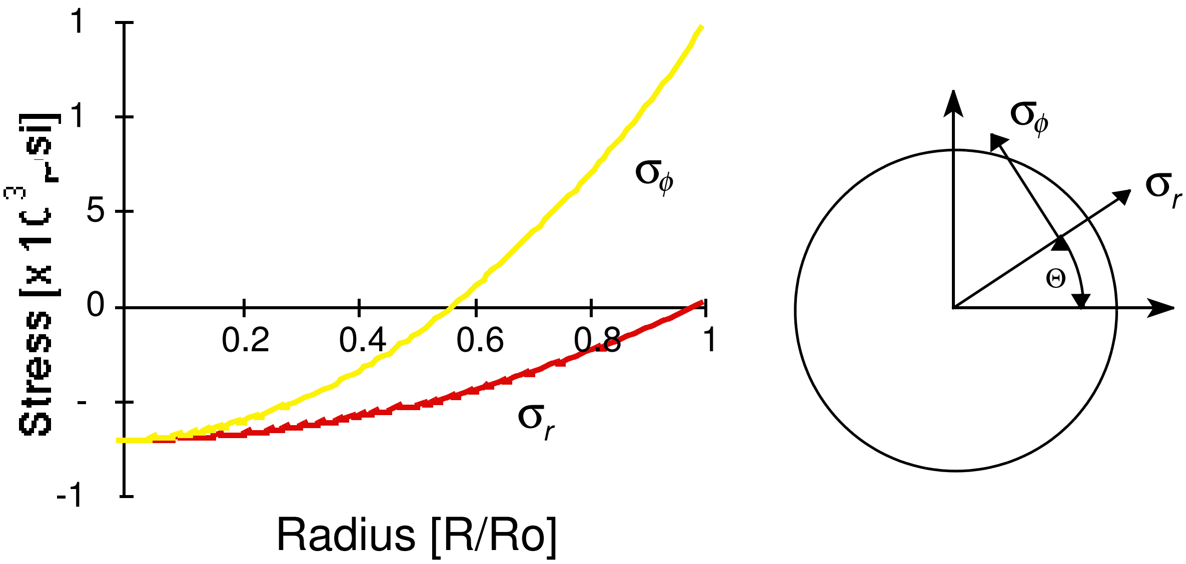


Figure 7.1-11. Radial and Tangential Stress Components in Thermally Loaded Nd:YAG Rod. The Center Temperature is 115°C and the Surface Temperature is 58°C.

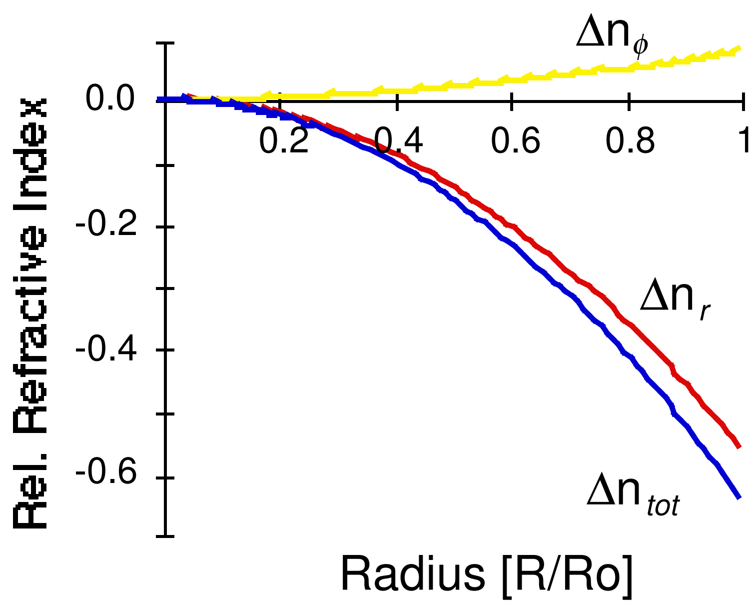


Figure 7.1-12. Stress Induced Change of Refractive Index

Super Gaussian Mode

A mode which can be described by the relationship I(x,y)=EXP(-x^n) where n>=2.

In the limit of increasing n's such a mode becomes a top hat distribution. As a consequence a super Gaussian mode has a higher efficiency in non-linear processes for a given peak power. Figure 7.1-13 shows how the Gaussian mode from the DPSS laser evolves into a super Gaussian mode as the pulse undergoes a saturated gain process.

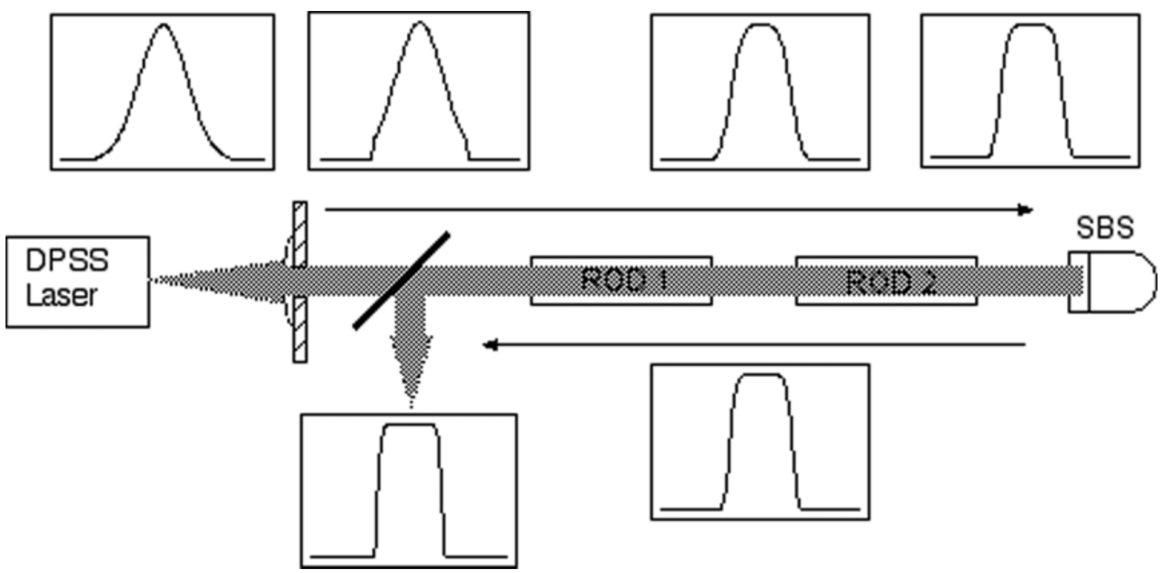


Figure 7.1-13. Transformation of Gaussian Beam from DPSS Laser into Top-Hat Mode

Thermally Induced Stress

As energy is deposited into the laser amplifier the Nd:YAG rods temperature increases. Even though this is far from the melting point of the material, it is still of interest because it gives rise to internal stress in the rod. One effect of this stress is to create a positive lens of the rod. This can be compensated for by using a corrective negative lens in the laser. A problem arises if the repetition rate or power load is changed since then the heat load varies and hence also the lensing effect. In the Infinity system this is automatically taken care of by the phase conjugate process. A second effect of the thermally induced stress is that the Nd:YAG rod becomes birefringent, i.e exhibits different refractive indices in different directions. A consequence of this is that polarized light entering the rod will exit depolarized. In the Infinity system this is compensated fore by rotating the polarization 90° between the rods. The depolarization the beam suffered when passing through the first rod is then compensated for when it passes through the second rod. It is important for this scheme to work that the beam path through the second rod is the same as through the first rod. This is accomplished using a relay imaging system between the rods (see above).

Transverse Electrical Mode

There are two principal types of modes in a laser cavity, called longitudinal (axial) and transverse (radial), respectively. The longitudinal mode determines bandwidth and coherence length of the pulse while the transverse determines beam divergence and beam diameter. In order to describe the mode structure in quantitative tearooms a nomenclature has been developed; a mode is said to be TEM[pqm] which is short for Transverse Electromagnetic Mode and the numbers pqm denotes the orders of the mode. The two fits letters, p and q, are connected to the number of the longitudinal mode. Usually it is desirable to have the laser run in the lowest possible transverse mode, which for most lasers means TEM_{00}. The reason for this is that this mode has the highest brightness, smallest beam diameter and follow Gaussian beam optics.

Unsaturated Gain

Also called small signal gain. The gain that is achieved when the amplified signal is still small compared to the saturated signal. This means that the extracted energy is smaller than the available stored energy.

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FSB's

NO.	DATE	TITLE
282	06/30/1995	New Installation Procedures for Infinity 40-100
287	11/01/1995	Intensity Fluctuations in the IR
290	11/14/1995	Infinity Installations
292	12/08/1995	DPSS Laser Electronics
293	12/08/1995	Infinity Noise Specifications
305	04/02/1996	Harmonic Mount Upgrades
315	9/30/1996	Part Number Change for Head and Power Supply Boards
319	10/23/1996	CE-MARK Infinity Power Supply Change
321	2/17/1997	New Part Numbers for Pre-mounted Optics
322	2/17/1997	Power Supply Control Board Part Numbers
324	2/25/1997	Faraday Isolator Design Change
333	9/17/1997	Infinity Packing Procedure
340	3/18/1998	Intermittent Shutdown of the Infinity Laser System
382	4/1/2000	Infinity Water Level Low Fault

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Schematics

DRAWING TITLE	REVISION	DRAWING NUMBER	SHEETS INCLUDED
Cable, D.I. Water Sensor	B	0166-630-00	1
Power Supply Control Board	A	0167-823-00	0
Power Supply Control Board	B	0168-863-00	12
Booster Board	D	0164-777-00	3
Photocell Board	C	0166-267-00	1
Signal Distribution Board	C	0166-343-00	2
Head Board #1	A	0168-755-00	1
Head Board #2	A	0168-757-00	3
Energy Modulator Board	D	0166-726-00	0
QS-DPSS Controller Board	D	0166-920-00	0
DPSS Q-Switch Power Supply	B	0166-918-00	1
QS-DPSS Prelase Autolock Board	A	0167-781-00	1
Infnity Triggering BD	D	0167-849-00	1

Power Supply Interconnect

Head Interconnect

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